UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement High Speed 1.2V MOSFET SPICE Model Document

Version 1.0 Phase 4
2015/04/08

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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2_p1	2010/05/11	Kelly Pao	1. Original Release 2. The model had been passed the verification of new silicon data .
0.2_p2	2010/08/11	Shih-Hsin Yeh	1. Remove the typo in the model card. 2. Copy Igate model from L110E model.
0.3_p1	2010/10/22	Kelly Pao	1. Removed the m in igate area formula.(area='w*0.9*l*0.9*m*1e4*tcig'--> area='w*0.9*l*0.9*1e4*tcig') 2. Updated the wmax,lmax
0.4_p1	2010/10/28	Yealing Lin	1. Add Twell NMOS model (copy from Twin well NMOS). 2. Modify Igate model: (a) m=m to m=1. (b) area='w*0.9*l*0.9*1e4*tcig'to area='w*0.9*(l*0.9+8e-9)*1e4*tcig'. (c) 'rsh*nrd/m' to 'rsh*nrd'. (d) 'rsh*nrs/m' to 'rsh*nrd'.
0.5_p1	2010/12/24	YZ Wang	1. Added thermal noise enhancement model. 2. Revised the target values (10um/0.116um) based on EDR revision. 3. Triple well NMOS model was copied from twin well NMOS model.
1.0_p1	2012/09/14	Laney Fan	1. Version up to 1.0 based on 6 pcs Si data.
1.0_p2	2013/04/15	YZ Wang	1. Modify the Verilog-A file.
1.0_p3	2015/02/05	Kelly Pao	1. Modify subckt naming of tn model.
1.0_p4	2015/04/08	Kelly Pao	1. Modify subckt naming of igate model.

1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110E-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.2V n-ch MOSFET
1.2V p-ch MOSFET
1.2V n-ch Triple well MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of this compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that could be used. One is binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is global model, which means one scalable parameter set is provided and it could cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for its many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Model usage

The compact model is provided in various formats. It's guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNSP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib" CaseName

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

- Add the following statement in the input file to the simulator.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

For 1.2V n-ch MOSFET: n_12_hsl110e_ig
For 1.2V p-ch MOSFET: p_12_hsl110e_ig
For 1.2V n-ch MOSFET: n_12bpw_hsl110e_ig (for Triple well model)

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range :(After 90% shrink)

For 1.2V n-ch MOSFET:

0.108 μm <= L_{DES} <= 45 μm
0.144 μm <= W_{DES} <= 90 μm

For 1.2V p-ch MOSFET:

0.108 μm <= L_{DES} <= 45 μm
0.144 μm <= W_{DES} <= 90 μm

Adopt multi-finger structures if WDES > 9um.
0.306um<=SA,SB<=2.2um ,SD>0.36um
Note:SA,SB=2.2um is the reference dimension of the STI effect.

Voltage range:

For 1.2V n-ch MOSFET:

0V <= V_{GS} <= 1.2V (*1.1)
0V <= V_{DS} <= 1.2V (*1.1)
-1.2V <= V_{BS} <= 0V

For 1.2V p-ch MOSFET:

0V >= V_{GS} >= -1.2V (*1.1)
0V >= V_{DS} >= -1.2V (*1.1)
1.2V >= V_{BS} >= 0V

Temperature range:

For both 1.2V n-ch and p-ch MOSFET:

-50 ℃ ~ +125 ℃

UMC L110E is UMC L130E 90% direct shrunk option.The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 8 nm) 9/0.116
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Sloting inserting	metal line width>5.556um	mwtal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

1.Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.

2.Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 1.2 V n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	**MOSFET AC Model**	**Junction DC Model**	**Junction AC Model**
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGMNS	QGMNS	QGMNS	QGMNS
Wafer Number	#11	#11	#11	#11

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008µm). The dimension is after 90% shrink and 12nm sizing channel length back.

Table 2-2 Electrical Parameters for NMOS

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	0.216	0.193	0.216	0.216
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/ μm	16.01	16.03	16.00	16.00
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.116 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	0.356	0.325	0.353	0.356
V_{TSAT}	V_{DS} = 1.2 V	V	0.250	0.253	0.252	0.250
Body Effect V_T shift	V_{DS} = 1.2 V V_{BS}= 0 ~ -1.2 V	V	--	0.025	0.032	0.029
DIBL V_T shift	V_{DS}= 0.05~1.2 V	V	0.106	0.071	0.101	0.106
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	680	660.03	665.35	680
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	4.0E-09	2.1E-09	2.6E-09	3.6E-09
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.05 V	V	0.175	0.158	0.175	0.175
V_{TSAT}	V_{DS}= 1.2 V	V	--	0.155	0.159	0.159
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	18.19	18.17	18.18	18.18
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μm / 0.116 μm)						
V_{TLIN}	V_{DS} = 0.05 V	V	0.301	0.296	0.296	0.300
V_{TSAT}	V_{DS}= 1.2 V	V	0.213	0.218	0.210	0.212
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	964.07	943.34	952.14	963.64
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	6.1E-09	5.0E-09	5.9E-09	6.8E-09
Capacitance						
CJS	V_J = 0 V	F/m^2	7.20E-04	7.03E-04	6.97E-04	7.20E-04
CJSWS	V_J = 0 V	F/m	9.94E-11	1.07E-10	1.06E-10	9.94E-11
CJSWGS	V_J = 0 V	F/m	3.03E-10	4.50E-10	3.03E-10	3.03E-10
C_{OVL}	V_{DS} = V_{GS} = -0.5 V	F/m	3.01E-10	2.49E-10	2.00E-10	3.01E-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10	1	0.8	0.5	0.24	0.18	0.15	0.13	0.116
10	X	X	X	X	X	X	X	X	X
1	X				X				X
0.8					X				X
0.72	X								
0.65					X				X
0.43	X		X	X	X	X	X	X	X
0.25	X		X		X	X	X	X	X
0.16	X		X		X		X		
0.14	X		X		X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	1.2	1.1
V_{GS} (V)	0	1.35	0.05
V_{BS} (V)	0	-1.2	-0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	1.35	0.05
V_{GS} (V)	0.6	1.2	0.15
V_{BS} (V)	0	-1.2	-1.2

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (L_{DES} *0.9+8nm and W_{DES} *0.9) respectively.

- ***I_D vs. V_{DS} Plots***

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

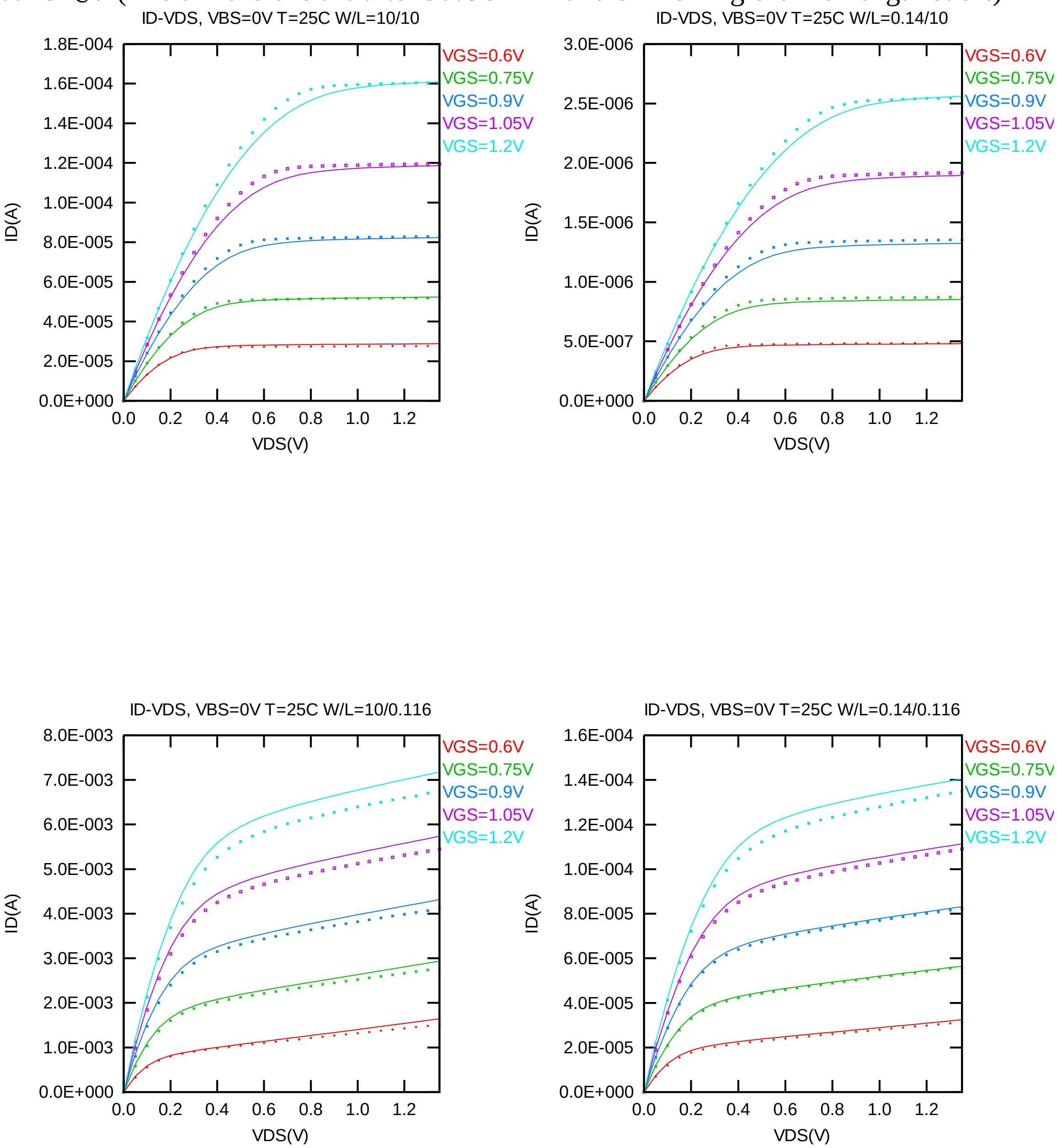


Figure 2-1 I_D vs. V_{DS} at V_{BS} = 0 V for NMOS

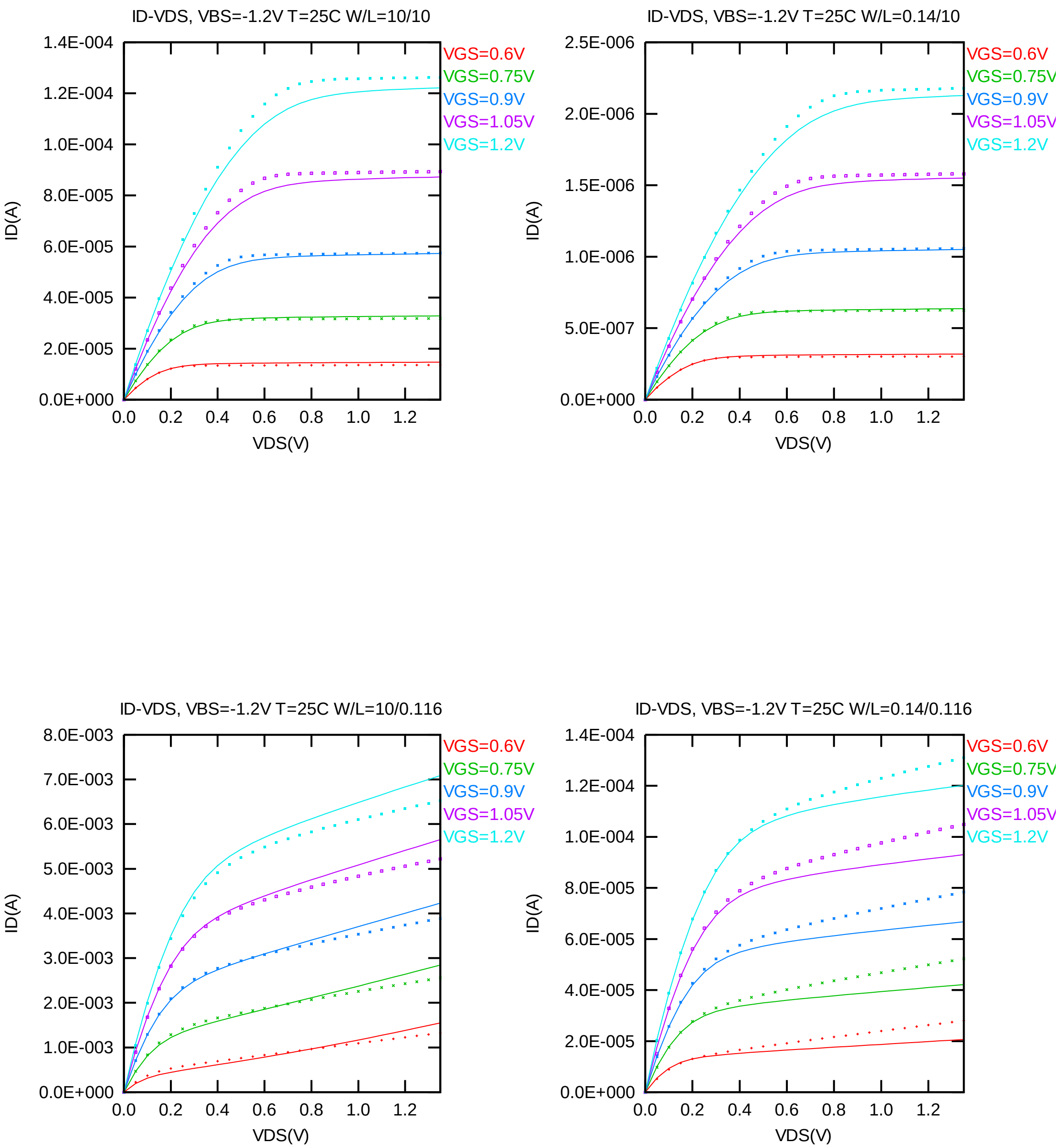


Figure 2-2 I_D vs. V_{DS} at V_{BS}=-1.2 V for NMOS

➢ G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

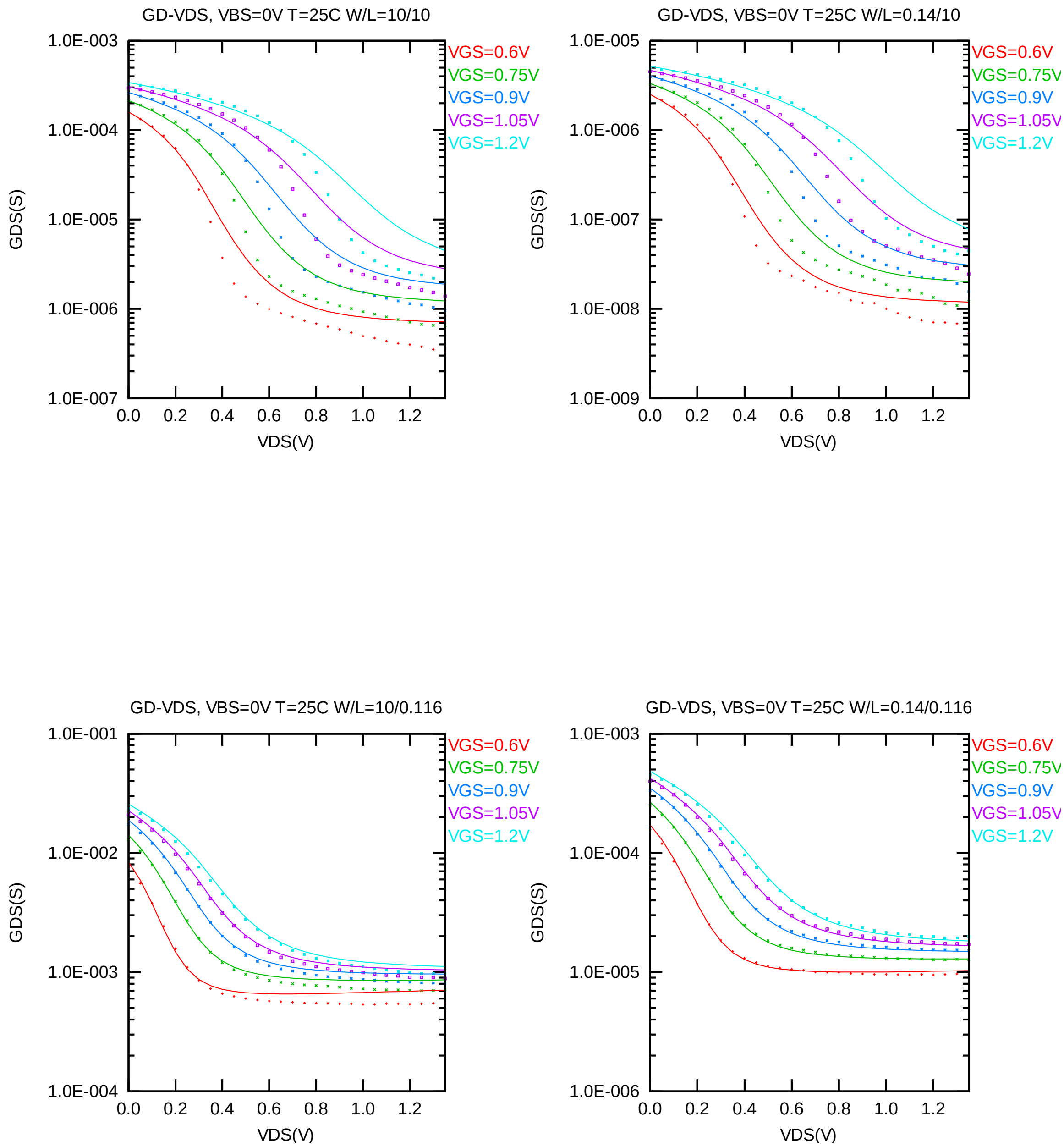


Figure 2-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for NMOS

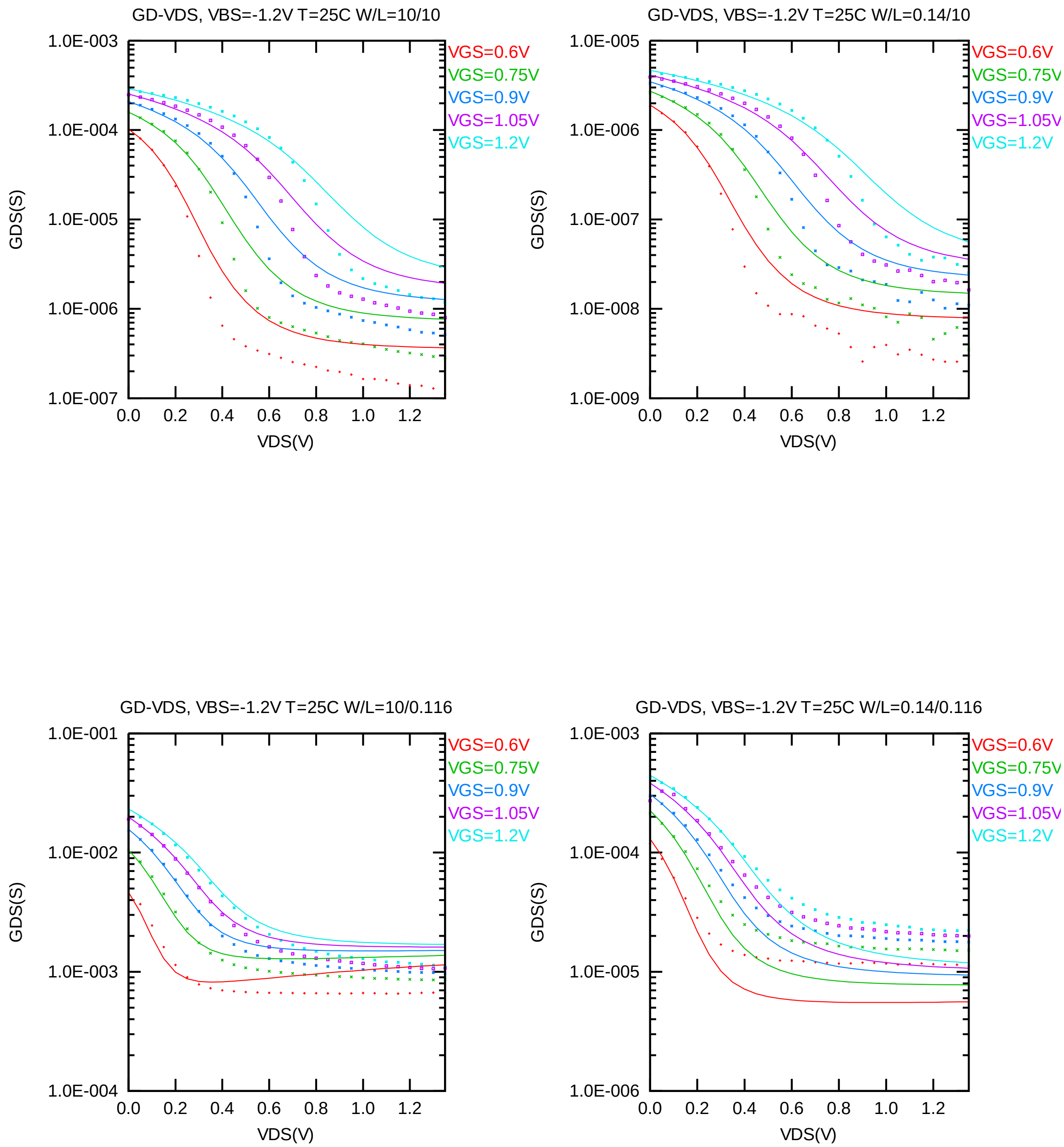


Figure 2-4 G_{DS} vs. V_{DS} at V_{BS} = -1.2 V for NMOS

➢ *I_D vs. V_{GS} Plots (linear scale)*

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

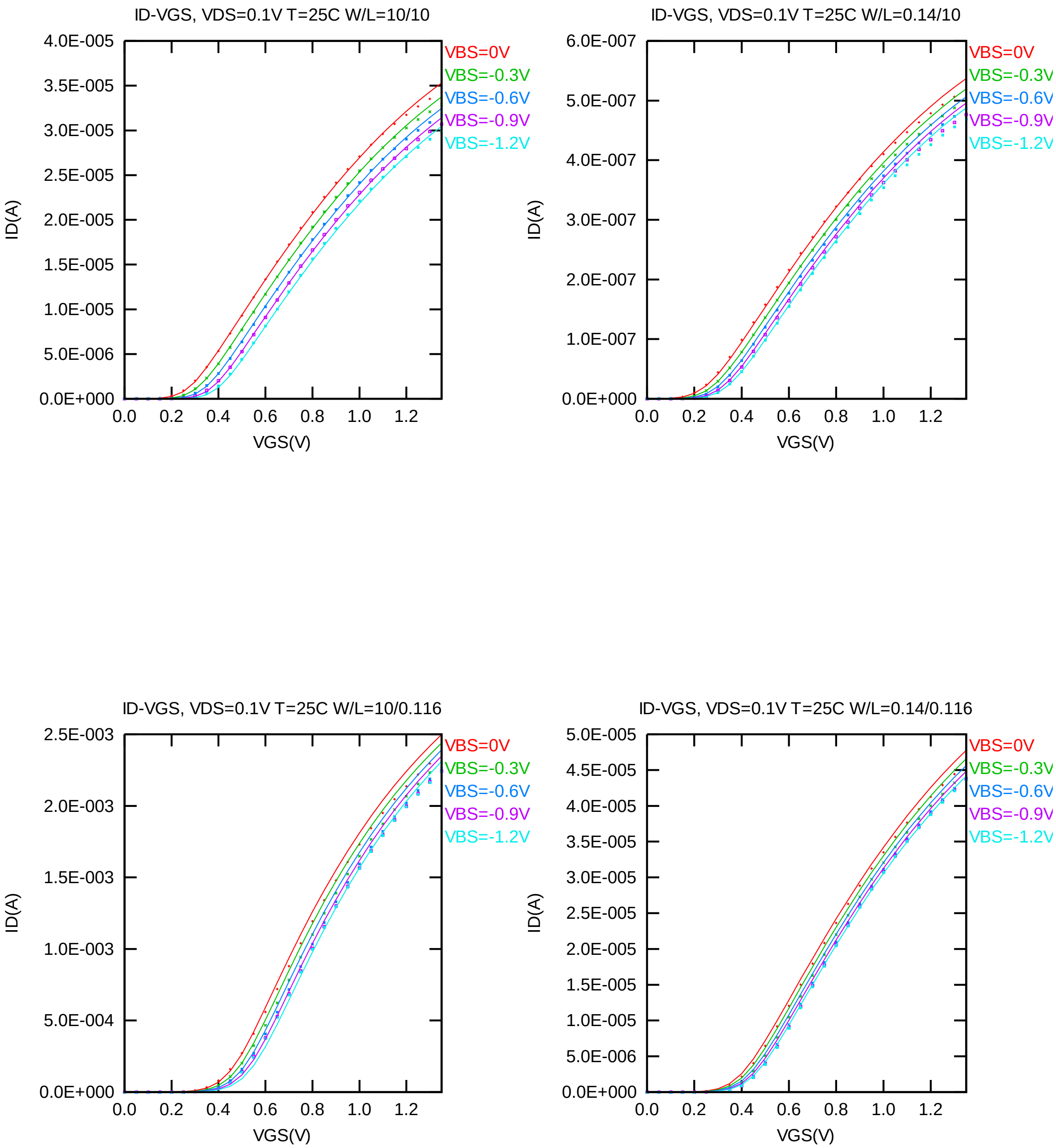


Figure 2-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

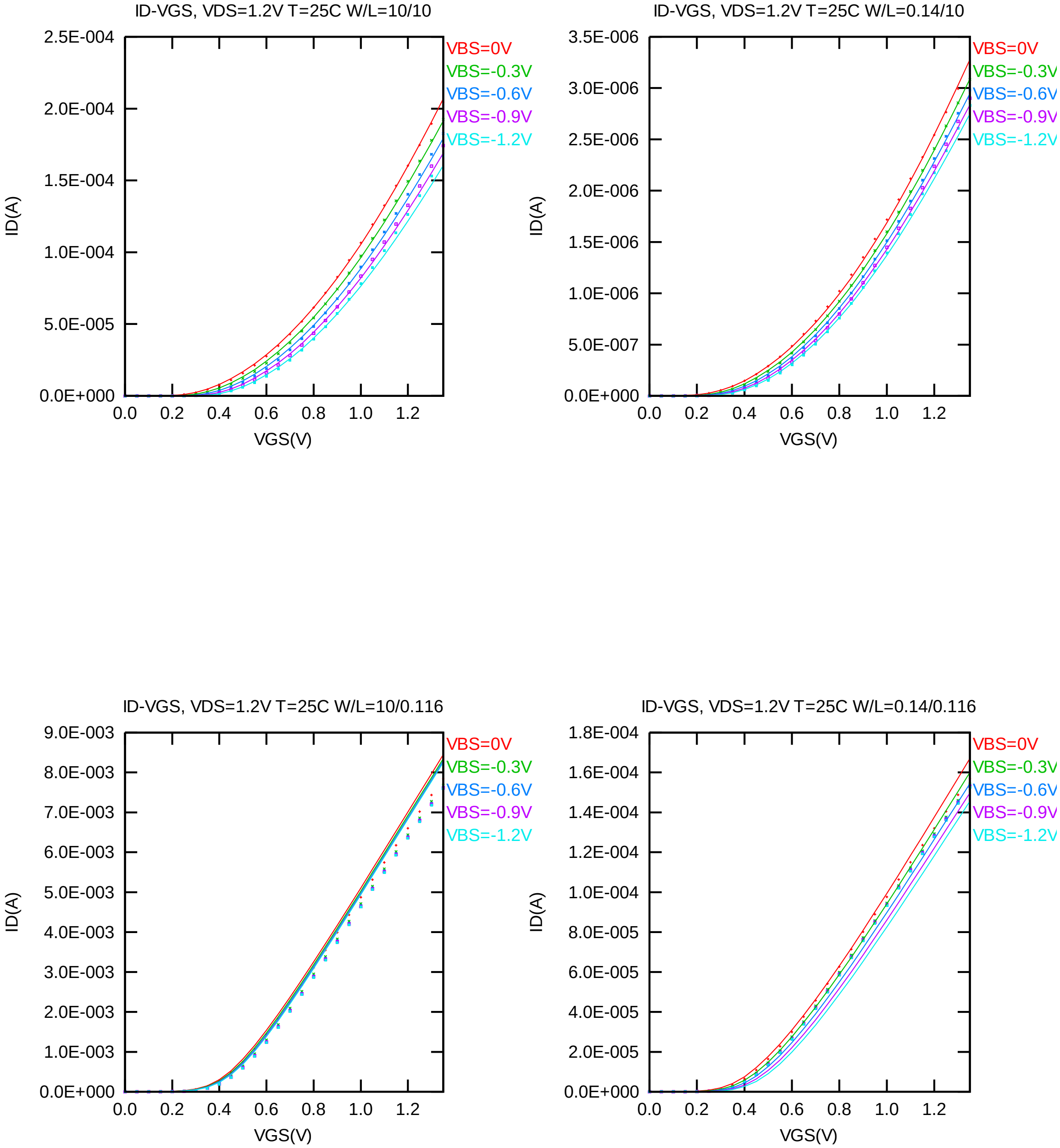


Figure 2-6 I_D vs. V_{GS} at V_{DS} = 1.2 V for NMOS

➢ I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

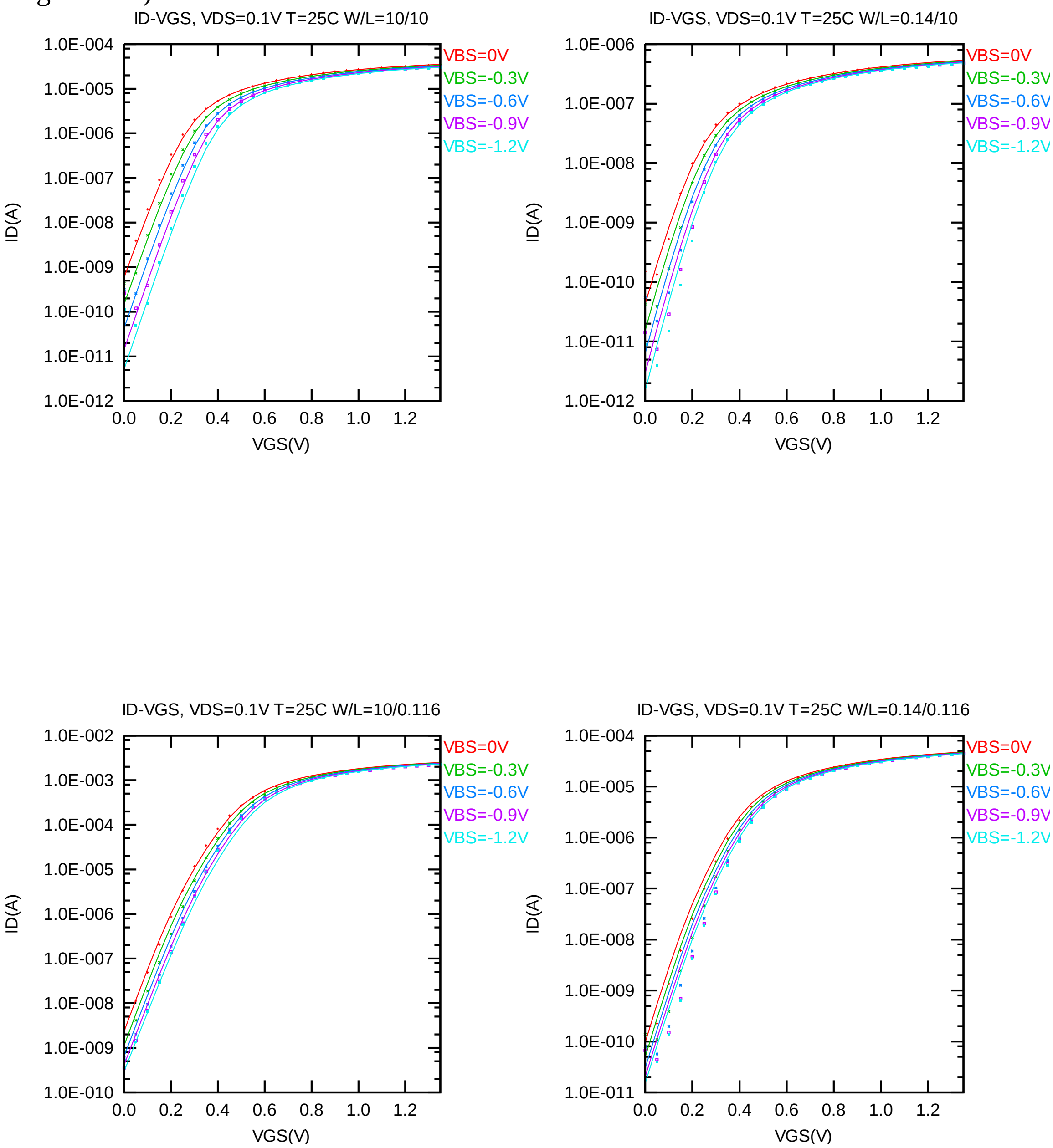


Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

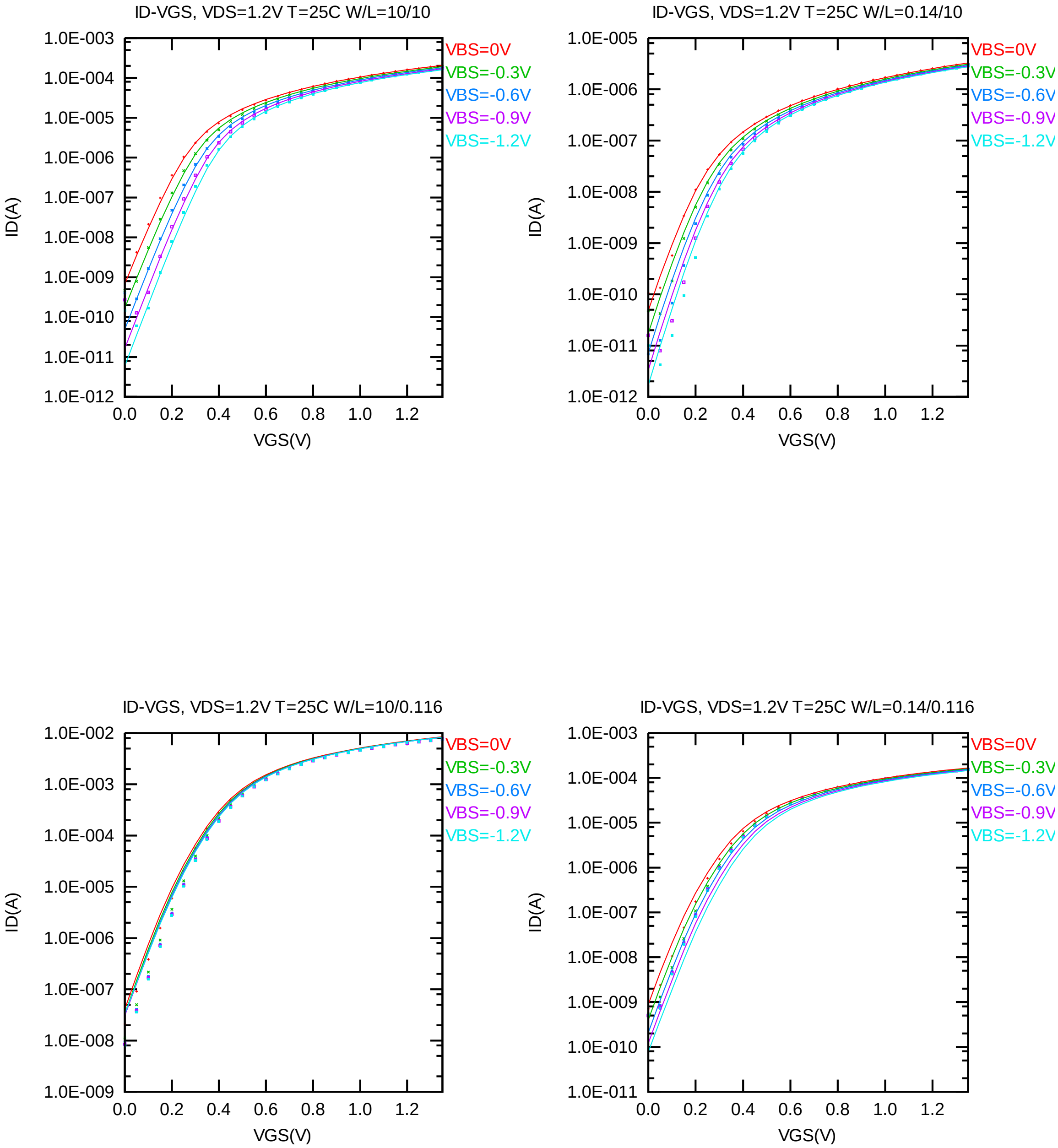


Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS} = 1.2V for NMOS

➢ G_M vs. V_{GS} Plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

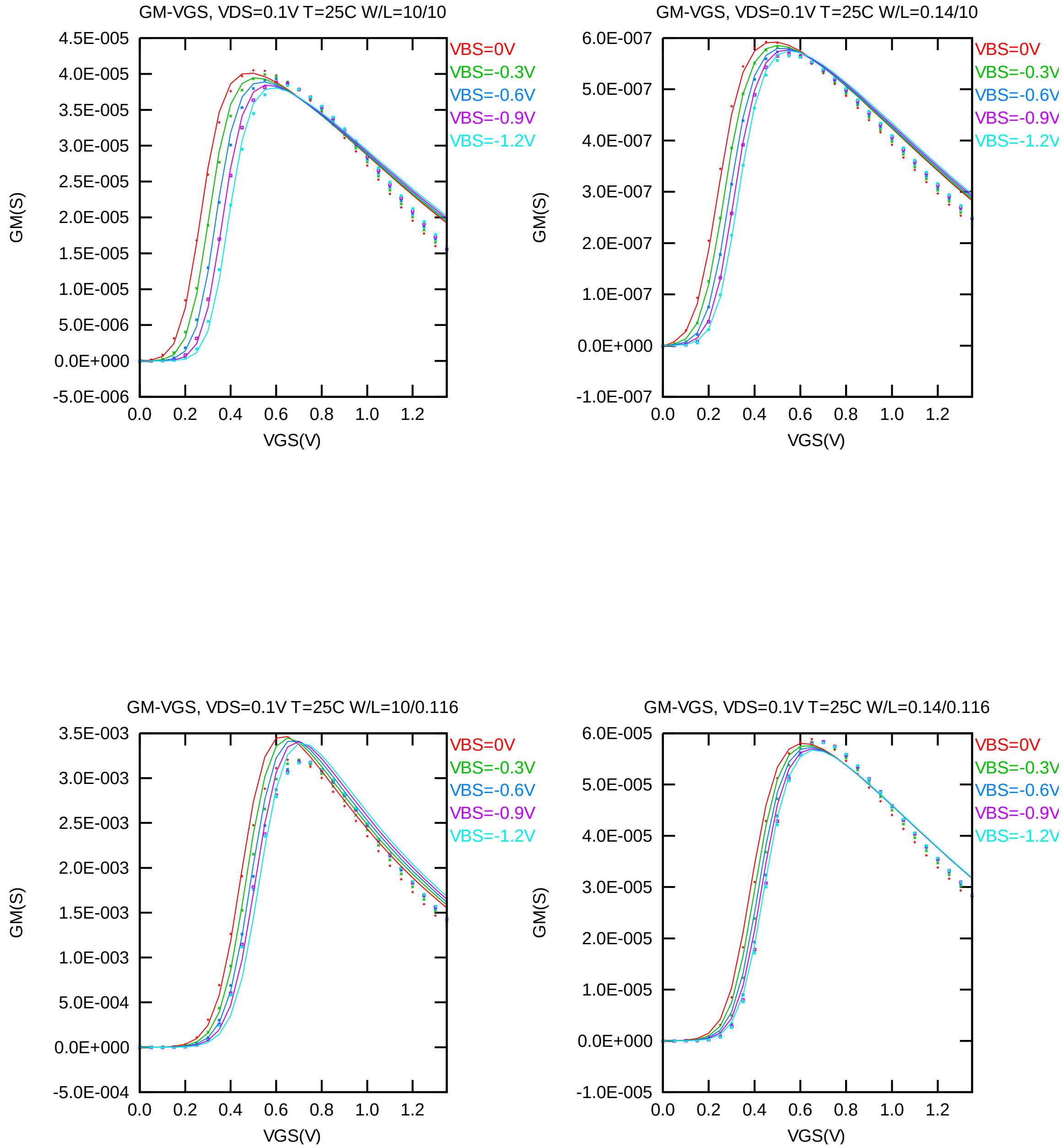


Figure 2-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for NMOS

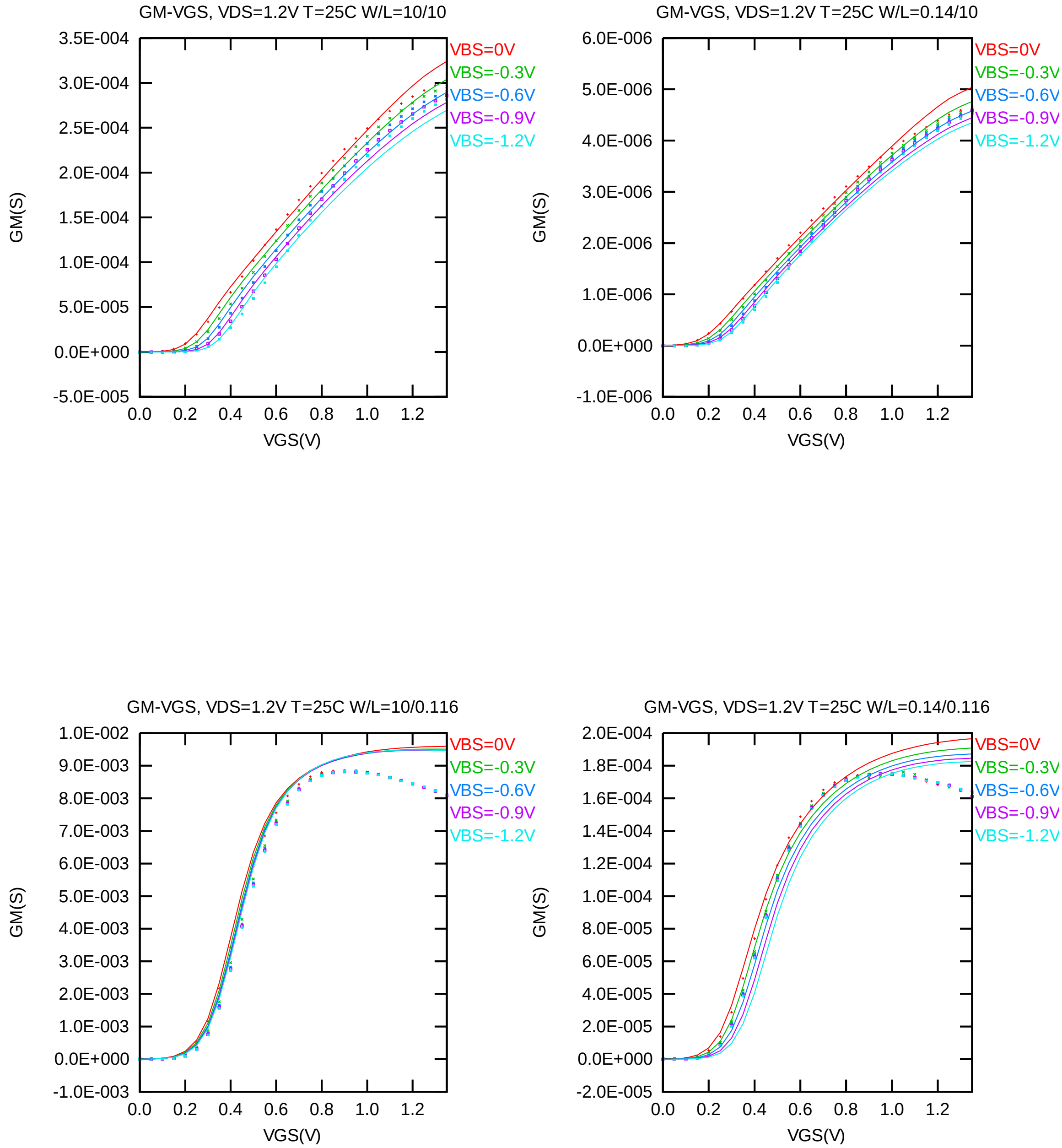


Figure 2-10 G_M vs. V_{GS} at V_{DS} = 1.2 V for NMOS

➢ *VT vs. LDES & VT vs. WDES*

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.08 μm) at 25 ℃. The plots show the physical dimension. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

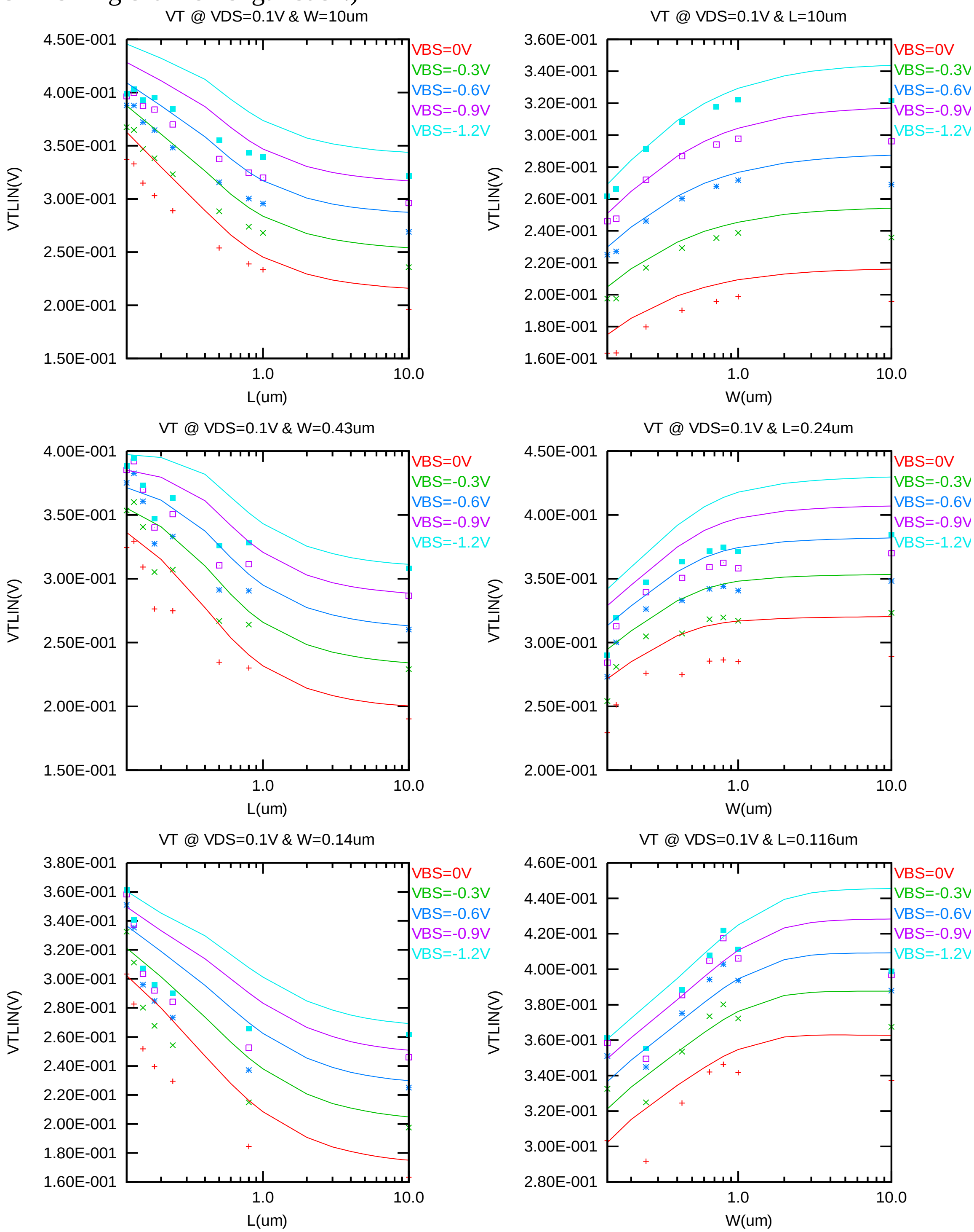


Figure 2-11 V_T vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for NMOS

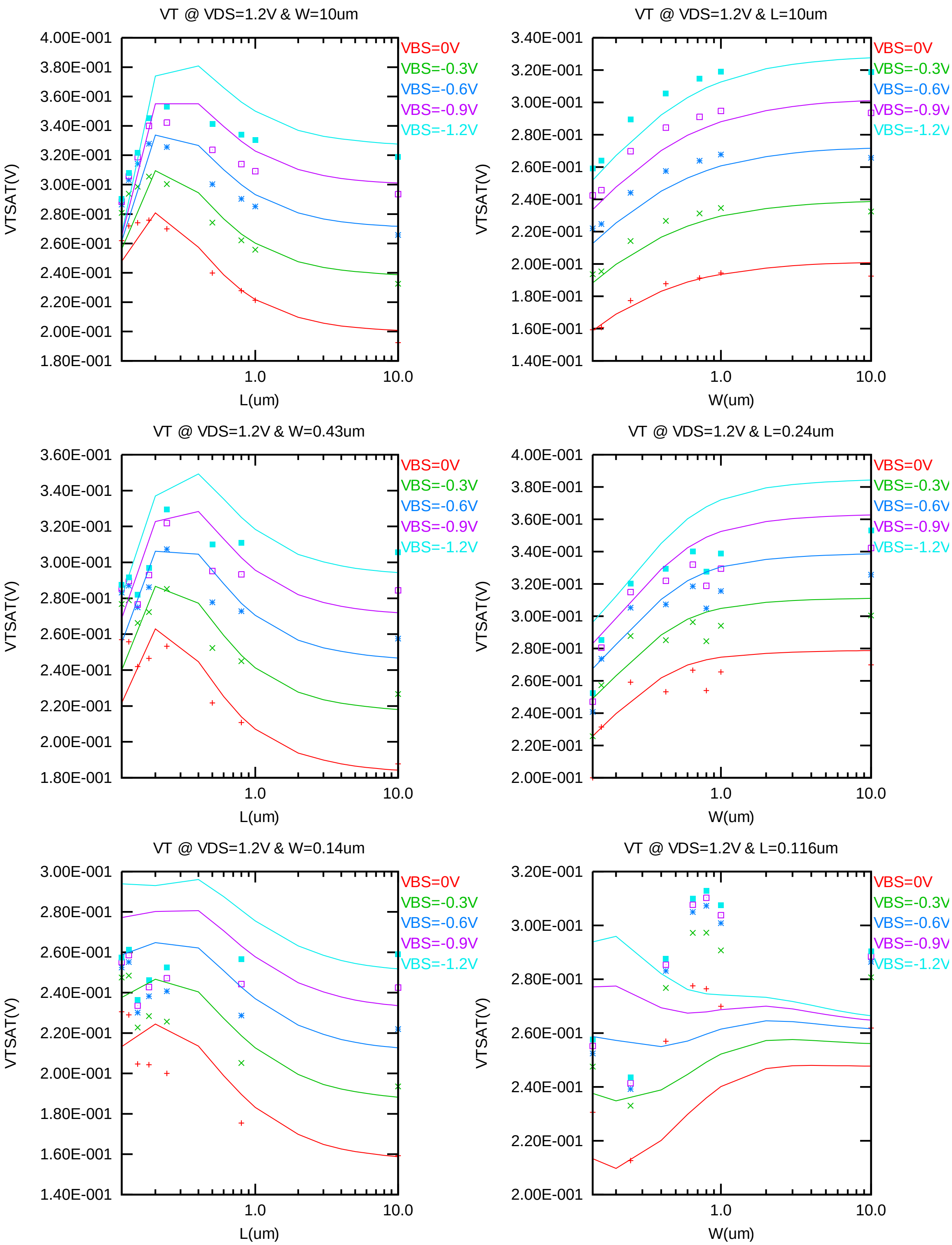


Figure 2-12 V_T vs. channel length & channel width for higt drain bias (V_{DS} = 1.2 V) for NMOS

➢ ***I***

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at V_{DS} = V_{GS} = 1.2V for V_{BS} = 0 V and V_{BS} = -1.2 V at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

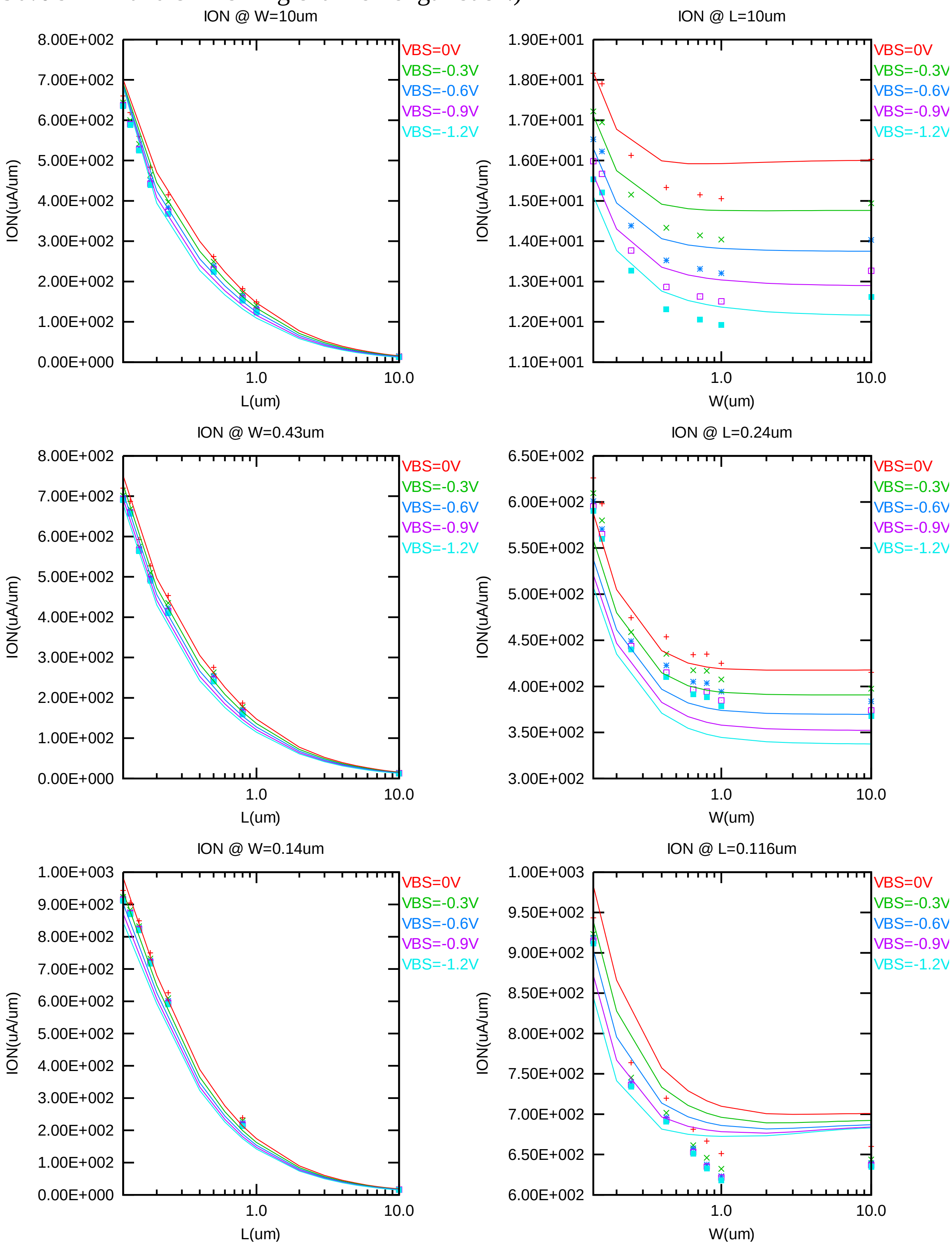


Figure 2-13 Saturation current I_{ON} vs channel length and channel width for NMOS

➢ ***V_T vs. Temperature***

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at I_D = 300nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008 µm). (The dimensions are after 90% shrink and 8nm sizing channel length back.)

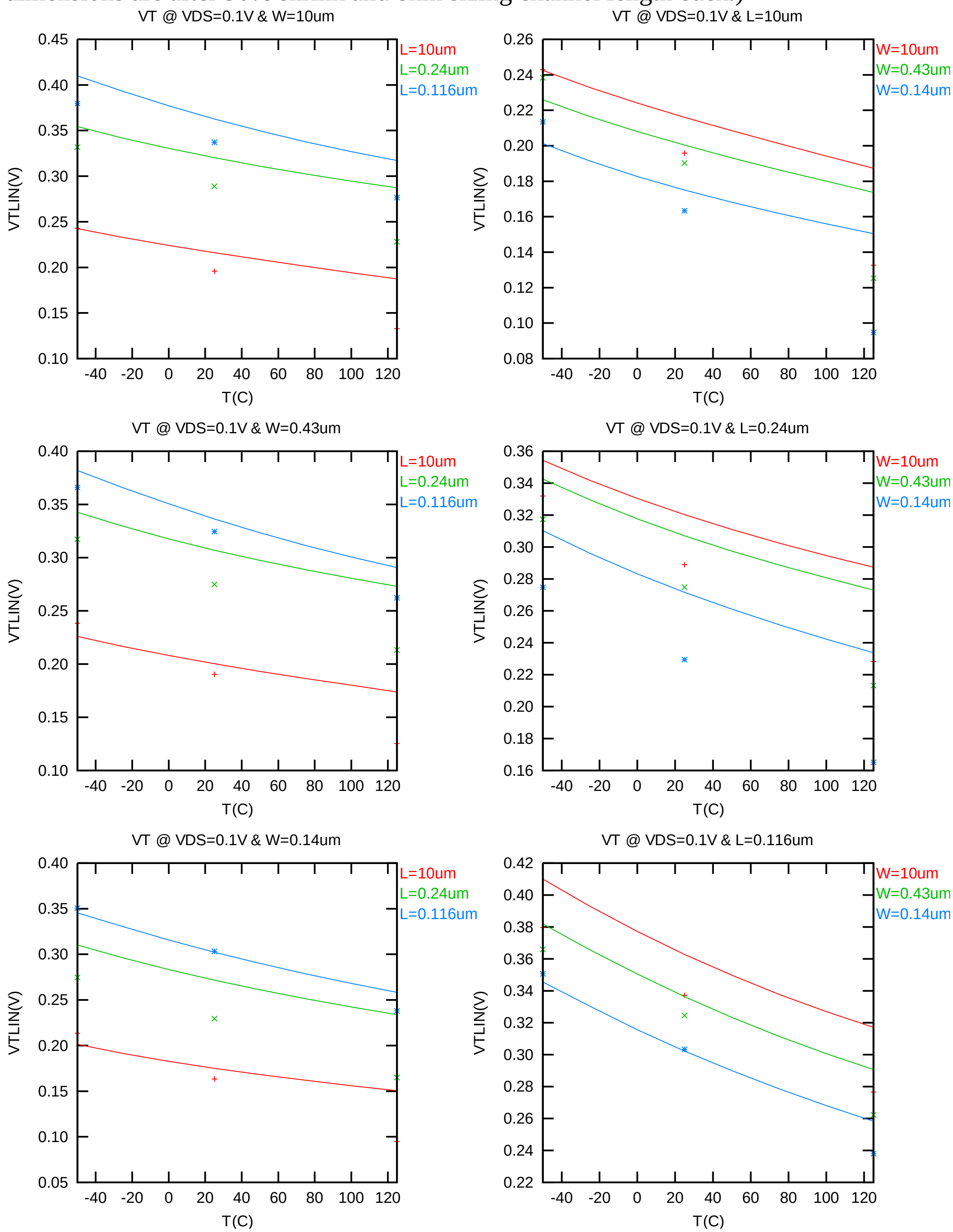


Figure 2-14 V_{TLIN} vs temperature for NMOS

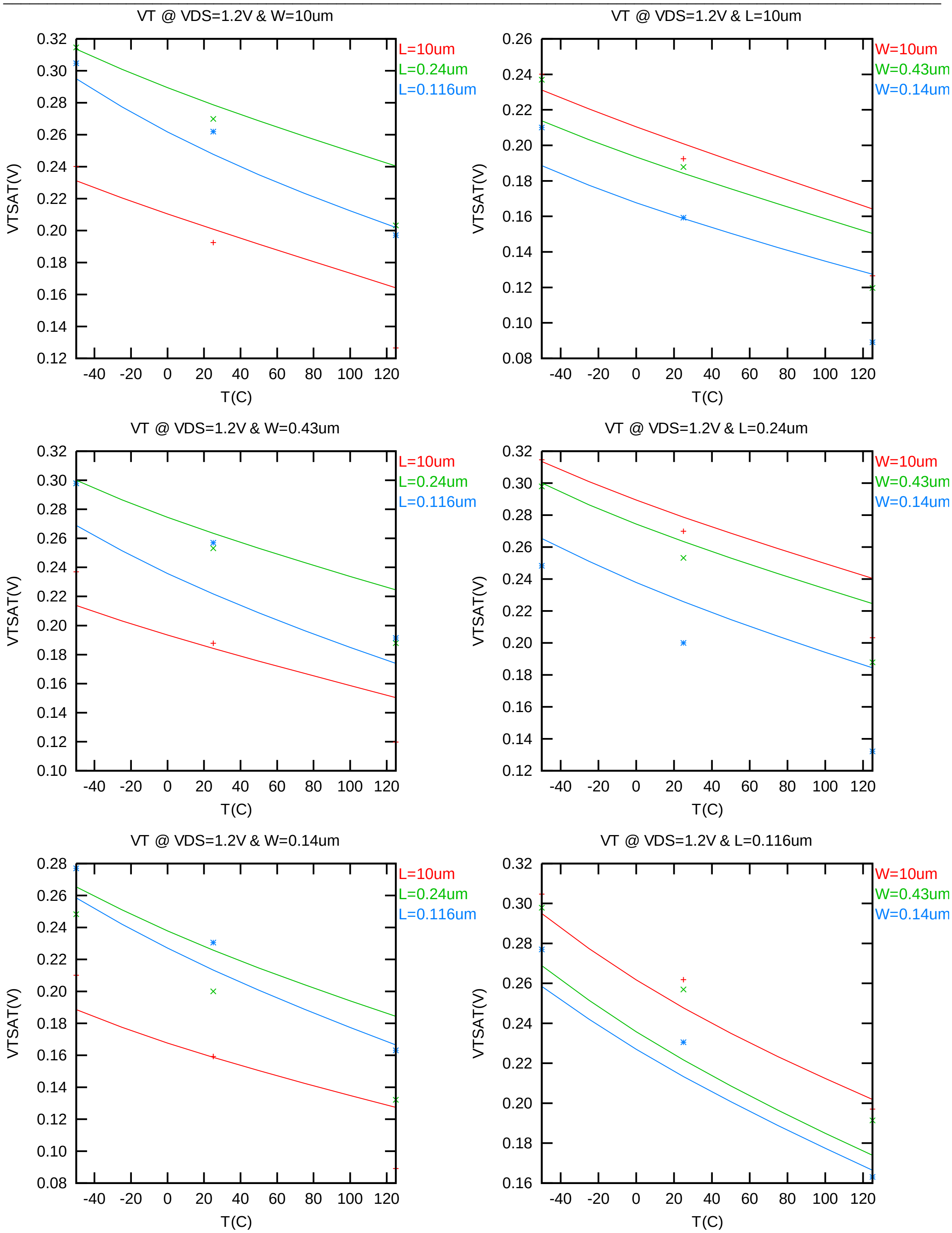


Figure 2-15 V_{TSAT} vs temperature for NMOS

➢ *I_{ON} vs Temperature*

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2$ V, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

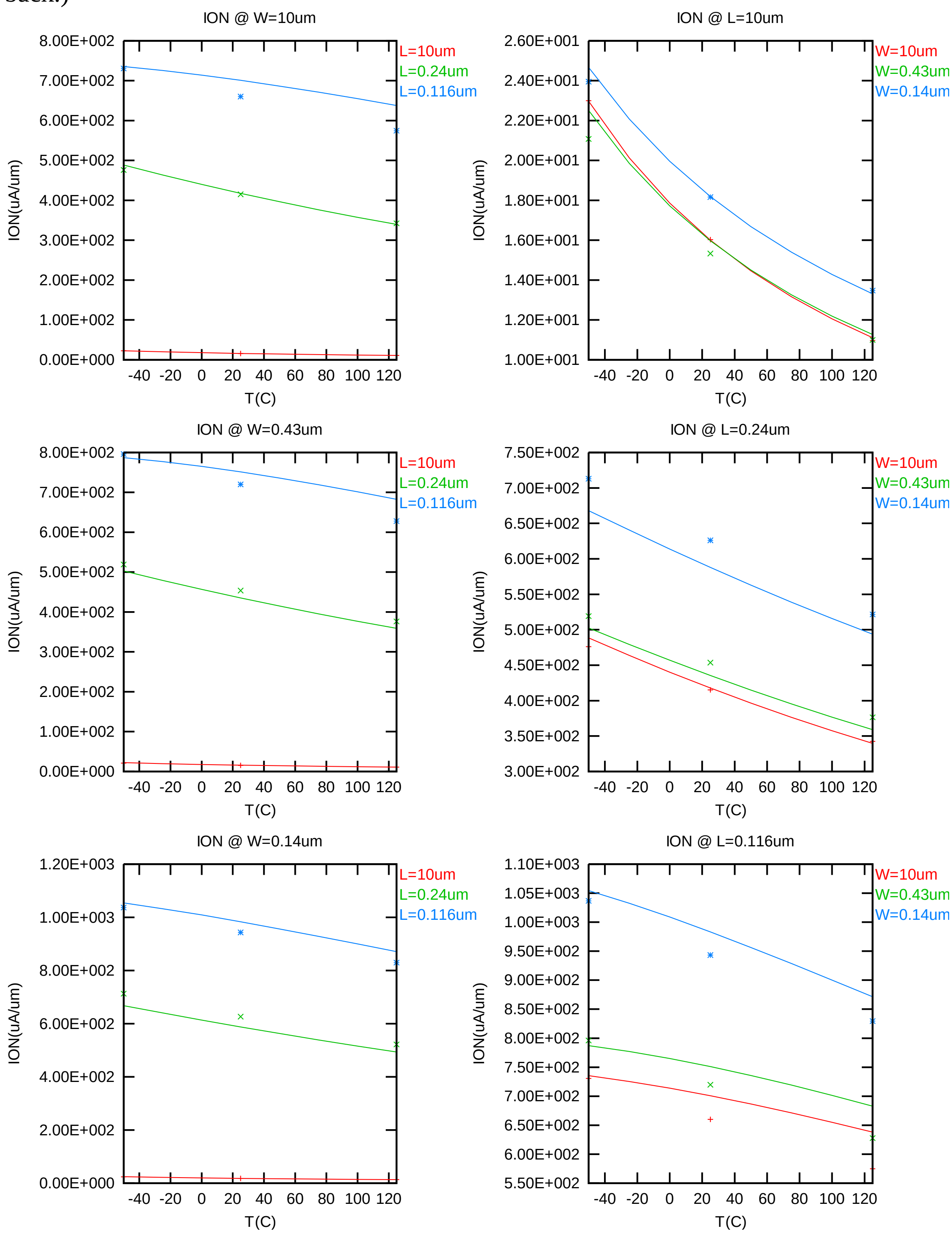


Figure 2-16 Saturation Current vs temperature for NMOS

2.3.4 I_{gate} Extraction plot

➢ ***I_{GATE} vs V_{GS} @ V_{BS} =0 @25C***

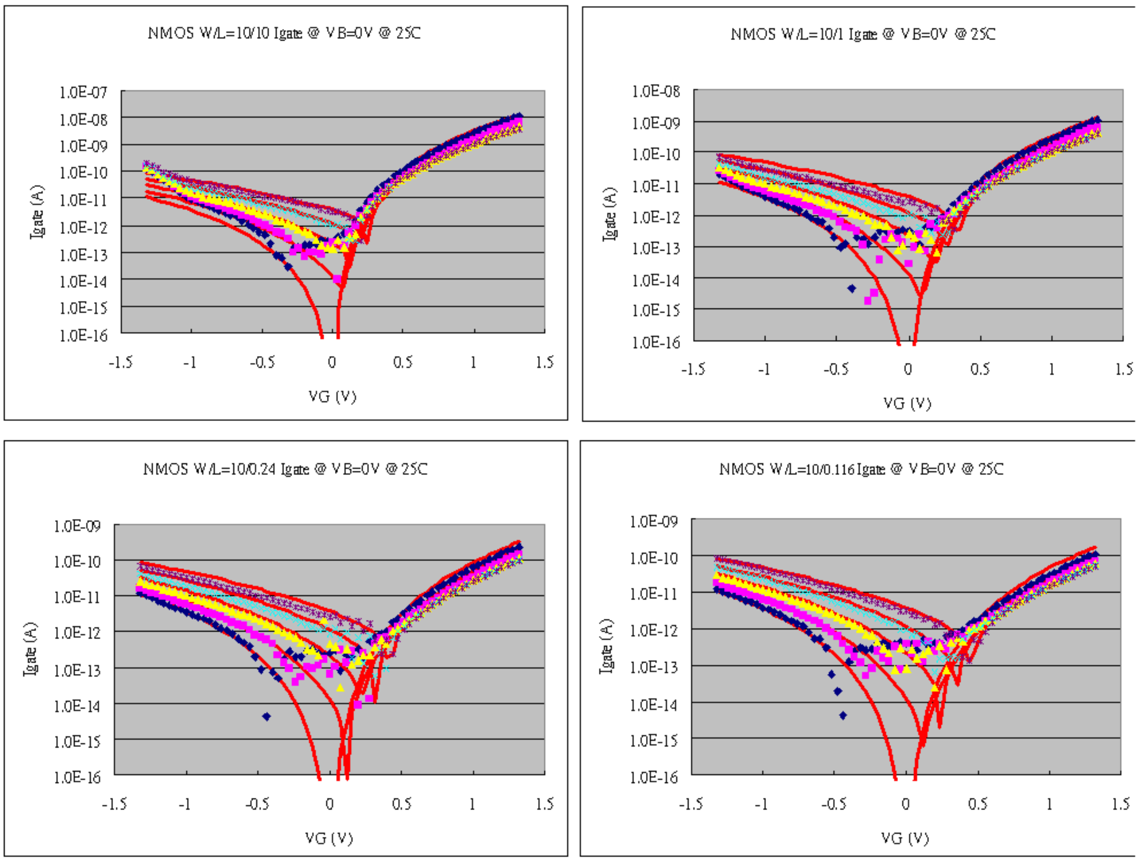


Figure 2-17 Measured and Simulated I_{GATE} for NMOS @ 25C

- ***I_{GATE} vs V_{GS} @ V_{BS} =0 @125C***

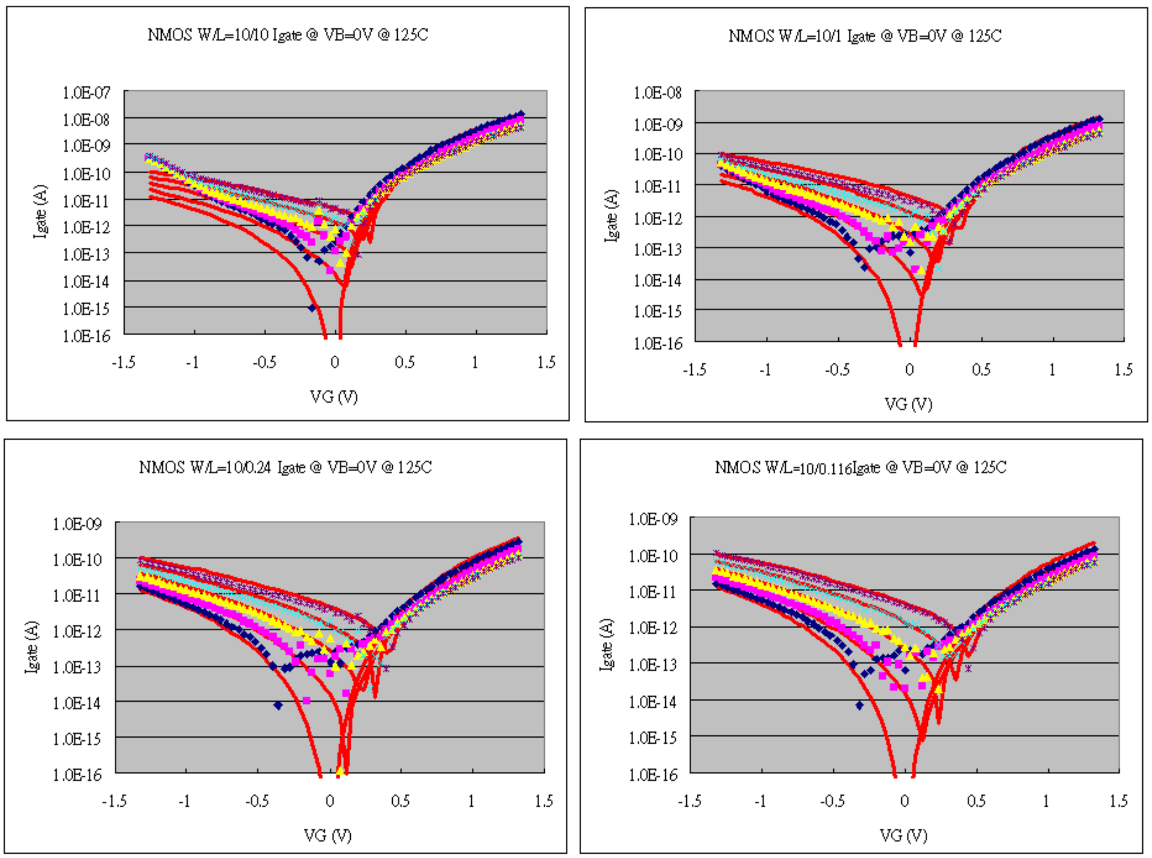


Figure 2-18 Measured and Simulated I_{GATE} for NMOS @ 125C

➢ ***I_{GATE} vs V_{GS} @ V_{BS}=0 @-50C***

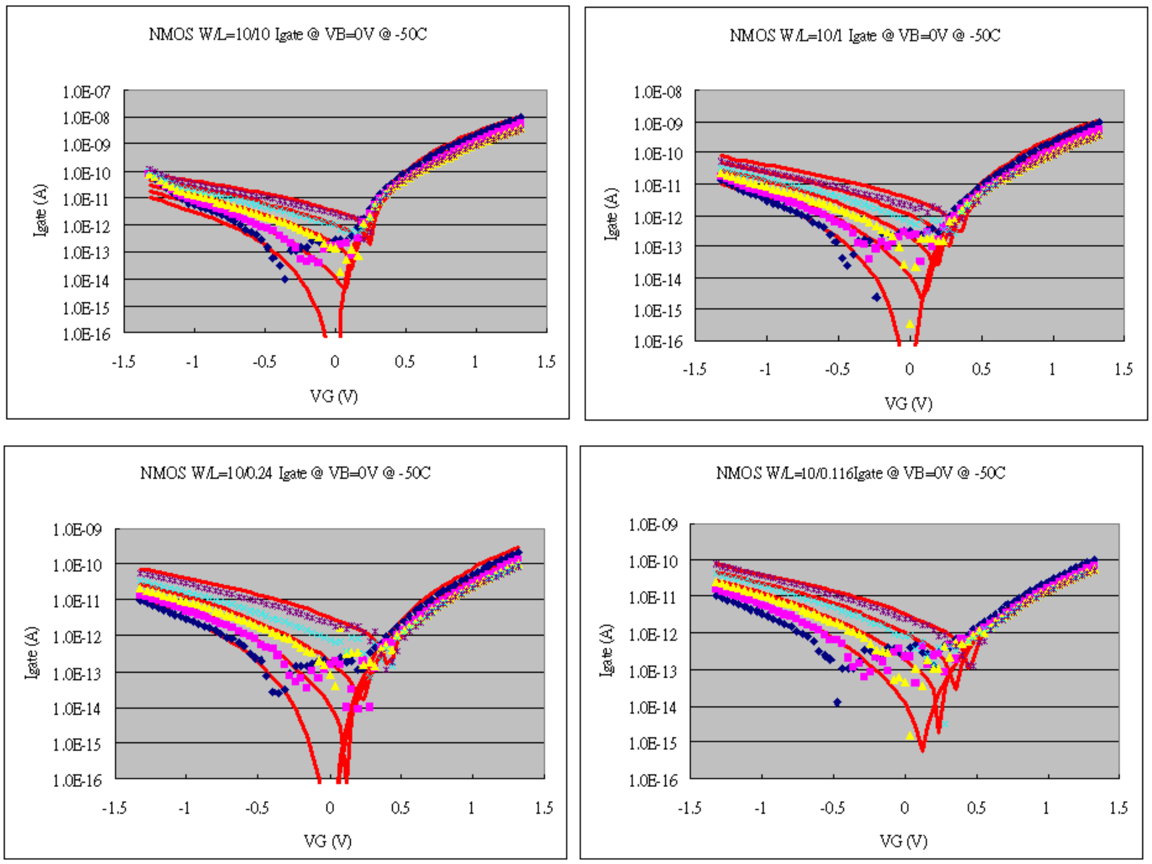


Figure 2-19 Measured and Simulated I_{GATE} for NMOS @ -50C

2.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

a. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
b. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \le V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \le V_{GS} \le V_{CC}$ (strong inversion)
I_D	$\le$75%	$\le$25%	$\le$7%
G_M	$\le$50%	$\le$20%	$\le$20%
G_{DS}	$\le$100%	$\le$50%	$\le$50%

➢ *I_D Accuracy*

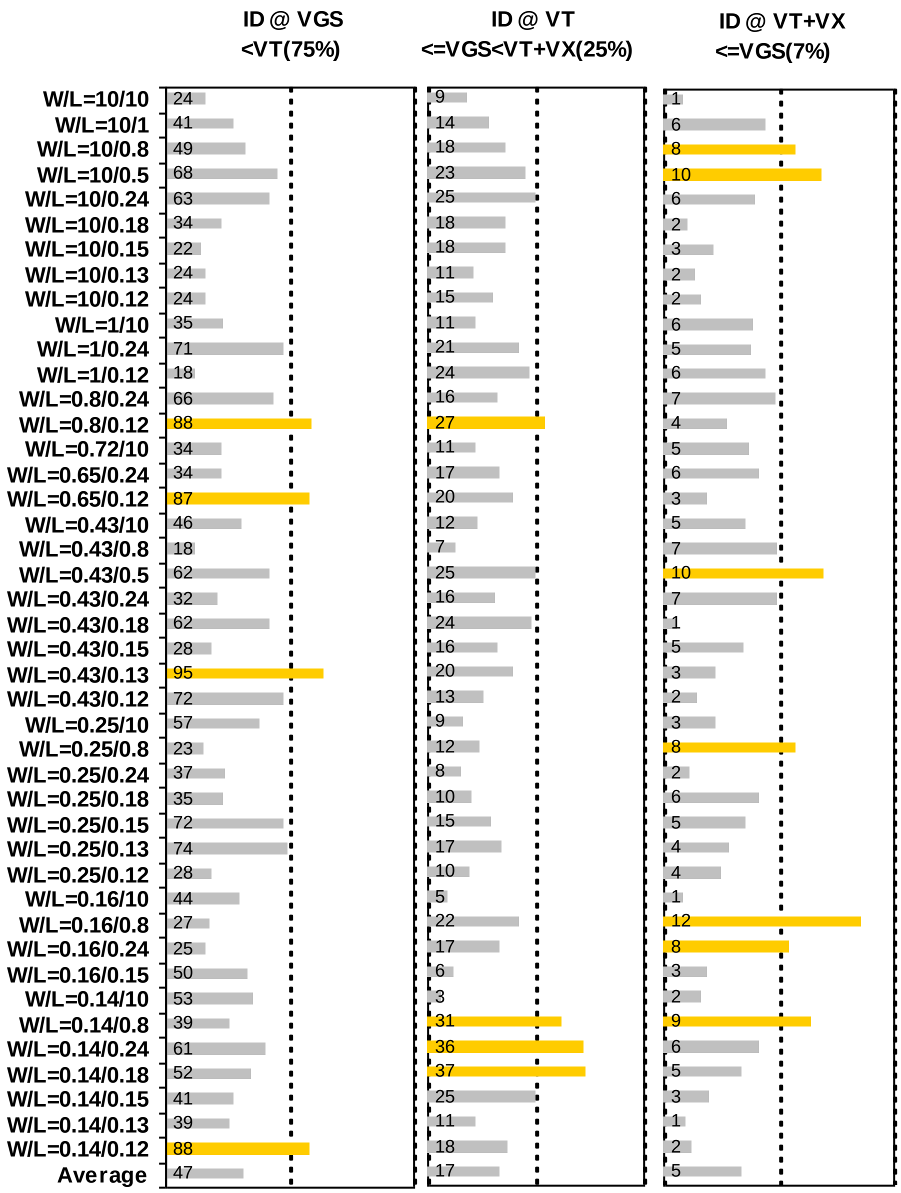


Figure 2-20 I_D accuracy at 25 ℃ for NMOS

➢ *G_M and G_{DS} Accuracy*

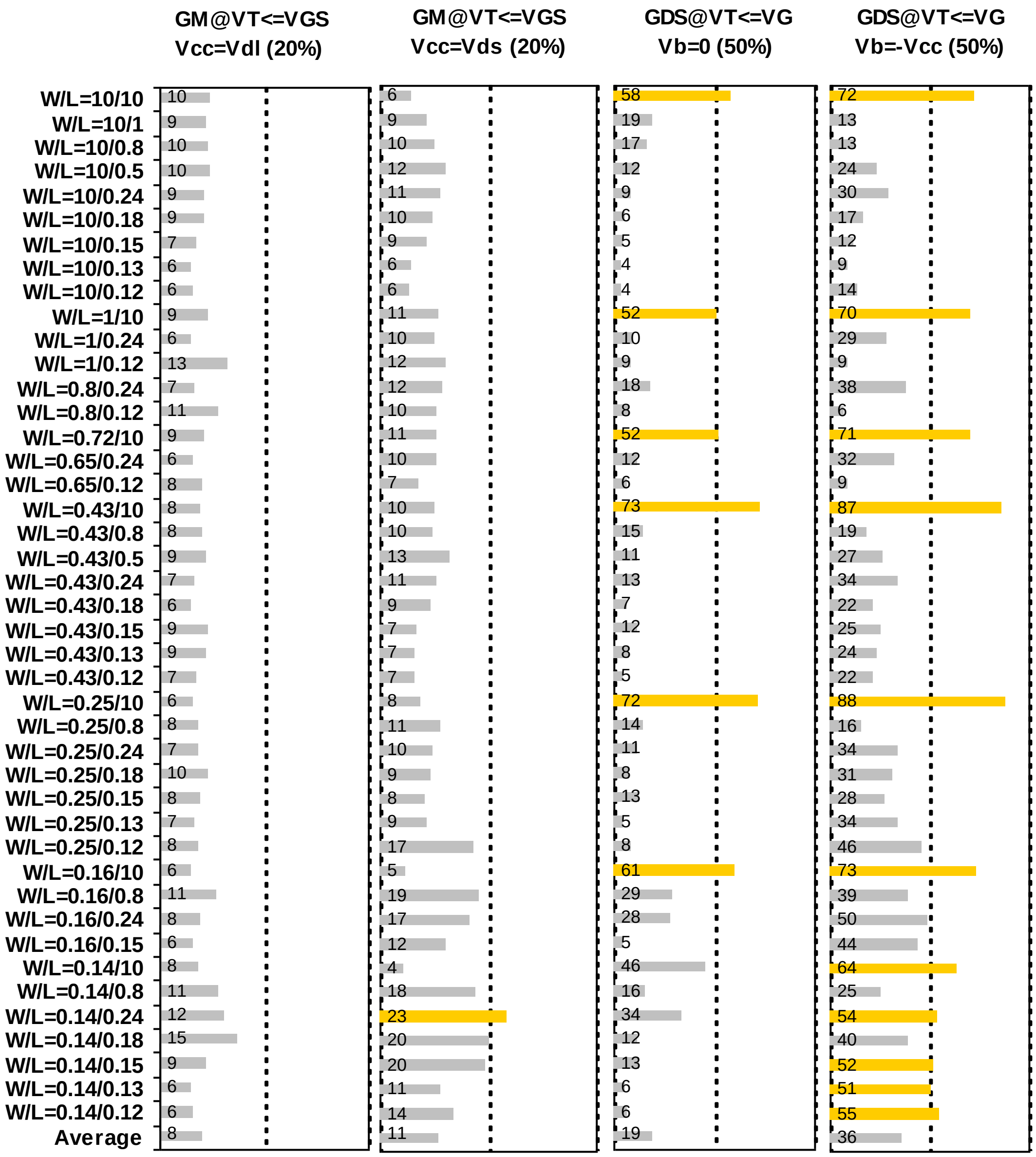


Figure 2-21 G_M and G_{DS} accuracy at 25 ℃ for NMOS

2.3.6 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structures for NMOS Cgg Capacitor structure

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components, C_{GDS}	200	100

2.3.7 Measurement conditions

Capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-7 capacitor measurement conditions for NMOS Cgg

V_{BIAS} [V]	From -1.35 to 1.35 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.3.8 Tox extraction

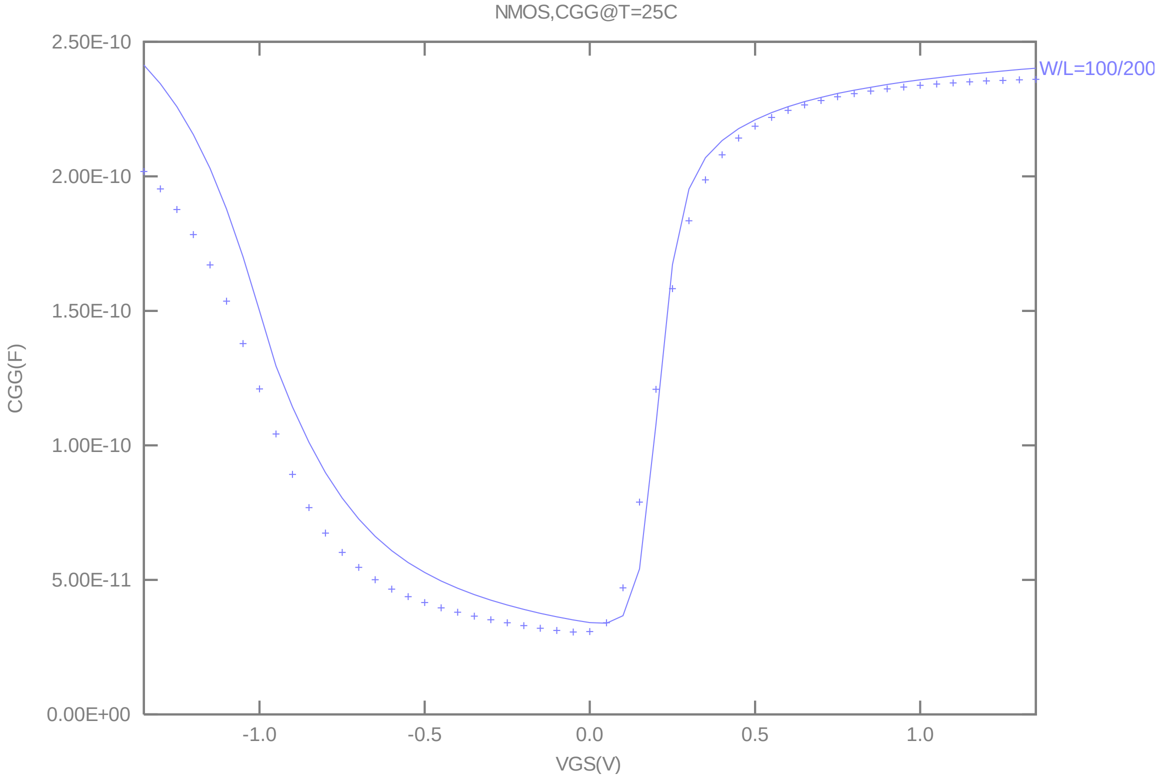


Figure 2-22 Measured and simulated Cgg curve

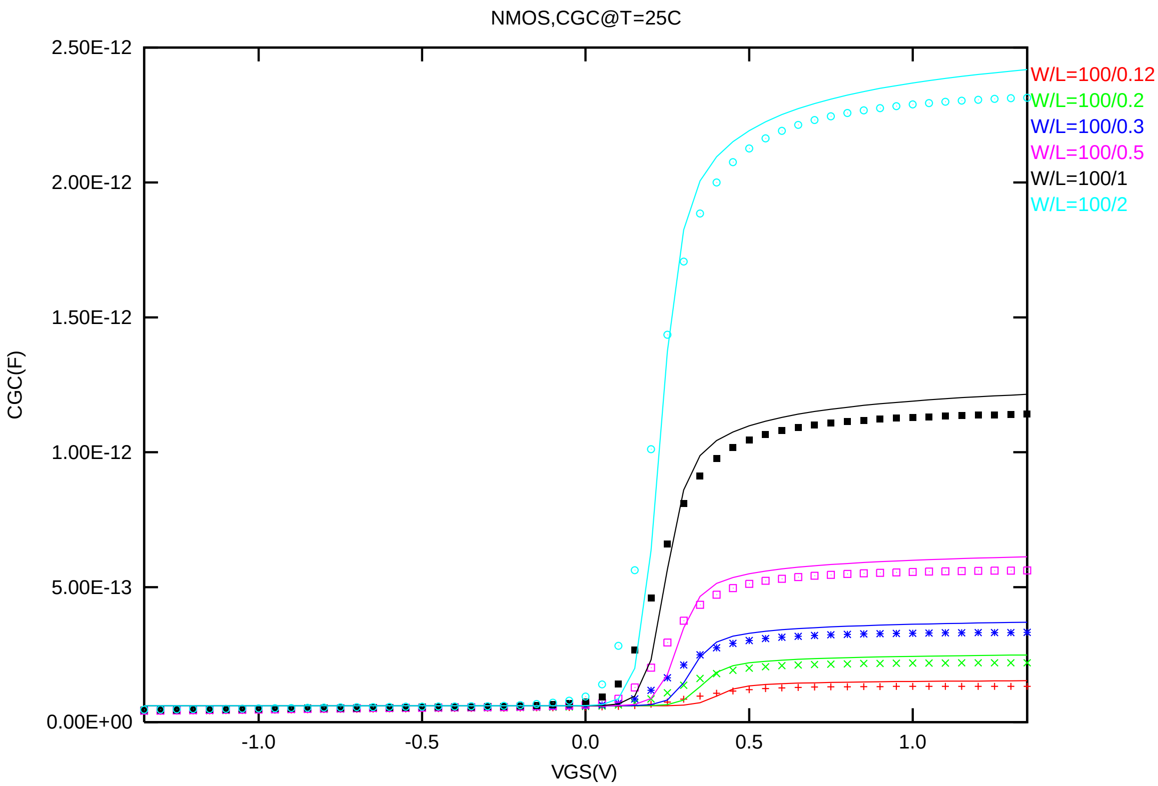


Figure 2-23 Measured and simulated Cgc curve

2.4 Parasitic source / drain junction capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-8 Junction capacitor structures for NMOS

Structures	**Area[m^2]**	**Perimeter[m]**	**Description**
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.00E-08	1.83E-02	Poly finger structure

2.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-9 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 1.2 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

2.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-10 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	7.20E-04	CJSW	F/m	9.94E-11	CJSWG	F/m	4.11E-10
PB	V	5.60E-01	PBSW	V	5.40E-01	PBSWG	V	1.11E+00
MJ	--	2.58E-01	MJSW	--	1.30E-01	MJSWG	--	7.17E-01
TCJ	1/K	9.56E-04	TCJSW	1/K	7.32E-04	TCJSWG	1/K	1.44E-03
TPB	V/K	1.63E-03	TPBSW	V/K	3.10E-04	TPBSWG	V/K	1.82E-03

➢ *CJ, CJSW vs. VBIAS Plots*

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

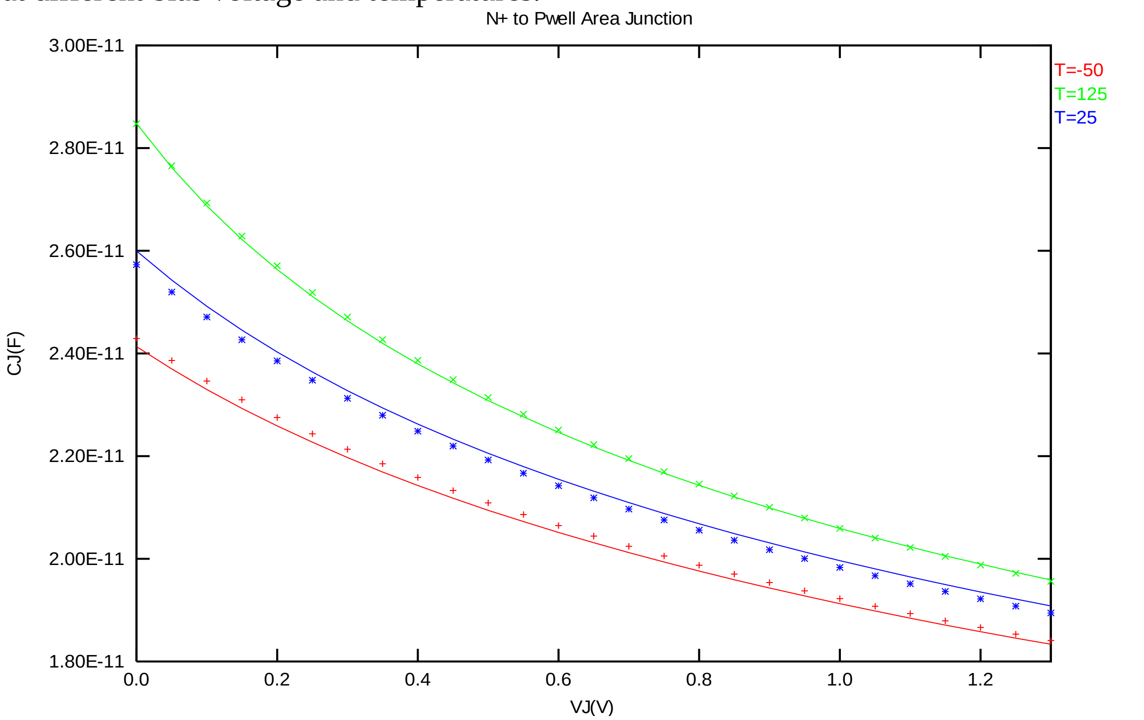


Figure 2-24 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

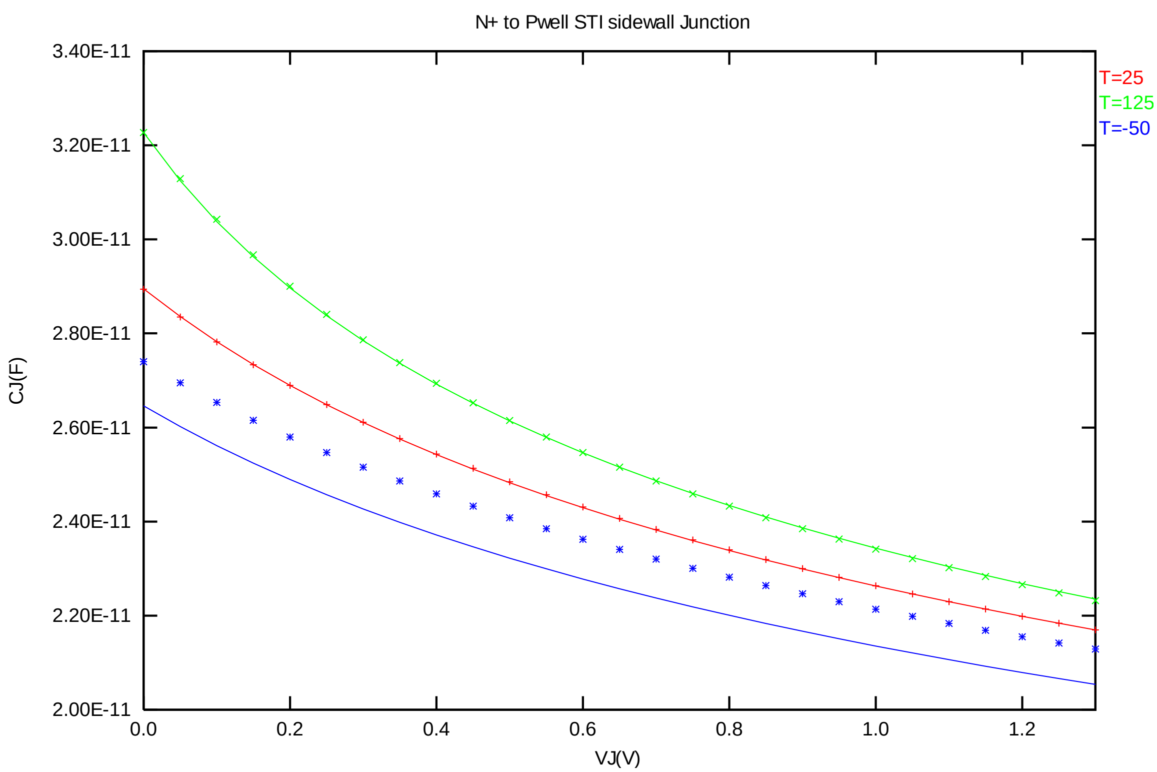


Figure 2-25 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

3 1.2 V p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGMNS	QGMNS	QGMNS	QGMNS
Wafer Number	#11	#11	#11	#11

3.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =-70nA* (W_{DES} *0.9)/(L_{DES} *0.9+0.008μm). The dimension is after 90% shrink and 8nm sizing channel length back.

Table 3-2 Electrical Parameters for PMOS

All parameters are given for T=25 ^oC ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = -0.05 V	V	-0.242	-0.223	-0.242	-0.242
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/ μm	-3.07	-3.08	-3.07	-3.06
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.116 μ m)						
V_{TLIN}	V_{DS} = -0.05 V	V	-0.284	-0.269	-0.285	-0.284
V_{TSAT}	V_{DS} = -1.2 V	V	-0.180	-0.177	-0.186	-0.180
Body Effect V_T shift	V_{DS} = -1.2 V V_{BS}= 0 ~ --1.2 V	V	--	-0.126	-0.142	-0.137
DIBL V_T shift	V_{DS}= -0.05~-1.2 V	V	-0.104	-0.092	-0.099	-0.104
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-290	-279.96	-279.27	-290
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-7.0E-09	-5.7E-09	-5.7E-09	-8.1E-09
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = -0.05 V	V	-0.210	-0.208	-0.210	-0.210
V_{TSAT}	V_{DS}= -1.2 V	V	--	-0.207	-0.204	-0.204
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-4.82	-4.90	-4.81	-4.80
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μm / 0.116 μm)						
V_{TLIN}	V_{DS} = -0.05 V	V	-0.300	-0.293	-0.284	-0.298
V_{TSAT}	V_{DS}= -1.2 V	V	-0.223	-0.219	-0.184	-0.221
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-375.03	-364.21	-362.71	-370.65
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-1.3E-09	-1.2E-09	-1.0E-09	-1.4E-09
Capacitance						
CJS	V_J = 0 V	F/m^2	9.69E-04	9.58E-04	9.36E-04	9.69E-04
CJSWS	V_J = 0 V	F/m	1.18E-10	1.21E-10	1.11E-10	1.18E-10
CJSWGS	V_J = 0 V	F/m	4.82E-10	7.03E-10	4.82E-10	4.82E-10
C_{OVL}	V_{DS} = V_{GS} = 0.5 V	F/m	3.09E-10	2.81E-10	2.70E-10	3.09E-10

3.3 MOSFET DC model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

Table 3-3 Geometry of PMOS devices measured for parameter extraction

W\L	10	1	0.8	0.5	0.24	0.18	0.15	0.13	0.12
10	X	X	X	X	X	X	X	X	X
1	X				X				X
0.8					X				X
0.72	X								
0.65					X				X
0.43	X		X	X	X	X	X	X	X
0.25	X		X		X	X	X	X	X
0.16	X		X		X		X		
0.14	X		X		X	X	X	X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-1.2	-1.1
V_{GS} (V)	0	-1.35	-0.05
V_{BS} (V)	0	1.2	0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-1.35	-0.05
V_{GS} (V)	-0.6	-1.2	-0.15
V_{BS} (V)	0	1.2	1.2

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to the physical dimensions (L_{DES} *0.9+8nm and W_{DES} *0.9) respectively.

➢ ***I_D vs. V_{DS} Plots***

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

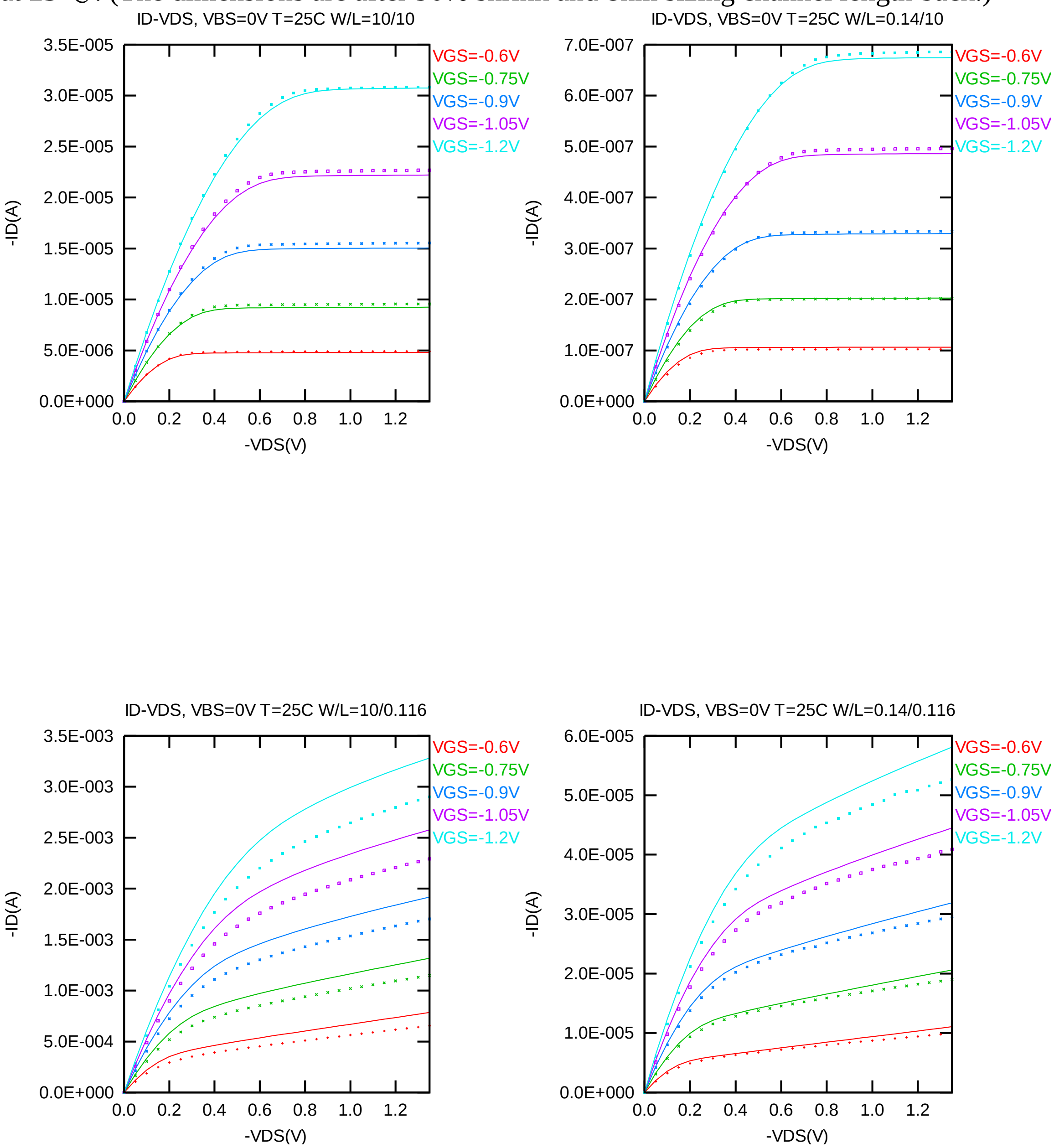


Figure 3-1 I_D vs. V_{DS} at V_{BS} = 0 V for PMOS

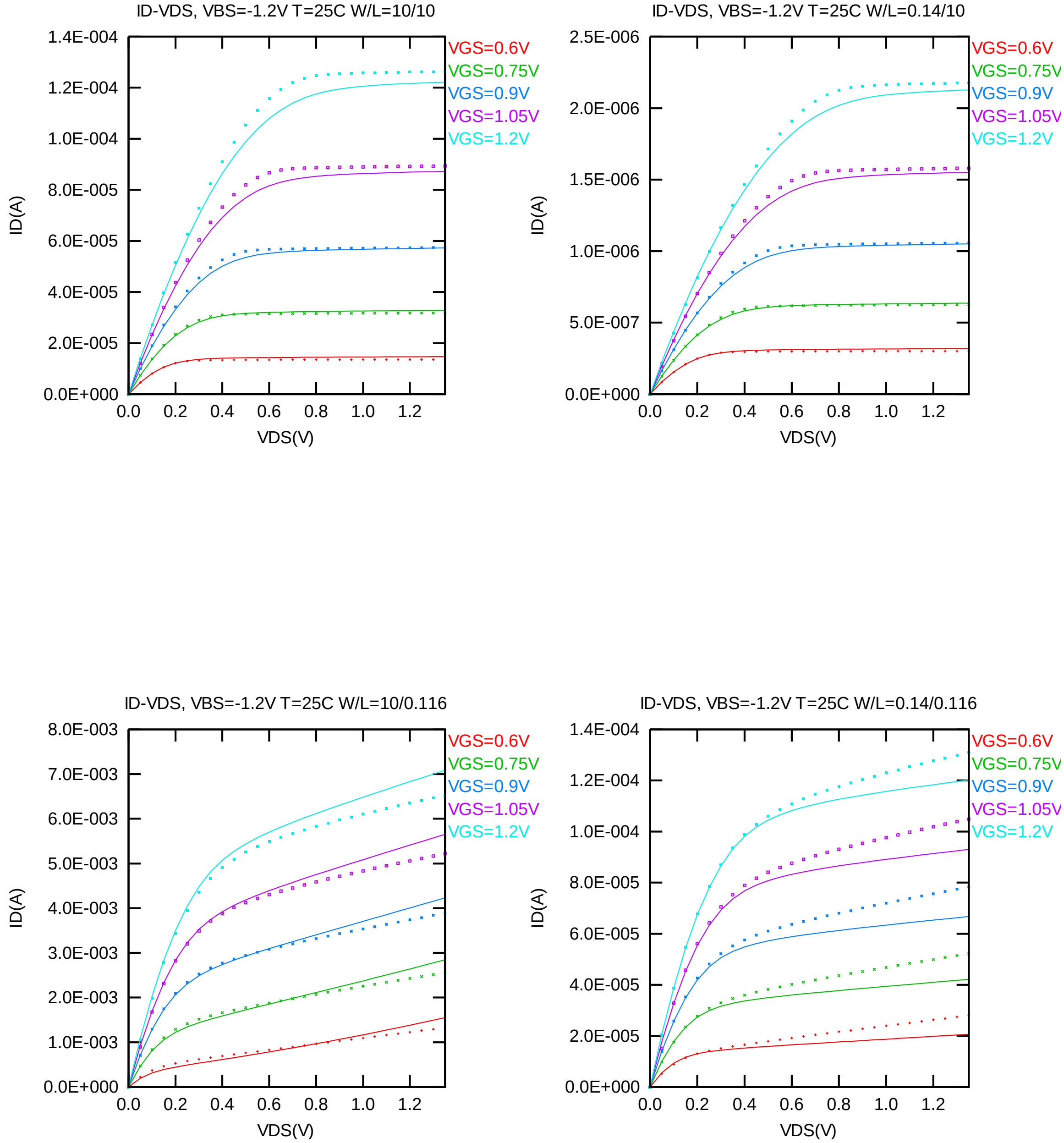


Figure 3-2 I_D vs. V_{DS} at V_{BS}=1.2 V for PMOS

➢ *G_{DS} vs. V_{DS} Plots*

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

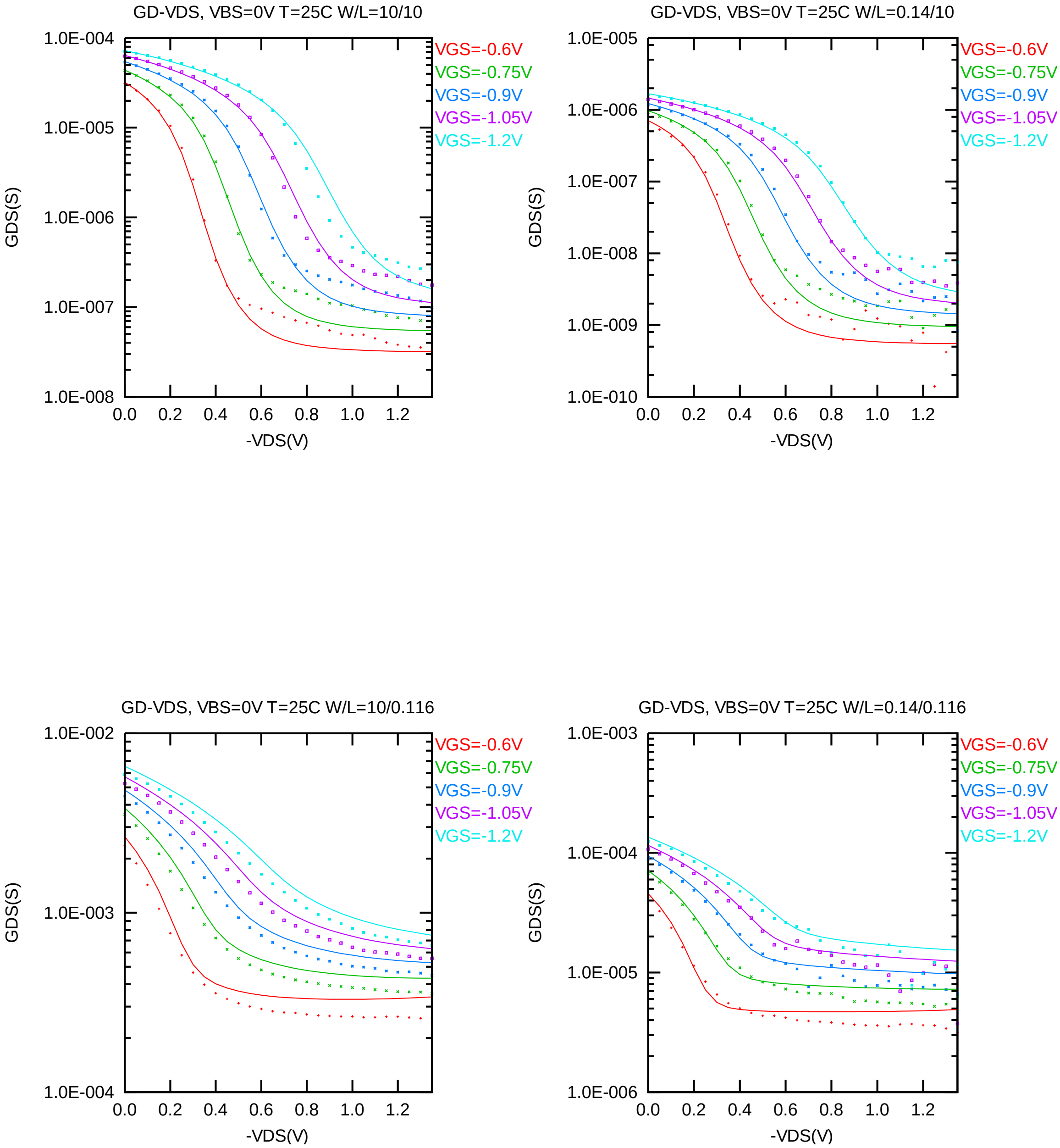


Figure 3-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for PMOS

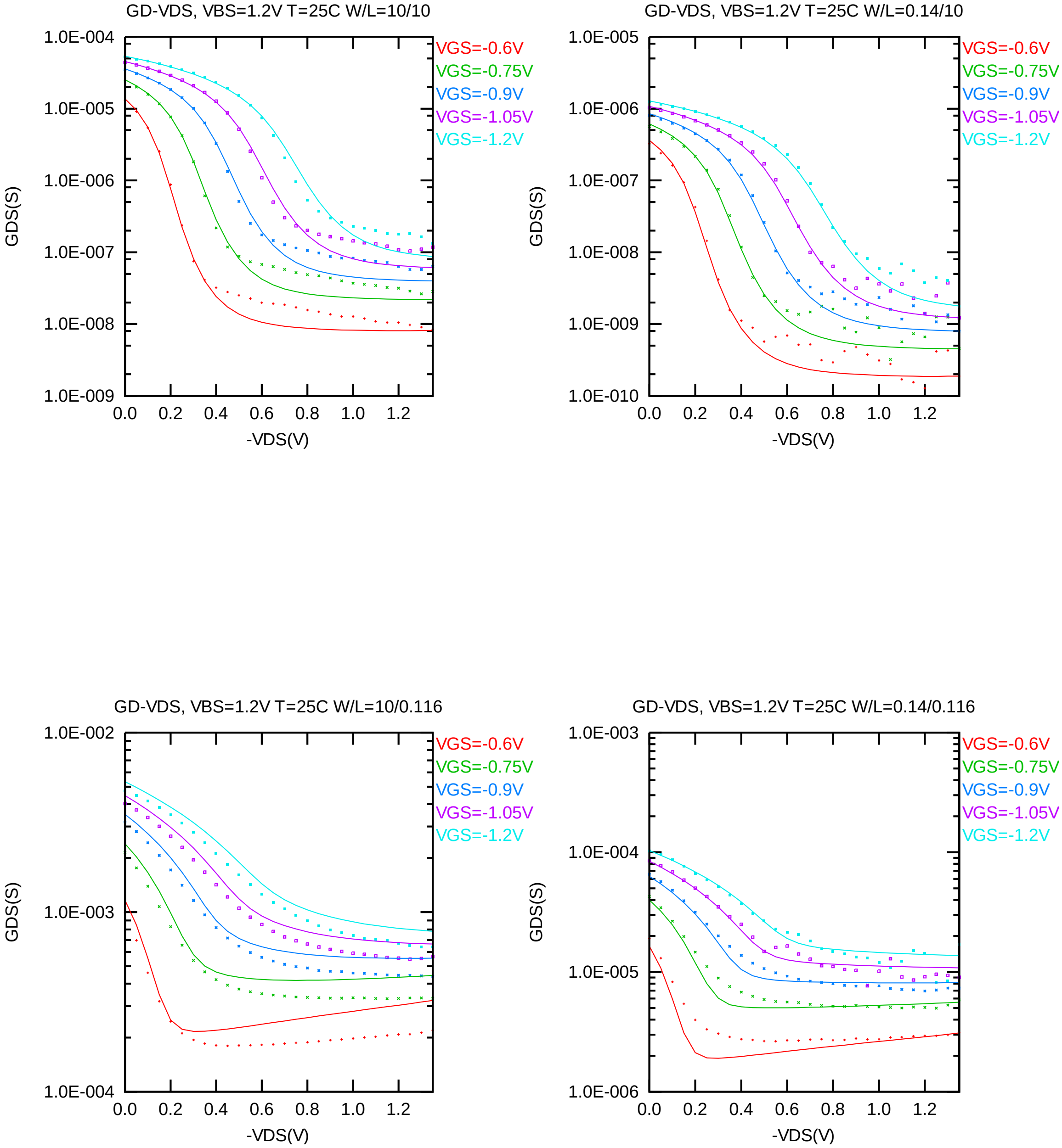


Figure 3-4 G_{DS} vs. V_{DS} at V_{BS} = 1.2V for PMOS

➢ *I_D vs. V_{GS} Plots (linear scale)*

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

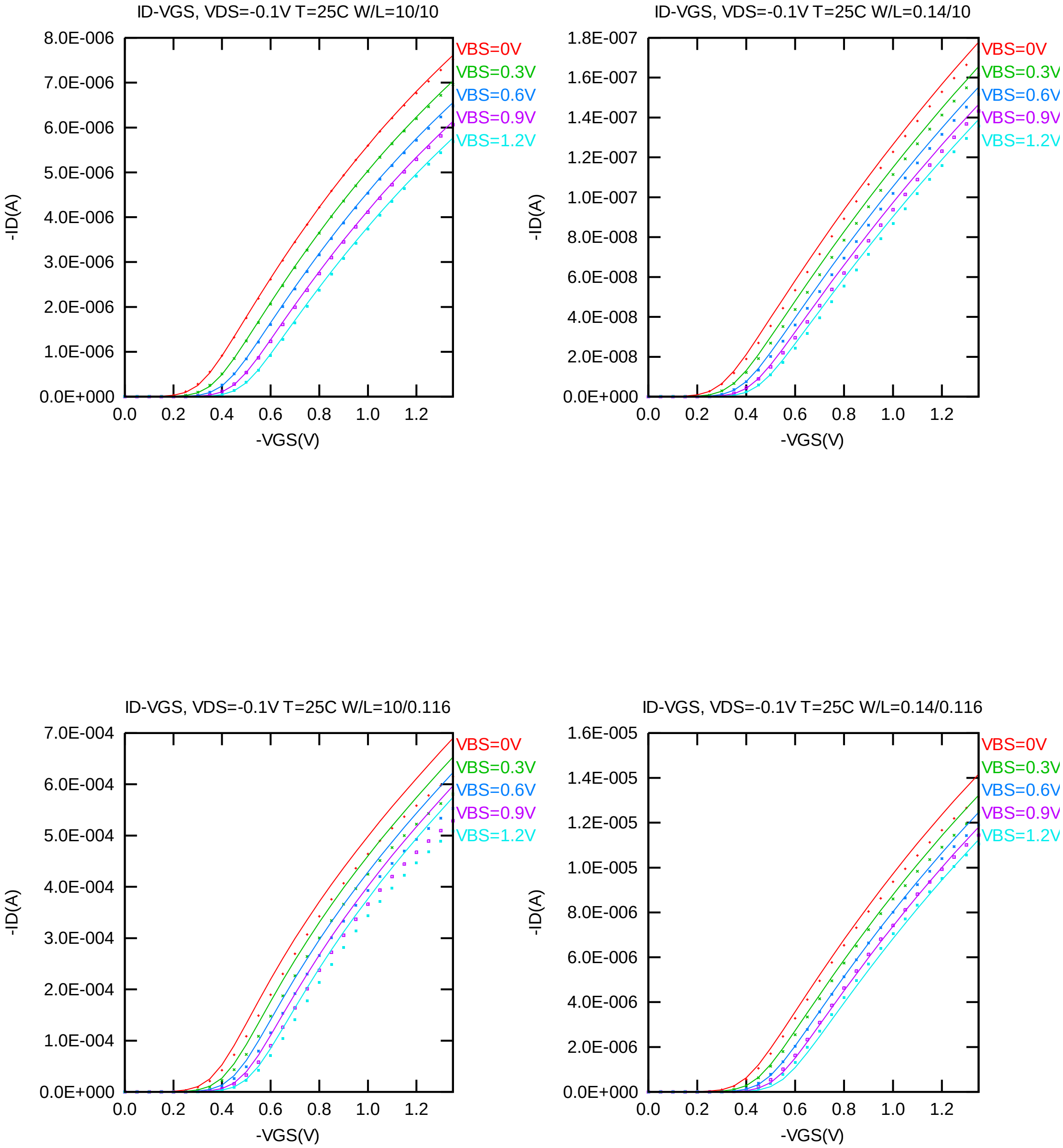


Figure 3-5 I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

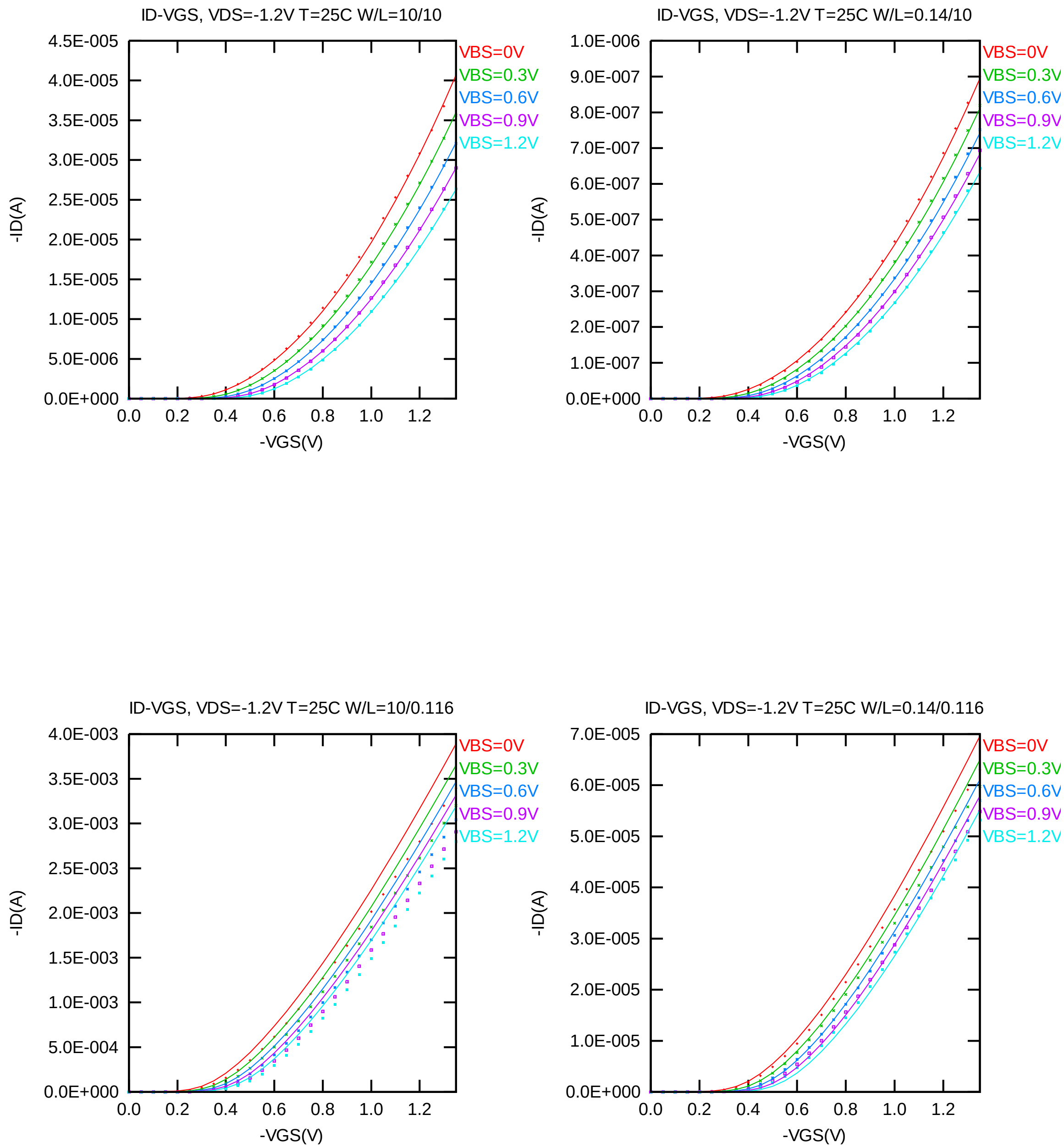


Figure 3-6 I_D vs. V_{GS} at V_{DS} = -1.2V for PMOS

➢ I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

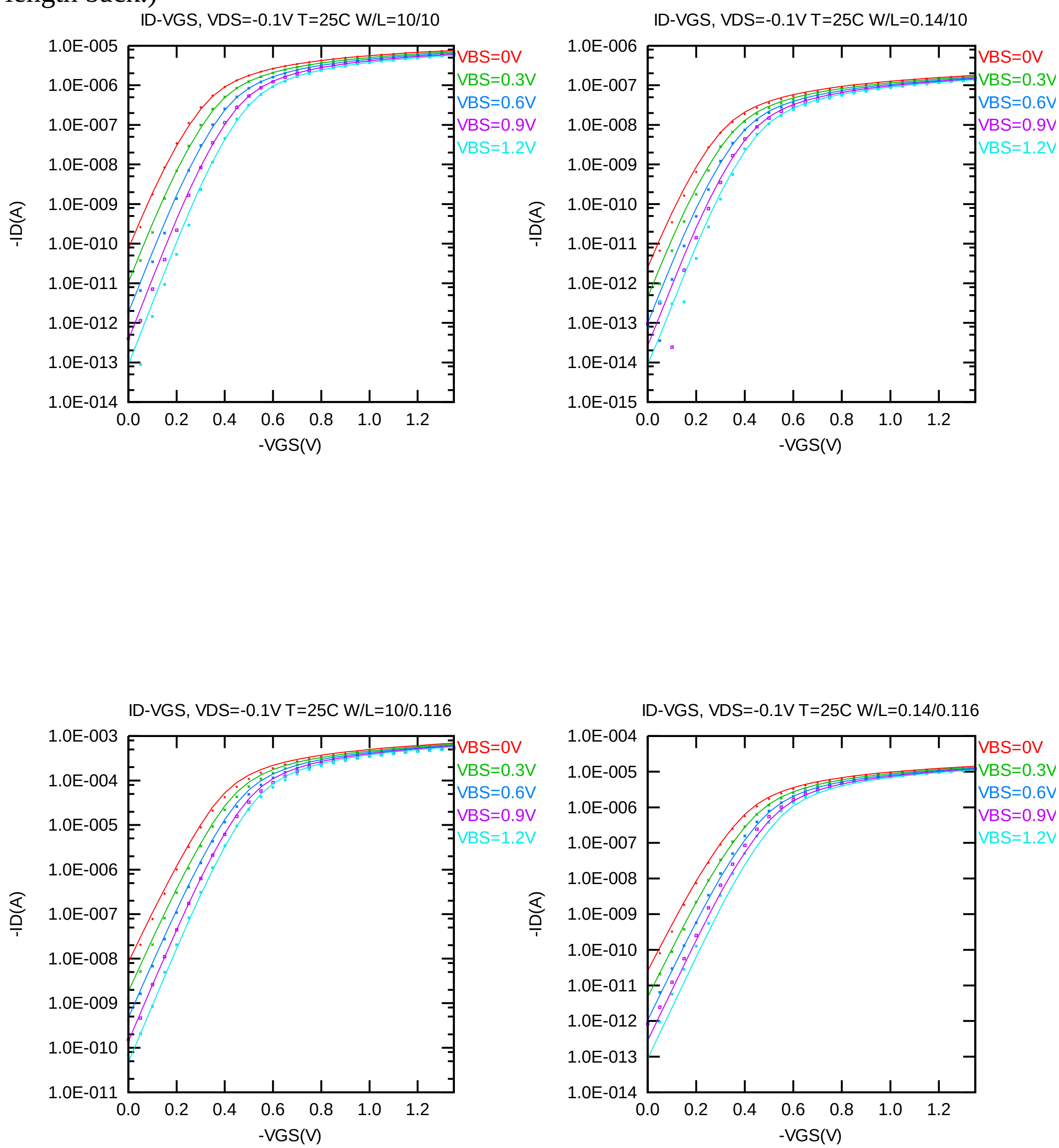


Figure 3-7 Log scaled I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

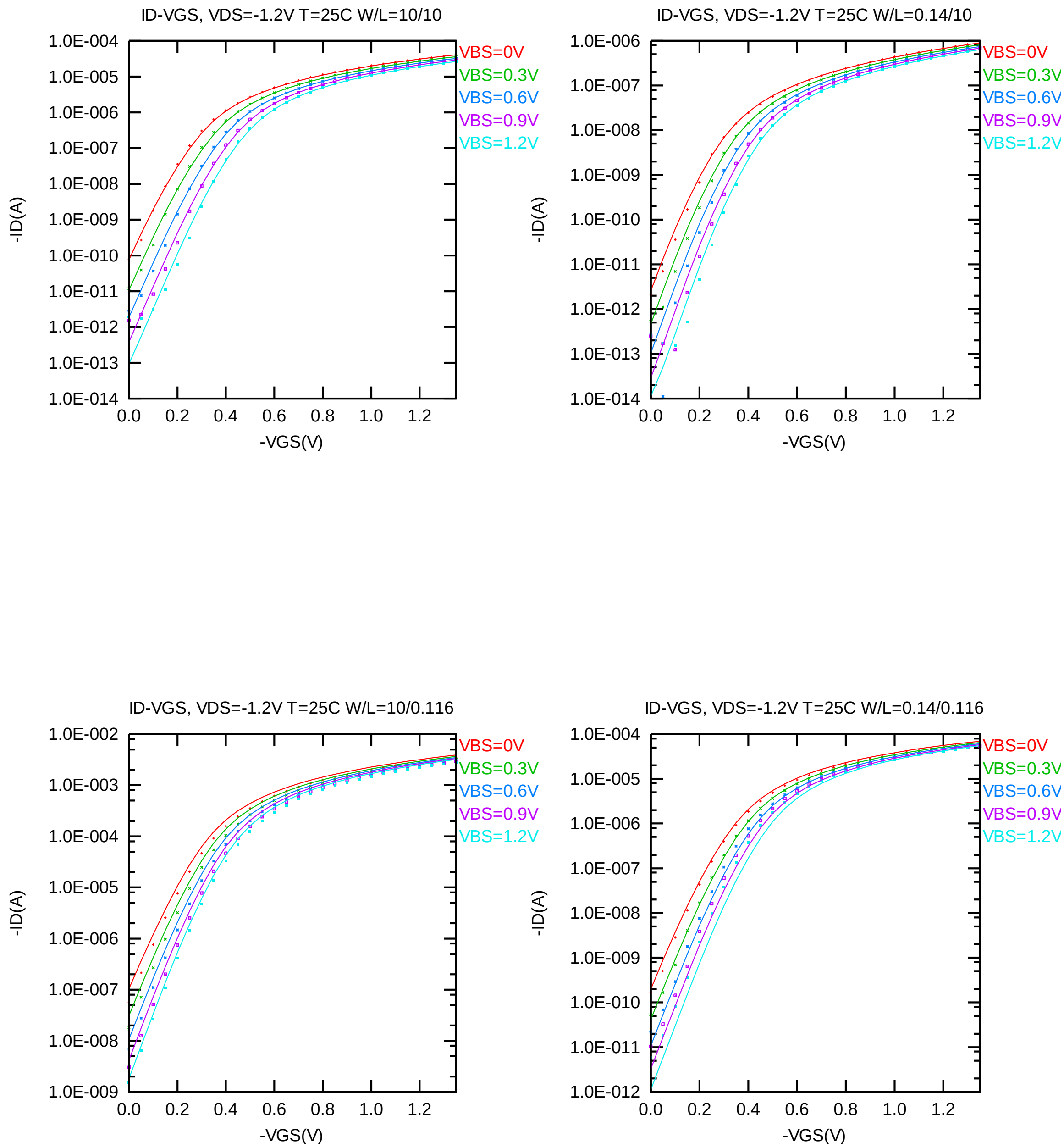


Figure 3-8 Log scaled I_D vs. V_{GS} at V_{DS} = -1.2 V for PMOS

➢ *G_M vs. V_{GS} Plots*

The following plots show the measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

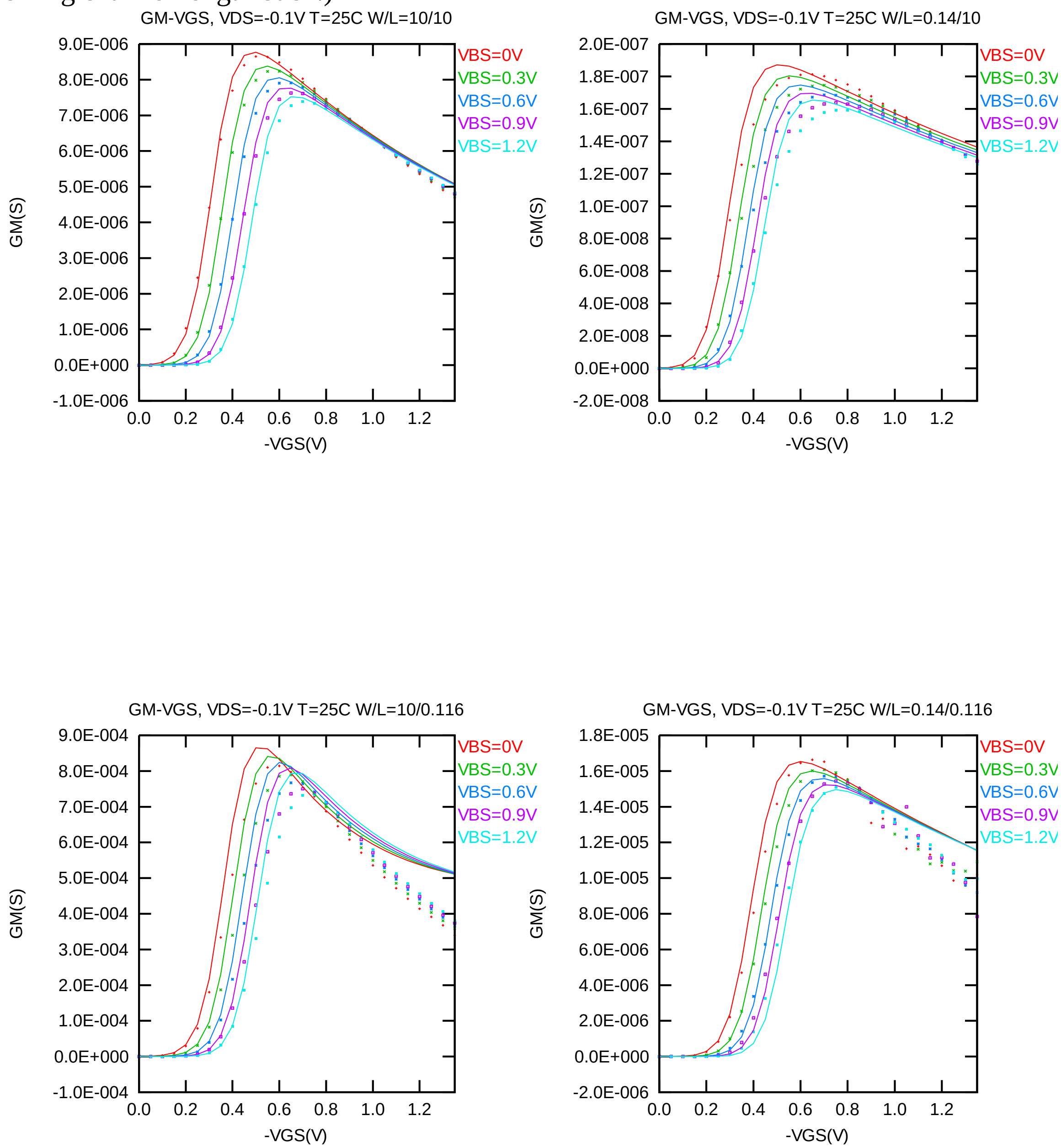


Figure 3-9 G_M vs. V_{GS} at V_{DS} = -0.1 V for PMOS

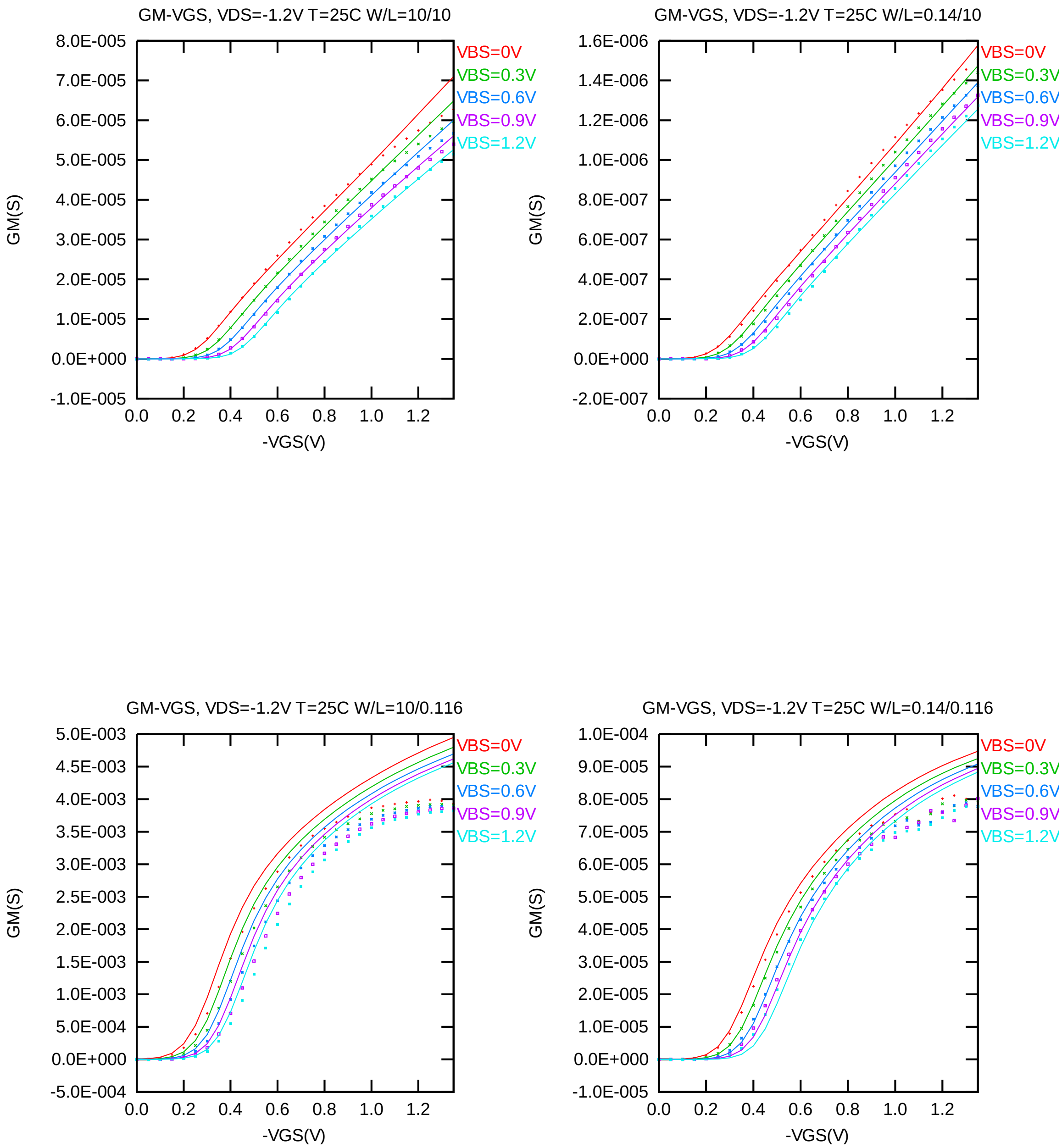


Figure 3-10 G_M vs. V_{GS} at V_{DS} = -1.2 V for PMOS

- ***V_T vs. L_{DES} & V_T vs. W_{DES}***

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008 μm) at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

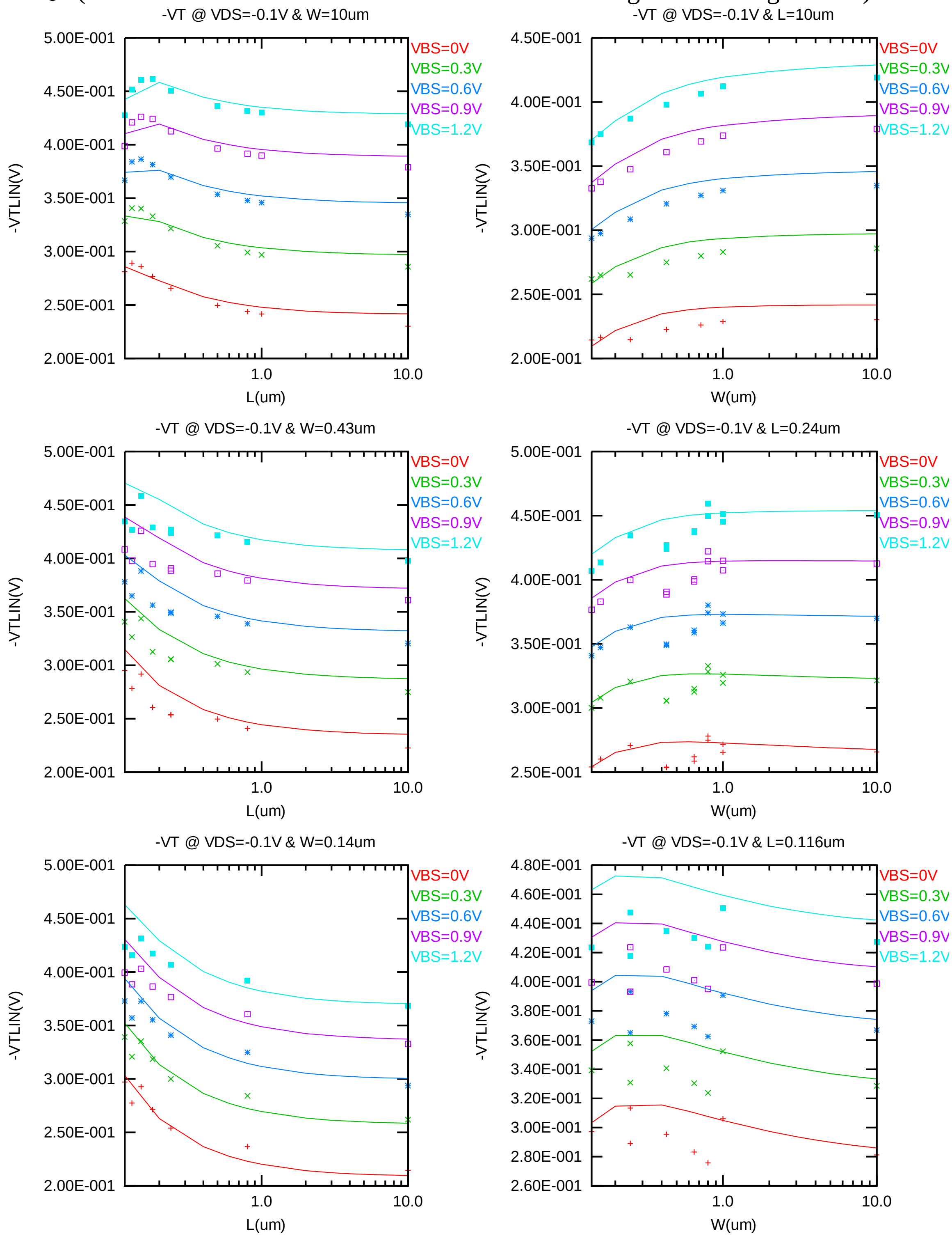


Figure 3-11 V_T vs. channel length & channel width for low drain bias (V_{DS} = -0.1 V) for PMOS

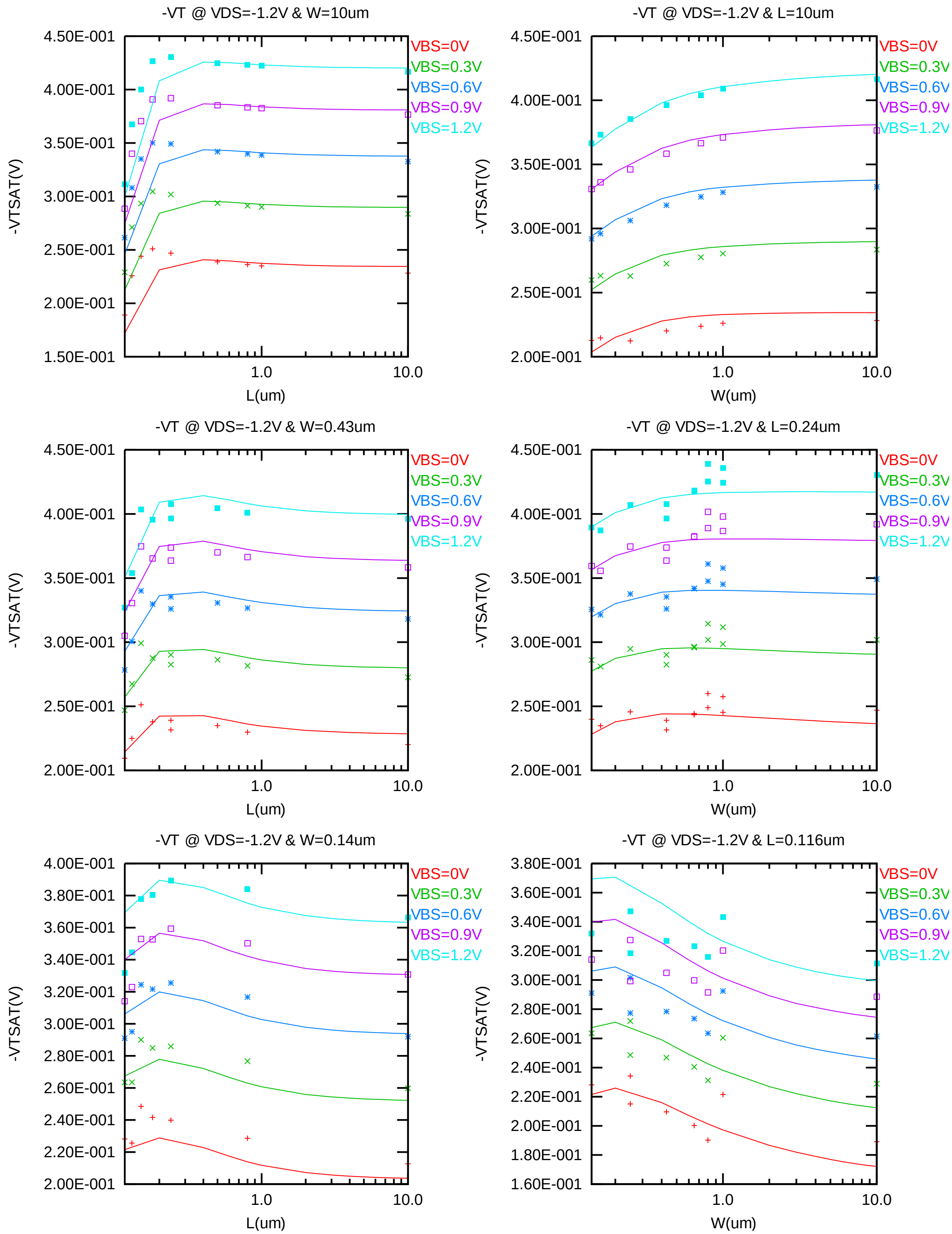


Figure 3-12 V_T vs. channel length & channel width for low drain bias (V_{DS} = -1.2 V) for PMOS

- ***I***

➢ ***I_{ON}* vs. *L_{DES}* & *I_{ON}* vs. *W_{DES}***

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -1.2$ for $V_{BS} = 0$ V to $V_{BS} = 1.2$V at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

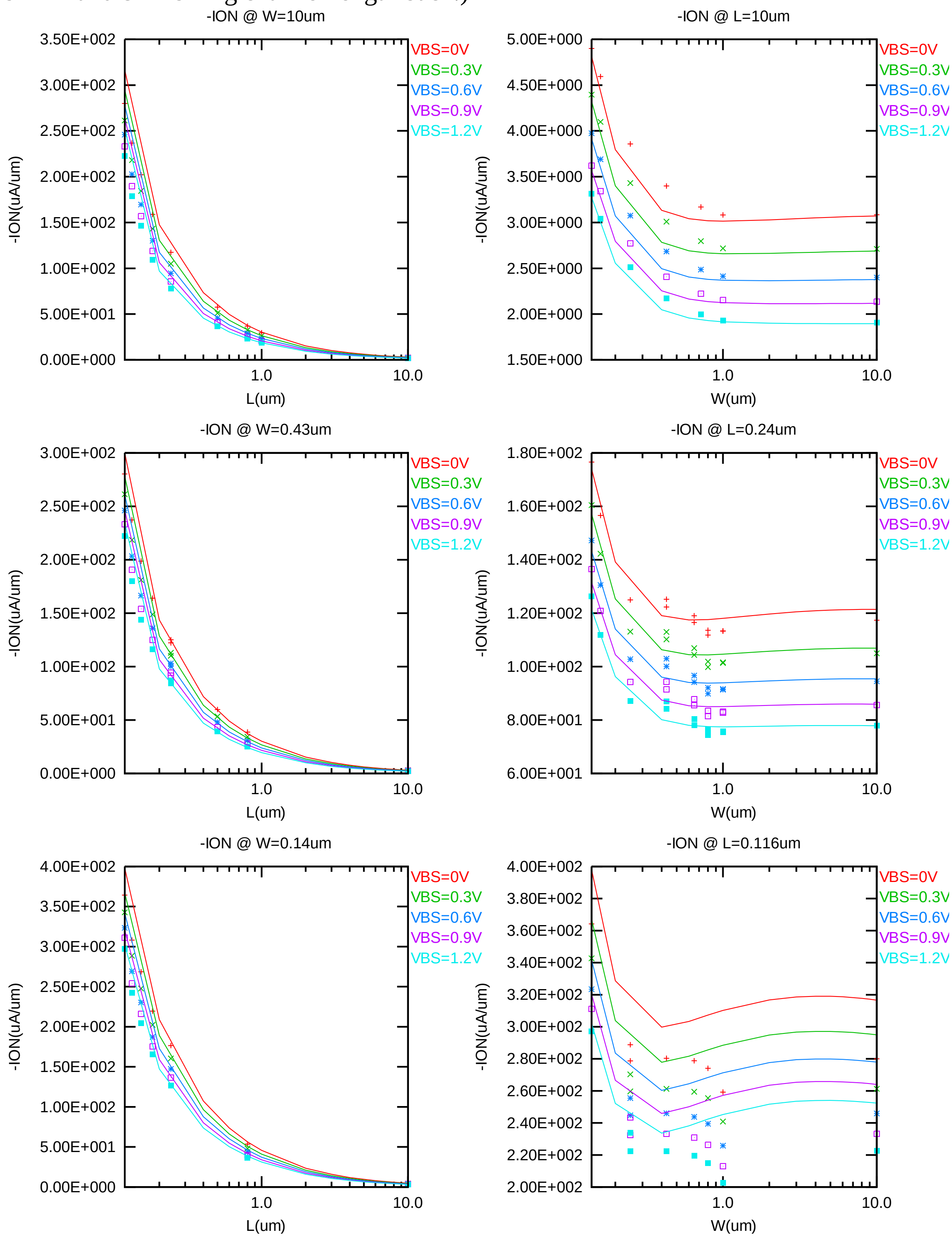


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

➢ *V_T vs. Temperature*

The following plots give the temperature dependence of the linear and saturation threshold voltage extracted at I_D = -70nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008 μm). (The dimensions are after 90% shrink and 8nm sizing channel length back.)

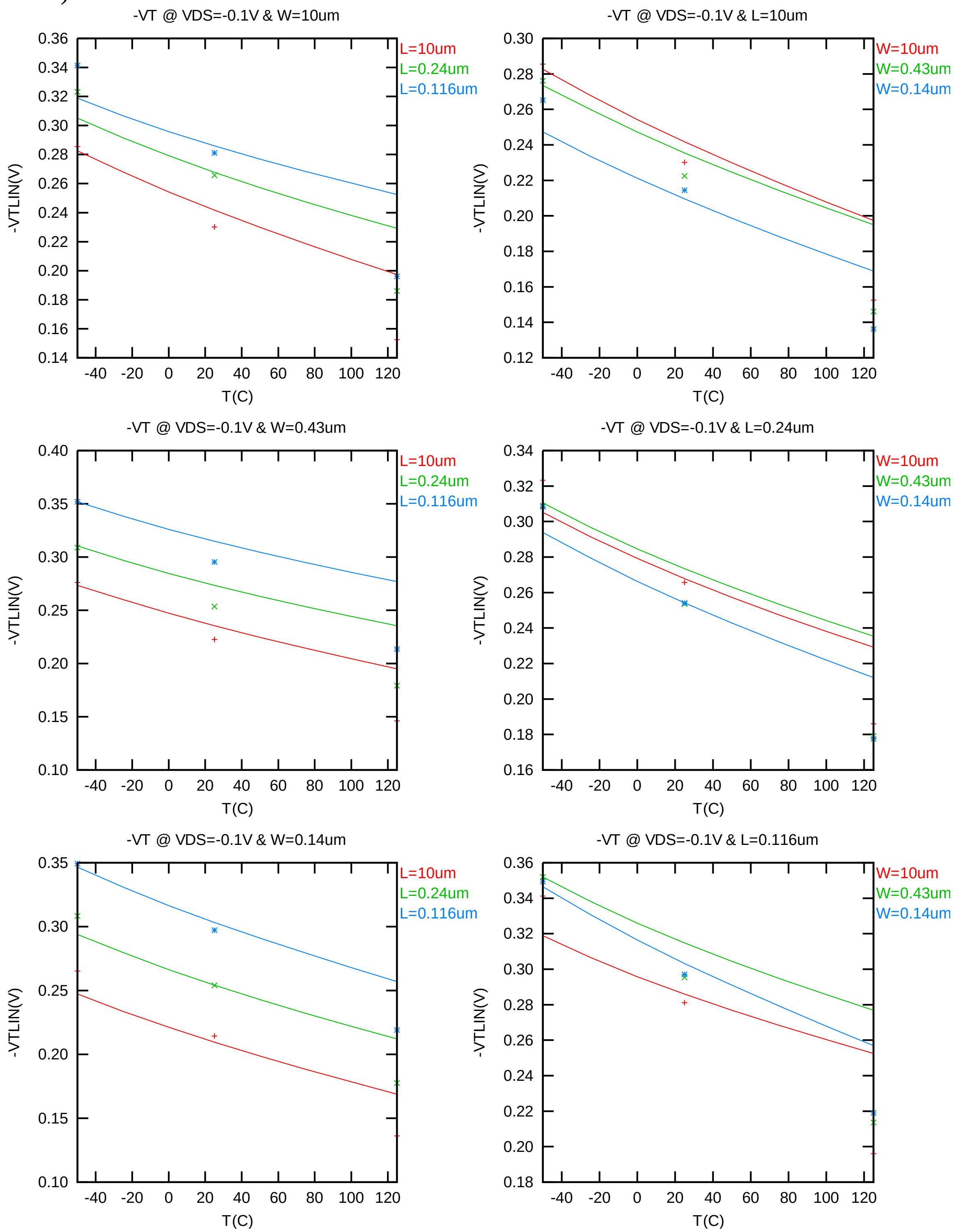


Figure 3-14 V_{TLIN} vs. temperature for PMOS

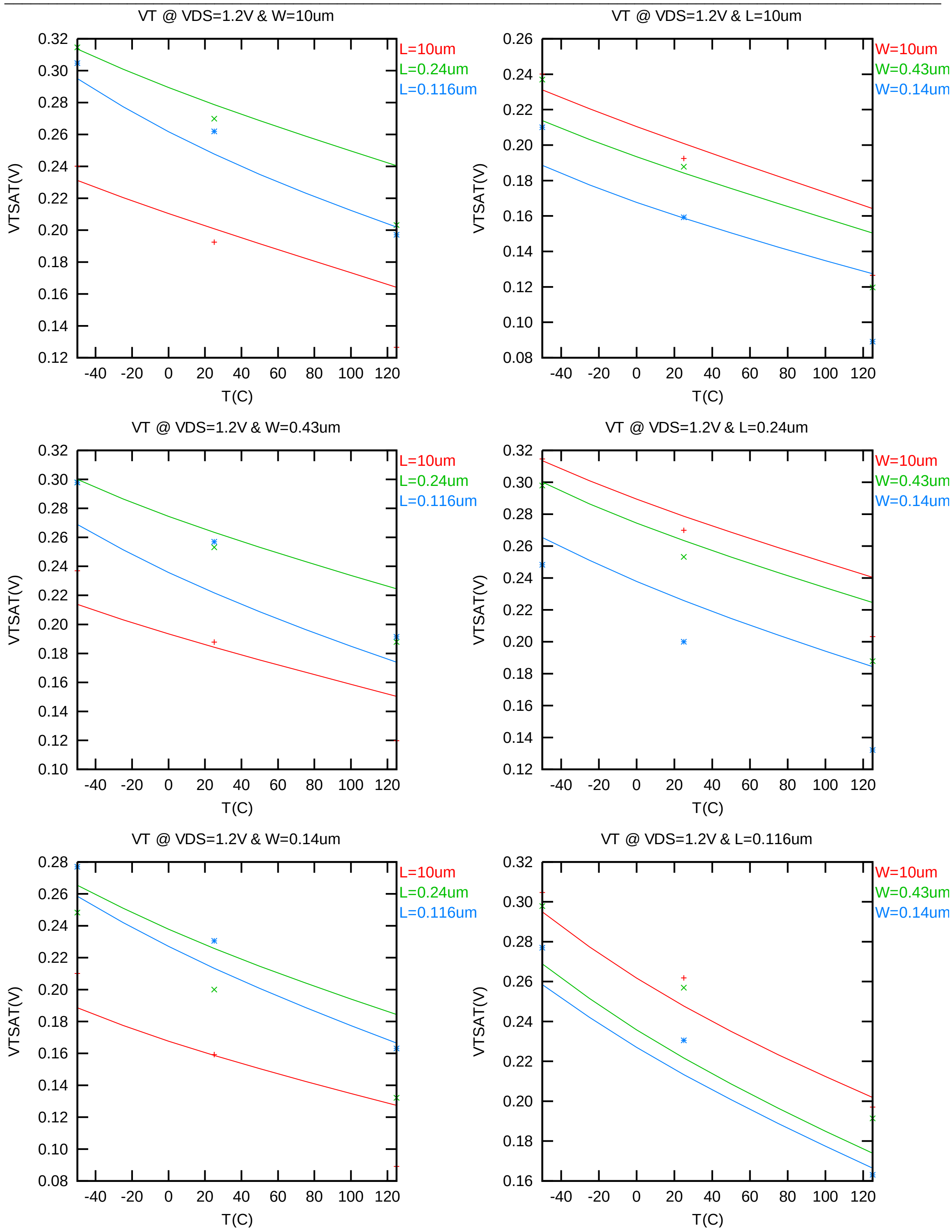


Figure 3-15 V_{TSAT} vs. temperature for PMOS

➢ *I_{ON} vs Temperature*

The following plot shows the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = -1.2$ V, $V_{BS} = 0$ V. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

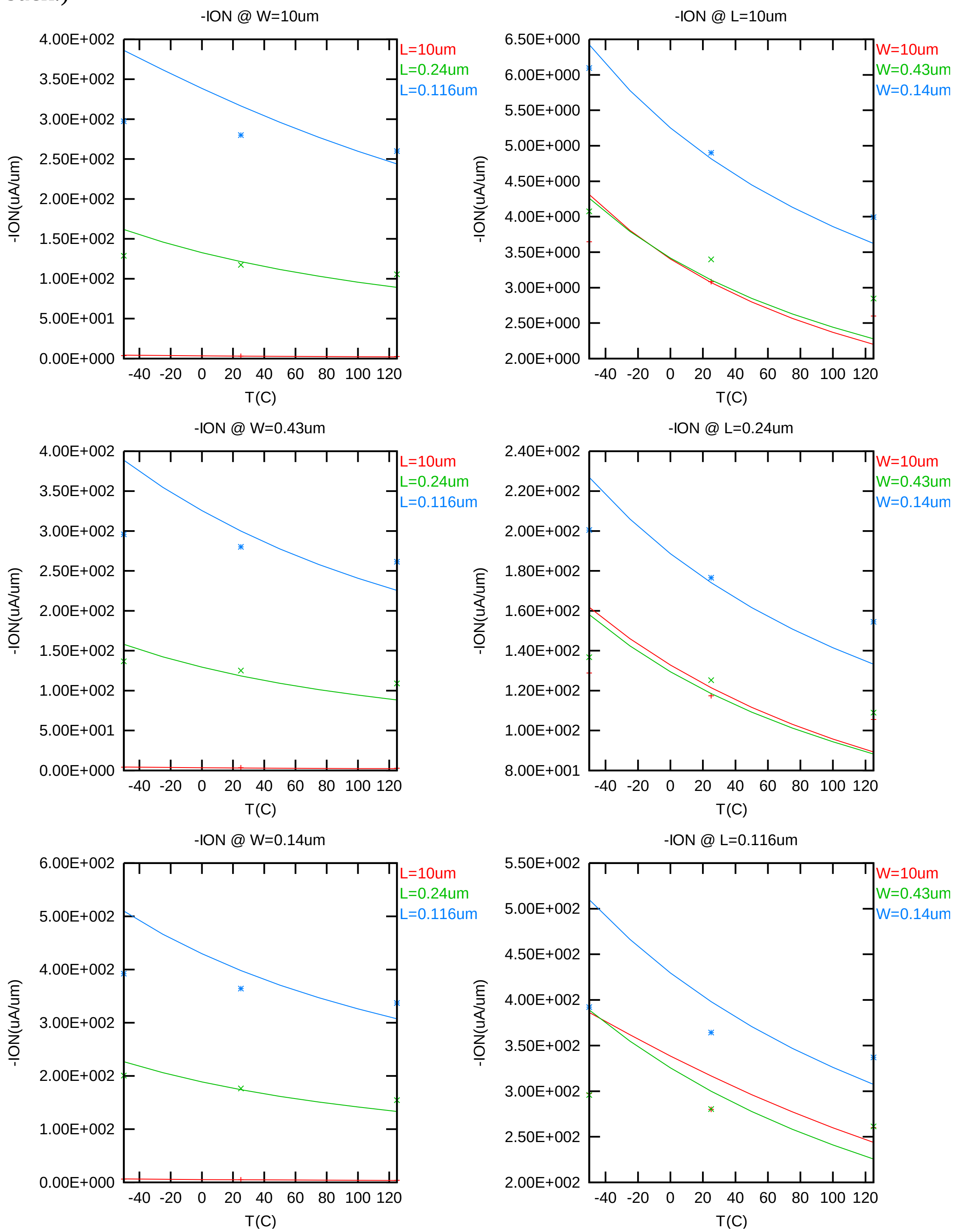


Figure 3-16 Saturation Current vs. temperature for PMOS

3.3.4 I_{gate} Extraction plot

- ***I_{GATE} vs V_{GS} @ V_{BS} =0 @25C***

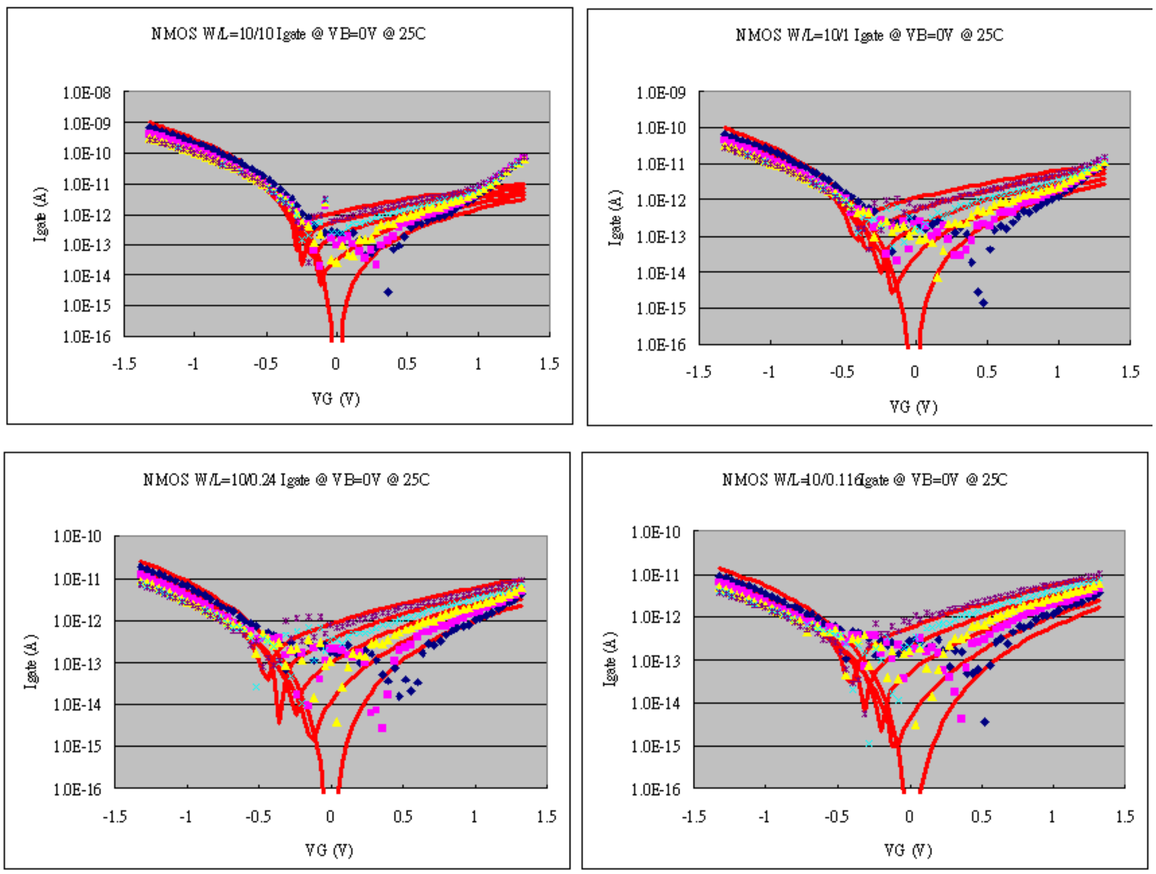


Figure 3-17 Measured and Simulated I_{GATE} for PMOS @25C

- ***I*$_{GATE}$ *vs V*$_{GS}$ *@ V*$_{BS}$ *=0 @125C***

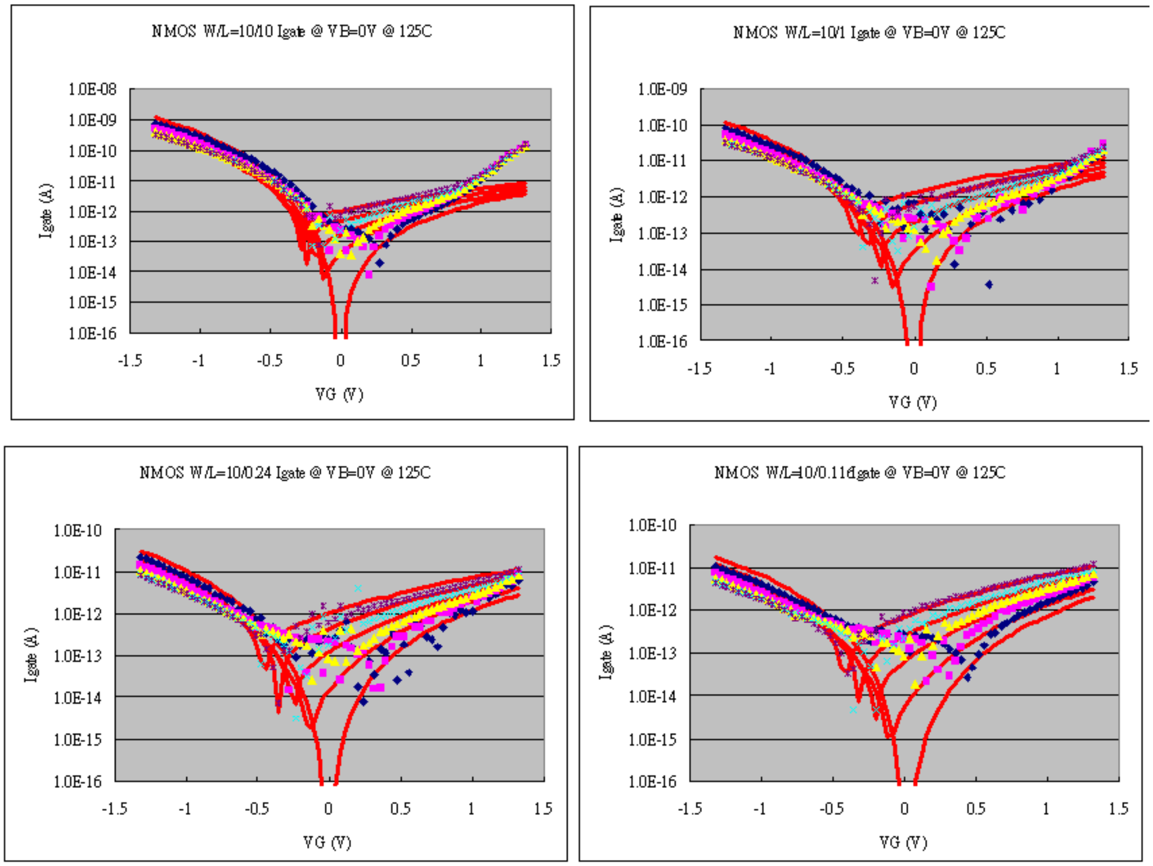


Figure 3-18 Measured and Simulated I$_{GATE}$ for PMOS @125C

- ***I_{GATE} vs V_{GS} @ V_{BS}=0 @-50C***

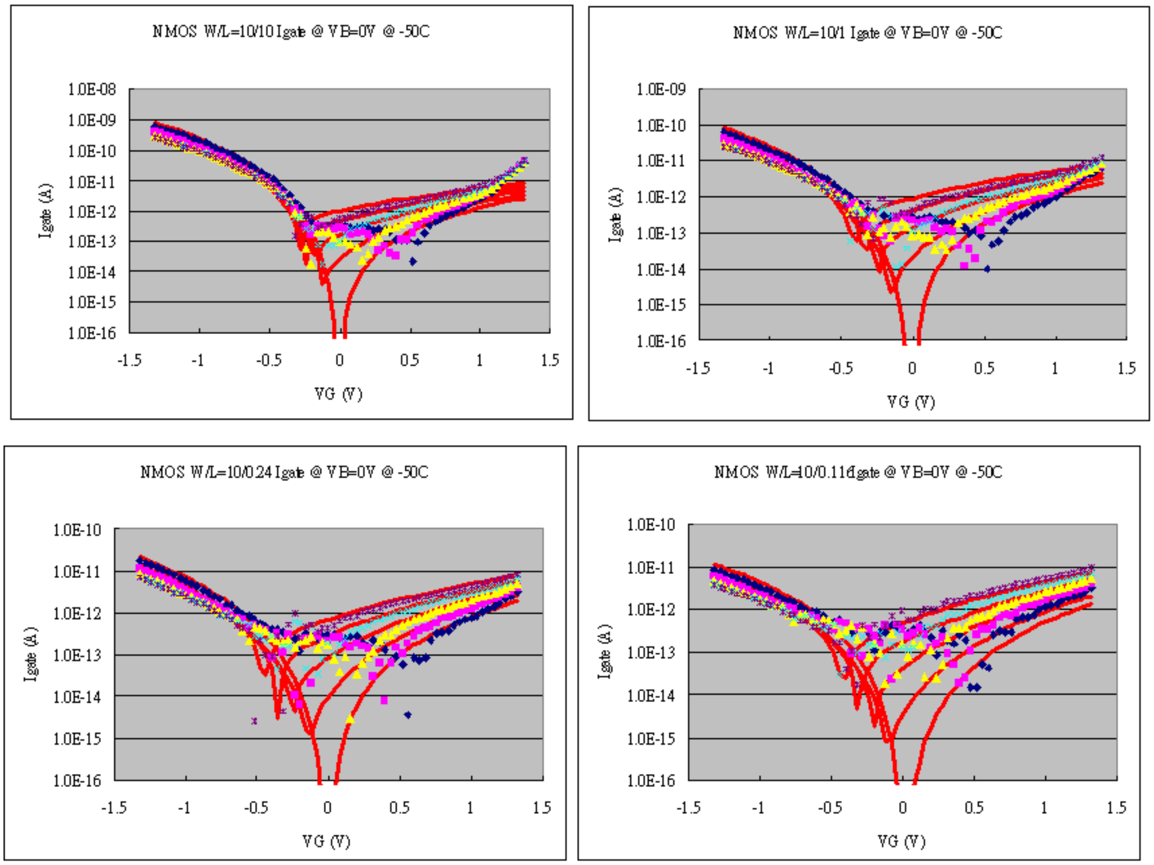


Figure 3-19 Measured and Simulated I_{GATE} for PMOS @-50C

3.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = -1.2$ V

2) G_M and G_{DS} accuracy criteria

a. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
b. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 3-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

➢ ***I_D Accuracy***

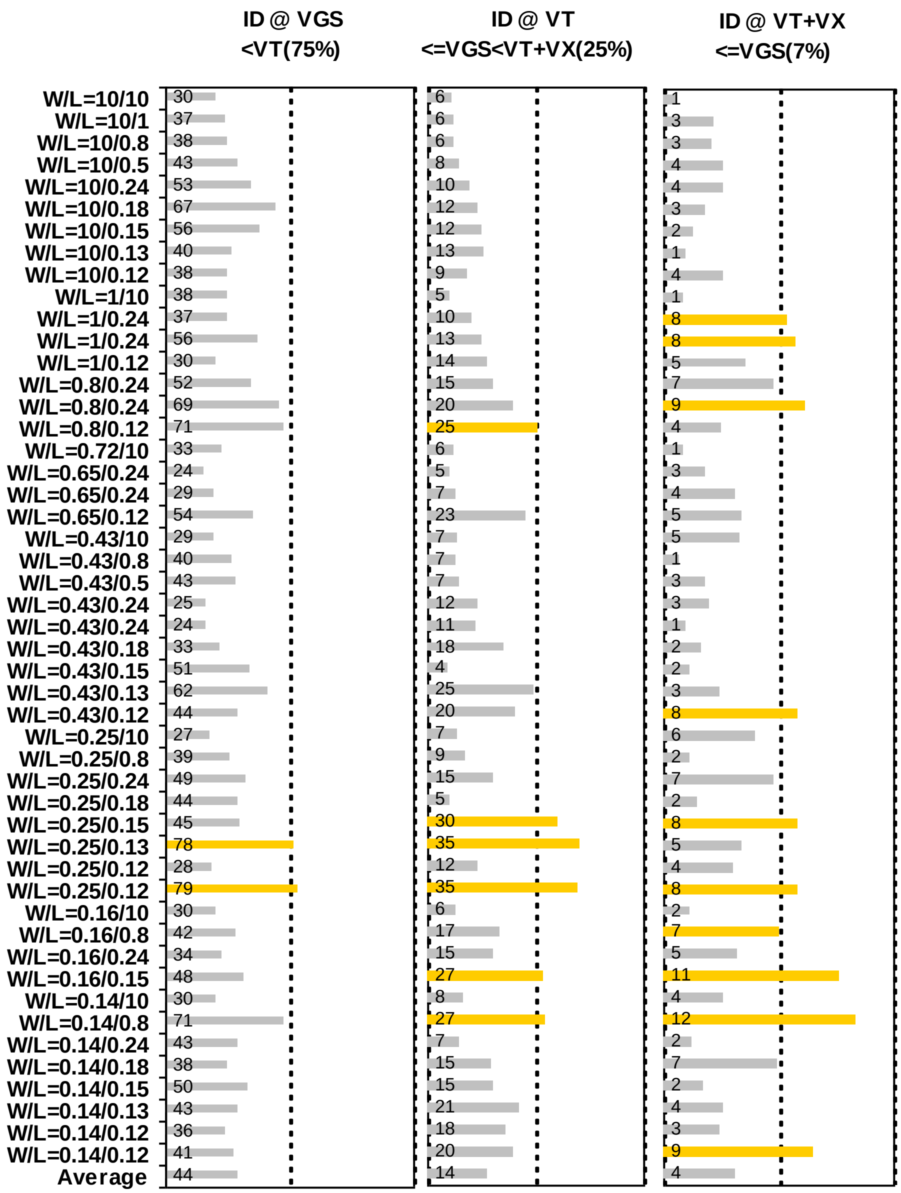


Figure 3-20 I_D accuracy at 25 ℃ for PMOS

➢ *G_M and G_{DS} Accuracy*

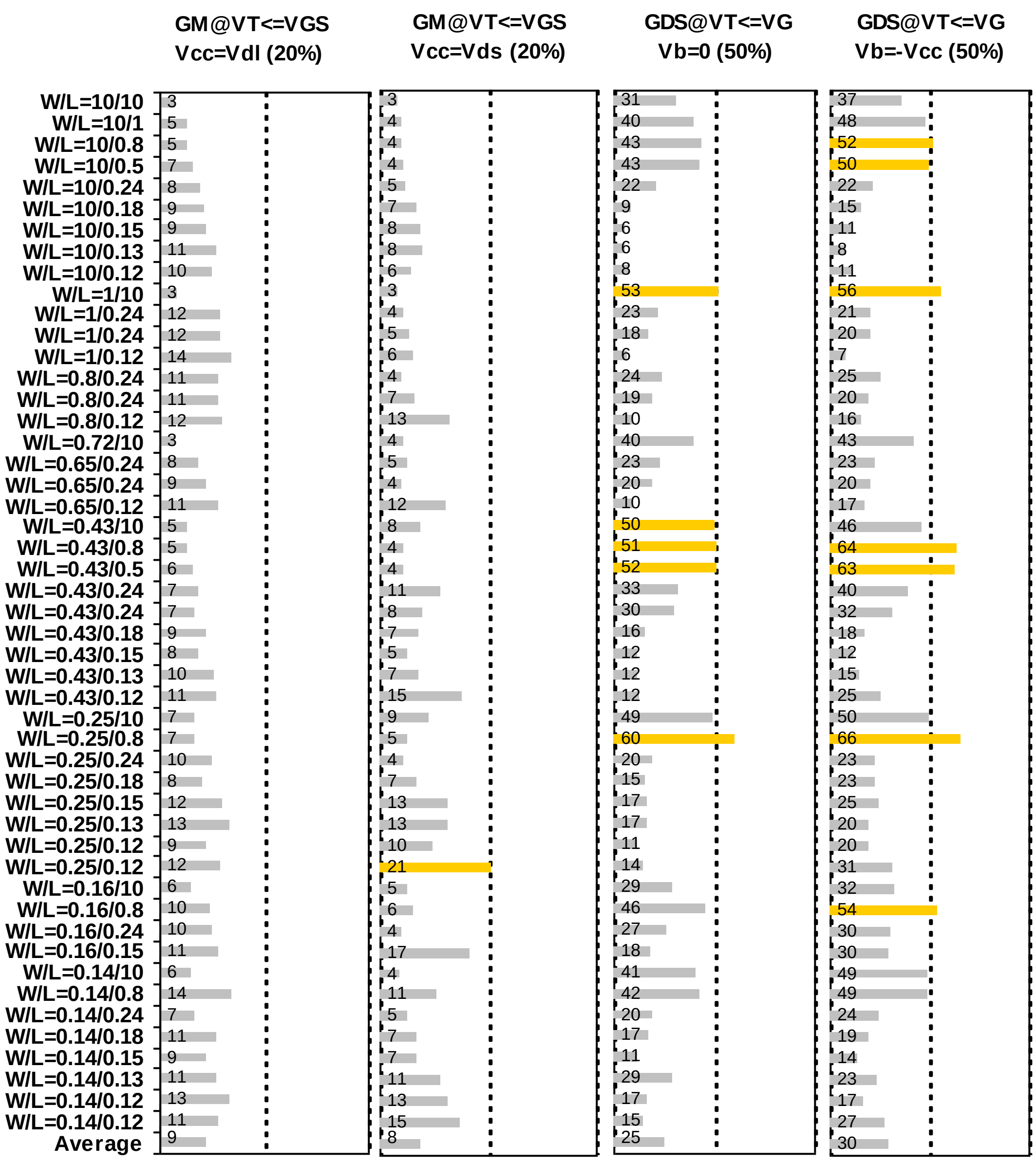


Figure 3-21 G_M and G_{DS} accuracy at 25 ℃ for PMOS

3.3.6 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Measurement structures for PMOS Cgg Capacitor structure

Structure	**L_{DES}[μm]**	**W_{DES}[μm]**
Gate capacitance components, C_{GDS}	200	100

3.3.7 Measurement conditions

Capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-7 capacitor measurement conditions for PMOS Cgg

V_{BIAS} [V]	From -1.35 to 1.35 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

3.3.8 Tox extraction

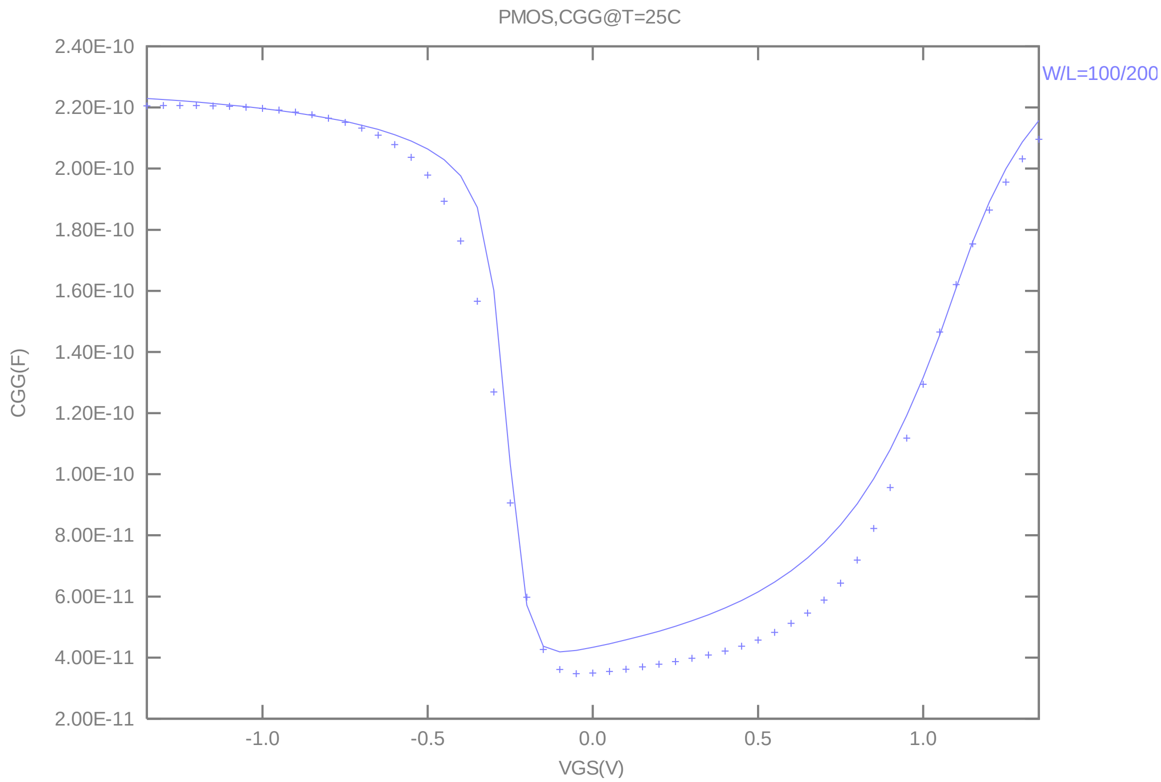


Figure 3-22 Measured and simulated Cgg curve

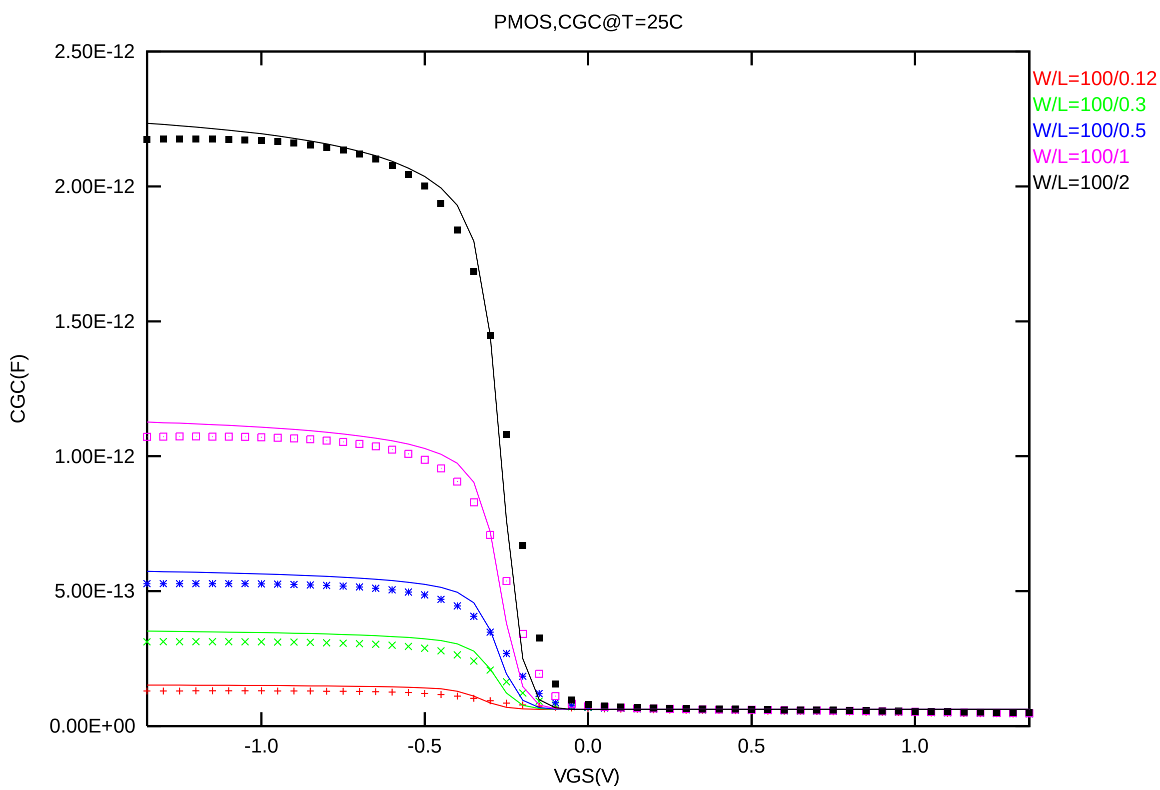


Figure 3-23 Measured and simulated Cgc curve

3.4 Parasitic source / drain junction capacitance model

3.4.1 Test devices

The following test structures were used for parameter extraction.

Table 3-8 Junction capacitor structures for PMOS

Structures	**Area[m^2]**	**Perimeter[m]**	**Description**
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.00E-08	1.83E-02	Poly finger structure

3.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-9 Junction capacitor measurement conditions for PMOS

V_{BIAS} [V]	VH: nwell & VL: p+ From 0 to 1.2 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

3.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 3-10 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	9.69E-04	CJSW	F/m	1.18E-10	CJSWG	F/m	5.31E-10
PB	V	6.87E-01	PBSW	V	4.50E-01	PBSWG	V	5.00E-01
MJ	--	3.08E-01	MJSW	--	1.00E-01	MJSWG	--	3.91E-01
TCJ	1/K	1.00E-03	TCJSW	1/K	5.24E-04	TCJSWG	1/K	5.29E-04
TPB	V/K	1.71E-03	TPBSW	V/K	2.59E-03	TPBSWG	V/K	9.24E-04

➢ *CJ, CJSW vs. VBIAS Plots*

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

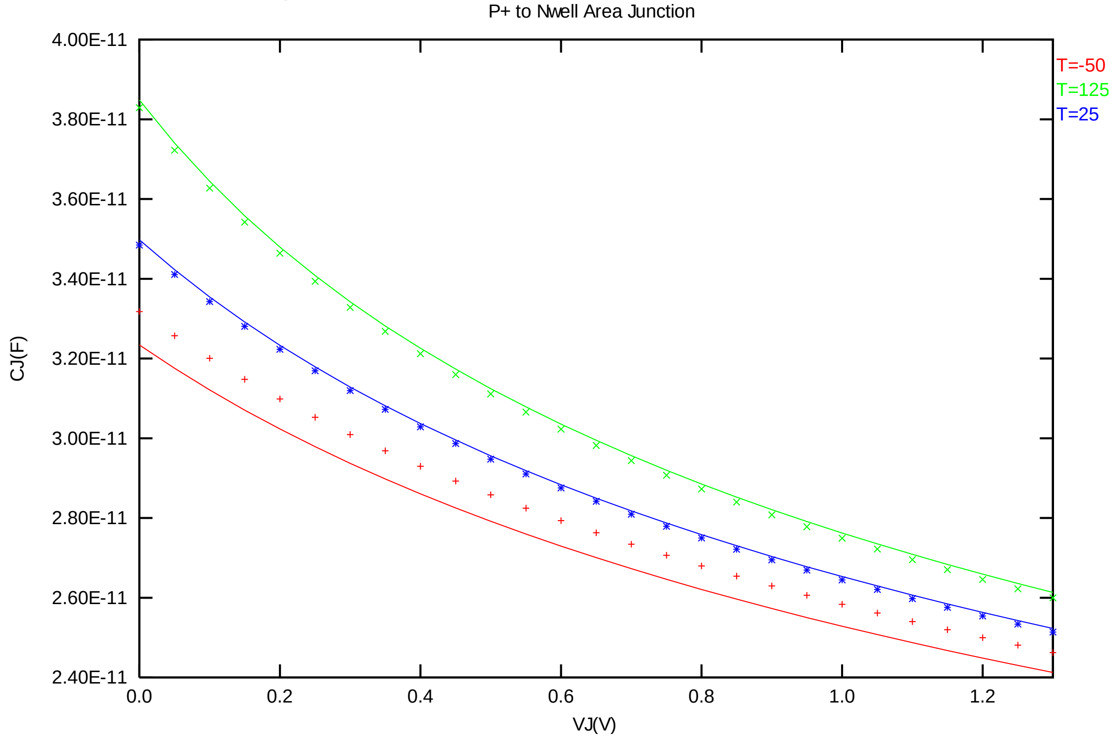


Figure 3-24 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

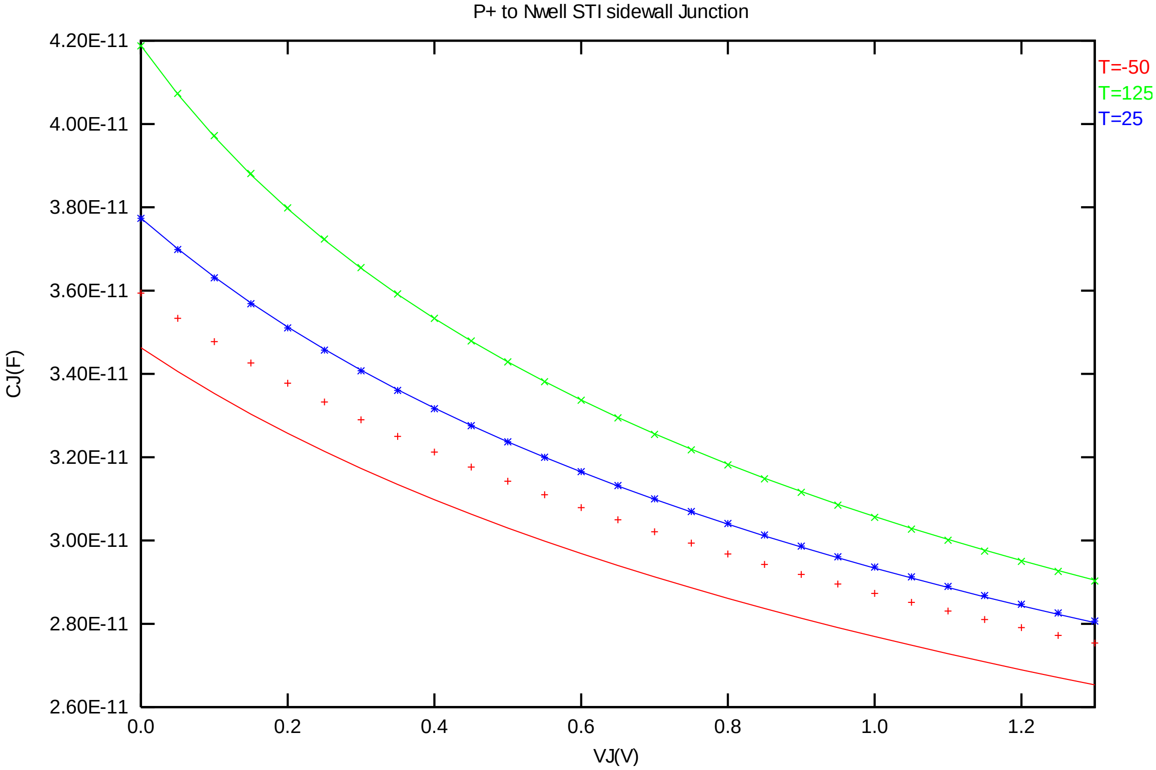


Figure 3-25 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

4 Dynamic Performance

4.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

Table 4-1 Ring Oscillator (Inverter) structure description for SA=0.34µm

R.O. (Inverter) Structure description	
Contact size	0.16 um X 0.16 um
Polysilicon gate to contact spacing	0.12 um
Contact to diffusion edge spacing	0.06 um
Rload	1E-5 Ohm
Lumped parasitic wiring capacitance	3.33E-15 F/stage
Inverter	
Logic gate type	Inverter
Channel length	0.116 um (with 8nm sizing back)
Channel width for n-ch MOSFET	2.5 um X 2
Channel width for p-ch MOSFET	3.75 um X 2

The simulation results at different temperature and worst-case models are listed below

Table 4-2 Ring oscillator simulation results (ps/stage)

Temp(C)	TT	FF	SS
-50	11.126	9.281	14.011
0	12.175	10.139	15.406
25	12.68	10.563	16.056
85	13.836	11.557	17.49
125	14.551	12.198	18.345

Vcc(V)	TT	FF	SS
0.90	19.163	14.726	26.731
1.00	16.213	12.886	21.706
1.10	14.171	11.561	18.387
1.20	12.68	10.563	16.056
1.32	11.355	9.652	14.04

5 Worst case model

5.1 Parameter Variations

Table 5-1 Corner parameters

	n_12_hsl110e	TT	SS	FF	SNFP	FNSP
1	tox	2.64E-09	+5.28E-11	-5.28E-11	+0.00E+00	+0.00E+00
2	wwl	-3.16E-22	-2.41E-23	+5.48E-23	-1.34E-23	+1.67E-23
3	xl	-3.00E-08	+2.32E-09	-5.60E-10	+1.15E-09	-1.15E-09
4	xw	1.00E-09	-1.84E-08	+1.22E-08	-8.05E-09	+6.68E-09
5	vth0	1.93E-01	+3.12E-02	-3.16E-02	+1.82E-02	-1.88E-02
6	pvth0	1.00E-15	+8.54E-16	-7.45E-16	+3.97E-16	-3.68E-16
7	k3	-2.95E+00	+2.35E+00	-2.83E+00	+1.29E+00	-1.33E+00
8	nlx	2.75E-07	+2.50E-08	-2.42E-08	+1.58E-08	-1.50E-08
9	vsat	1.03E+05	+6.50E+02	-1.52E+02	-7.50E+02	+4.20E+02
10	pvsat	3.36E-10	+4.84E-11	+1.70E-11	+1.85E-11	-3.20E-12
11	rdsw	4.40E+01	+9.47E+00	-8.80E+00	+3.54E+00	-2.60E+00
12	u0	3.72E-02	-1.49E-03	+1.20E-03	-8.82E-04	+7.98E-04
13	ags	7.26E-01	+1.09E-01	-1.05E-01	+5.11E-02	-5.07E-02
14	wags	5.06E-09	+6.32E-09	-6.88E-09	+3.62E-09	-3.52E-09
15	cj	7.20E-04	+10.0%	-10.0%	+10.0%	-10.0%
16	cjsw	9.94E-11	+10.0%	-10.0%	+10.0%	-10.0%
17	cjswg	3.03E-10	+10.0%	-10.0%	+10.0%	-10.0%
18	cgdo	1.90E-10	-10.0%	+10.0%	+0.0%	+0.0%
19	cgso	1.90E-10	-10.0%	+10.0%	+0.0%	+0.0%

	p_12_hsl110e	TT	SS	FF	SNFP	FNSP
1	tox	2.83E-09	+5.28E-11	-5.28E-11	+0.00E+00	+0.00E+00
2	wwl	-6.20E-23	+3.34E-23	-2.16E-23	-1.18E-23	+9.90E-24
3	xl	-2.10E-08	+3.68E-09	-7.43E-10	-6.35E-10	+1.33E-09
4	xw	4.00E-09	-2.20E-08	+5.10E-09	+3.34E-09	-7.20E-09
5	vth0	-2.51E-01	-2.93E-02	+3.01E-02	+1.89E-02	-1.85E-02
6	pvth0	-6.44E-16	-5.39E-16	+4.68E-16	+2.40E-16	-2.56E-16
7	k3	-1.22E+00	+2.02E+01	-2.14E+01	-1.03E+01	+1.04E+01
8	nlx	3.87E-08	+1.30E-08	-1.26E-08	-6.48E-09	+6.75E-09
9	vsat	1.22E+05	+2.91E+04	-7.84E+03	-4.16E+03	+8.60E+03
10	pvsat	-5.42E-10	+0.00E+00	-4.28E-10	-3.20E-10	+0.00E+00
11	rdsw	2.14E+02	+4.28E+01	-4.28E+01	-2.14E+01	+2.14E+01
12	u0	1.16E-02	-7.64E-04	+6.33E-04	+3.89E-04	-4.20E-04
13	ags	6.82E-01	+1.47E-01	-1.41E-01	-6.79E-02	+6.96E-02
14	wags	-2.53E-08	+3.75E-09	-1.08E-08	-5.21E-09	+3.38E-09
15	pags	-3.33E-14	+5.41E-14	+0.00E+00	+0.00E+00	+2.45E-14
16	cj	9.69E-04	+10.0%	-10.0%	-10.0%	+10.0%
17	cjsw	1.18E-10	+10.0%	-10.0%	-10.0%	+10.0%
18	cjswg	4.82E-10	+10.0%	-10.0%	-10.0%	+10.0%
19	cgdo	2.00E-10	-10.0%	+10.0%	+0.0%	+0.0%
20	cgso	2.00E-10	-10.0%	+10.0%	+0.0%	+0.0%

➢ ***Threshold voltage in linear region*** V_{TLIN}

The threshold voltage extracted at ID =300nA* (W_{DES} *0.9)/(L_{DES} *0.9+0.008 μ m) for NMOS and ID =-70nA* (W_{DES} *0.9)/(L_{DES} *0.9+0.008 μ m) for PMOS at 25 ℃. (The dimensions are after 90% shrink and 8nm sizing channel length back.)

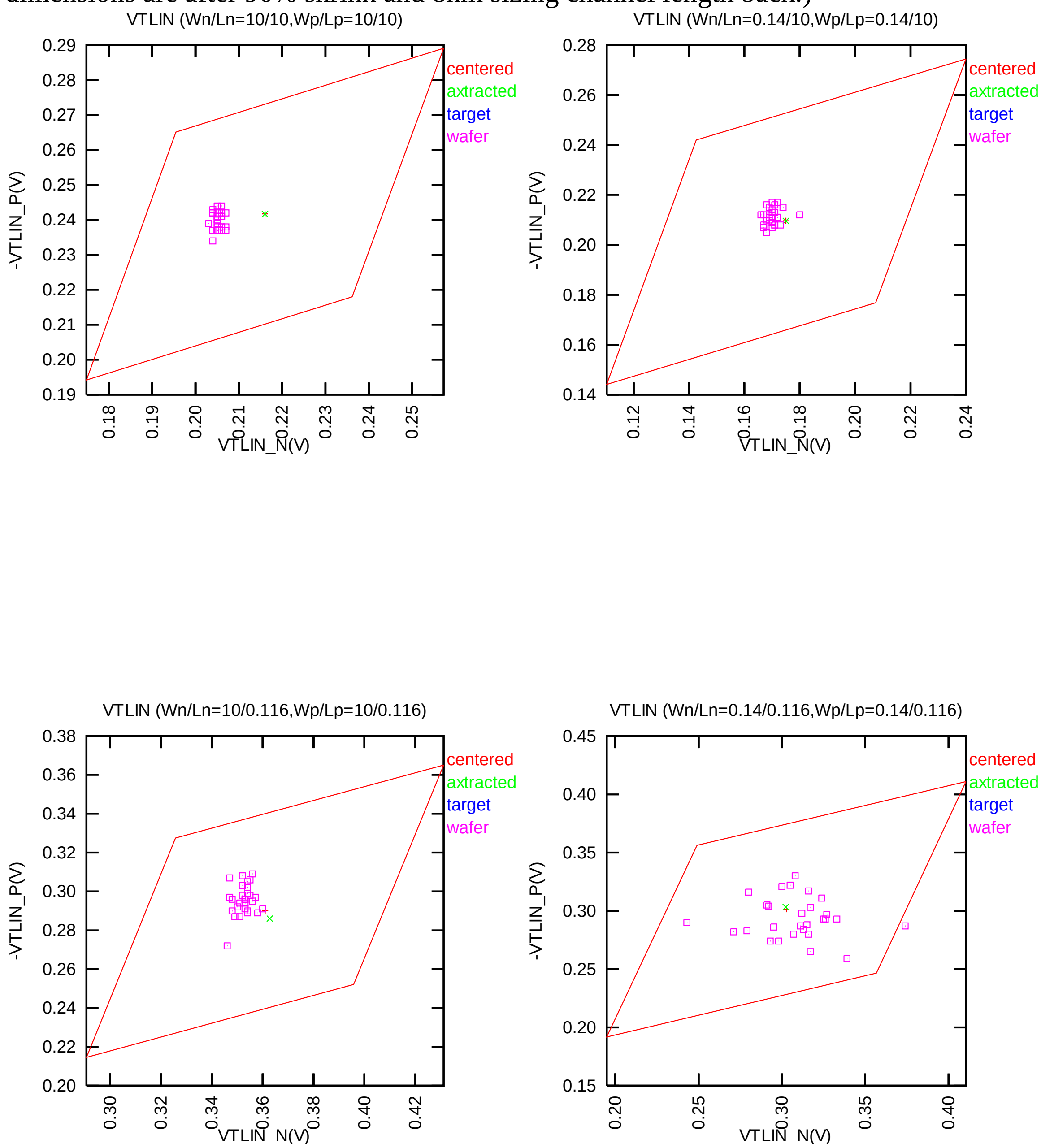


Figure 5-1 Corner graph of threshold voltage at low drain bias

➢ *Saturation Current I_{ON}*

(The dimensions are after 90% shrink and 8nm sizing channel length back.)

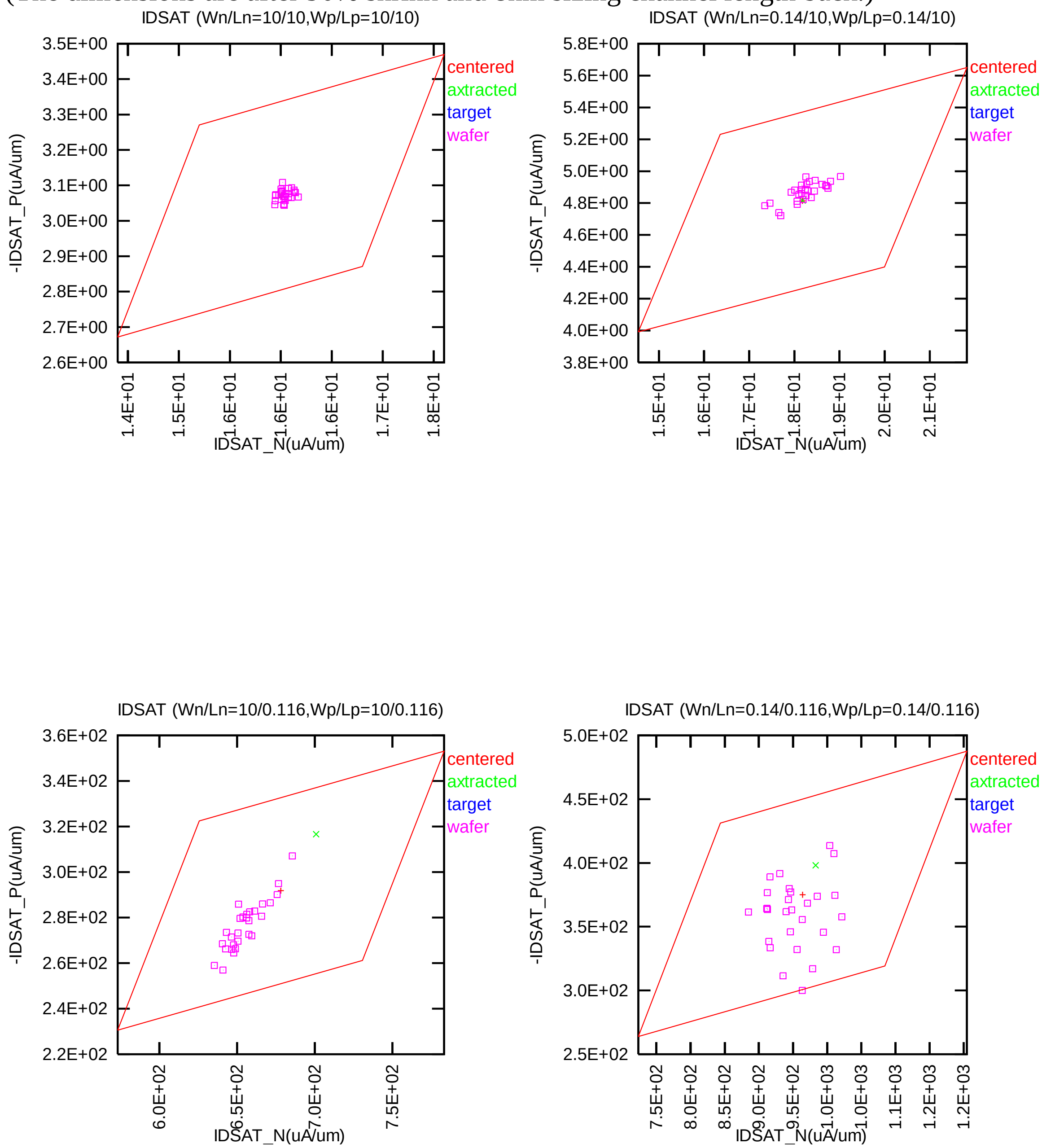


Figure 5-2 Corner graph of on-current

- ***DEVICE PERFORMANCE LOOK UP TABLE (SIMULATION RESULTS FROM HSPICE VER.05.3):***

** The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 8nm. It means: The EDR value of size W/L : 10um/0.12um would be the same as the model card simulation result in size of W/L : 10um/0.108um(After 90% shrink without 8nm sizing channel length back).

Table 5-2 Corner window of NMOS Vtln

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.261	0.249	0.216	0.201	0.164	0.141
SS	9/9	0.301	0.290	0.258	0.242	0.207	0.183
FF	9/9	0.220	0.208	0.175	0.160	0.121	0.098
SNFP	9/9	0.280	0.269	0.236	0.221	0.185	0.161
FNSP	9/9	0.240	0.229	0.196	0.180	0.142	0.119
TT	9/0.108	0.420	0.314	0.356	0.250	0.288	0.180
SS	9/0.108	0.489	0.386	0.427	0.323	0.361	0.255
FF	9/0.108	0.350	0.245	0.286	0.180	0.216	0.109
SNFP	9/0.108	0.454	0.351	0.391	0.287	0.324	0.218
FNSP	9/0.108	0.385	0.278	0.321	0.213	0.252	0.142
TT	0.144/9	0.220	0.207	0.176	0.160	0.128	0.105
SS	0.144/9	0.283	0.270	0.241	0.223	0.196	0.171
FF	0.144/9	0.156	0.144	0.112	0.096	0.060	0.038
SNFP	0.144/9	0.251	0.238	0.209	0.192	0.162	0.138
FNSP	0.144/9	0.188	0.176	0.144	0.128	0.094	0.071
TT	0.144/0.108	0.362	0.275	0.301	0.211	0.233	0.140
SS	0.144/0.108	0.468	0.388	0.409	0.327	0.345	0.258
FF	0.144/0.108	0.255	0.164	0.193	0.098	0.122	0.023
SNFP	0.144/0.108	0.415	0.332	0.355	0.269	0.289	0.199
FNSP	0.144/0.108	0.309	0.219	0.247	0.154	0.177	0.081

Table 5-3 Corner window of PMOS Vtln

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-VTH (V)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.298	0.293	0.242	0.235	0.177	0.165
SS	9/9	0.345	0.340	0.290	0.282	0.226	0.213
FF	9/9	0.251	0.246	0.194	0.187	0.128	0.117
SNFP	9/9	0.274	0.270	0.218	0.211	0.153	0.141
FNSP	9/9	0.321	0.316	0.265	0.258	0.201	0.189
TT	9/0.108	0.336	0.235	0.285	0.180	0.225	0.113
SS	9/0.108	0.412	0.316	0.361	0.262	0.303	0.197
FF	9/0.108	0.261	0.160	0.209	0.104	0.147	0.035
SNFP	9/0.108	0.299	0.196	0.246	0.141	0.186	0.073
FNSP	9/0.108	0.374	0.275	0.323	0.221	0.264	0.155
TT	0.144/9	0.269	0.265	0.211	0.205	0.142	0.133
SS	0.144/9	0.334	0.330	0.276	0.270	0.209	0.199
FF	0.144/9	0.204	0.201	0.145	0.140	0.076	0.066
SNFP	0.144/9	0.237	0.233	0.178	0.172	0.109	0.100
FNSP	0.144/9	0.301	0.298	0.243	0.237	0.176	0.165
TT	0.144/0.108	0.357	0.283	0.299	0.222	0.232	0.147
SS	0.144/0.108	0.466	0.400	0.410	0.340	0.345	0.269
FF	0.144/0.108	0.247	0.170	0.189	0.108	0.121	0.030
SNFP	0.144/0.108	0.302	0.225	0.244	0.164	0.176	0.087
FNSP	0.144/0.108	0.412	0.341	0.354	0.280	0.288	0.207

Table 5-4 Corner window of NMOS Ion

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	3.74	24.73	3.12	17.76	2.68	12.85
SS	9/9	2.79	22.01	2.42	15.97	2.15	11.71
FF	9/9	4.87	27.48	3.93	19.54	3.27	13.98
SNFP	9/9	3.27	23.33	2.77	16.86	2.42	12.30
FNSP	9/9	4.26	26.15	3.49	18.65	2.95	13.40
TT	9/0.108	104.36	744.50	113.38	679.90	118.56	590.49
SS	9/0.108	51.89	632.36	65.56	574.79	76.20	498.91
FF	9/0.108	170.76	857.27	171.99	784.99	168.56	681.30
SNFP	9/0.108	76.01	688.58	87.96	627.30	96.35	544.60
FNSP	9/0.108	136.41	800.02	141.82	732.50	143.08	636.64
TT	0.144/9	4.36	26.34	3.68	20.00	3.20	15.26
SS	0.144/9	2.69	20.62	2.38	15.99	2.17	12.50
FF	0.144/9	6.52	32.19	5.30	24.02	4.44	17.96
SNFP	0.144/9	3.48	23.42	3.00	18.00	2.67	13.91
FNSP	0.144/9	5.35	29.30	4.43	22.01	3.78	16.60
TT	0.144/0.108	175.48	1054.03	174.60	951.81	171.40	817.71
SS	0.144/0.108	65.65	794.86	77.83	715.07	87.69	617.19
FF	0.144/0.108	323.39	1310.21	302.39	1188.19	278.30	1019.65
SNFP	0.144/0.108	114.56	924.10	121.73	833.40	126.40	717.85
FNSP	0.144/0.108	245.59	1182.92	235.17	1070.07	222.31	918.13

Table 5-5 Corner window of PMOS Ion

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA/um)	-VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.52	4.05	0.53	3.39	0.56	3.00
SS	9/9	0.35	3.48	0.38	2.95	0.42	2.64
FF	9/9	0.74	4.63	0.71	3.84	0.72	3.35
SNFP	9/9	0.62	4.35	0.61	3.62	0.63	3.17
FNSP	9/9	0.44	3.76	0.45	3.17	0.49	2.83
TT	9/0.108	47.23	314.63	49.47	290.44	52.52	270.94
SS	9/0.108	24.76	251.87	27.94	228.76	31.56	212.30
FF	9/0.108	75.69	378.47	75.99	351.61	77.54	327.91
SNFP	9/0.108	60.80	346.54	62.12	321.08	64.47	299.33
FNSP	9/0.108	35.14	282.97	38.04	259.68	41.55	242.12
TT	0.144/9	0.82	6.20	0.82	5.20	0.86	4.61
SS	0.144/9	0.48	5.04	0.52	4.29	0.58	3.88
FF	0.144/9	1.27	7.39	1.20	6.11	1.20	5.32
SNFP	0.144/9	1.02	6.80	1.00	5.65	1.02	4.96
FNSP	0.144/9	0.64	5.61	0.66	4.74	0.72	4.25
TT	0.144/0.108	48.22	435.57	50.75	363.18	54.99	312.47
SS	0.144/0.108	14.44	296.93	19.02	253.65	24.62	228.09
FF	0.144/0.108	105.15	578.09	98.95	472.31	94.35	378.72
SNFP	0.144/0.108	74.06	506.67	73.02	417.81	73.79	346.71
FNSP	0.144/0.108	28.47	365.65	32.86	308.52	38.50	271.24

Table 5-6 Corner window of NMOS Ioff

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	8.67E-12	1.15E-11	5.00E-10	6.71E-10	9.47E-09	1.33E-08
SS	9/9	1.46E-12	1.94E-12	1.32E-10	1.76E-10	3.55E-09	4.95E-09
FF	9/9	5.20E-11	6.91E-11	1.88E-09	2.53E-09	2.47E-08	3.50E-08
SNFP	9/9	3.56E-12	4.73E-12	2.58E-10	3.46E-10	5.84E-09	8.16E-09
FNSP	9/9	2.15E-11	2.86E-11	9.80E-10	1.32E-09	1.54E-08	2.17E-08
TT	9/0.108	1.07E-11	5.26E-10	1.67E-09	3.19E-08	6.79E-08	6.44E-07
SS	9/0.108	5.32E-13	2.83E-11	1.75E-10	3.57E-09	1.26E-08	1.26E-07
FF	9/0.108	2.08E-10	8.37E-09	1.52E-08	2.46E-07	3.44E-07	2.88E-06
SNFP	9/0.108	2.32E-12	1.16E-10	5.30E-10	1.04E-08	2.90E-08	2.81E-07
FNSP	9/0.108	4.98E-11	2.39E-09	5.25E-09	9.76E-08	1.58E-07	1.46E-06
TT	0.144/9	1.90E-12	2.51E-12	3.91E-11	5.27E-11	3.60E-10	5.09E-10
SS	0.144/9	1.44E-13	1.92E-13	5.53E-12	7.42E-12	8.16E-11	1.14E-10
FF	0.144/9	2.53E-11	3.35E-11	2.64E-10	3.57E-10	1.50E-09	2.17E-09
SNFP	0.144/9	5.08E-13	6.73E-13	1.45E-11	1.95E-11	1.71E-10	2.40E-10
FNSP	0.144/9	7.10E-12	9.40E-12	1.04E-10	1.40E-10	7.45E-10	1.06E-09
TT	0.144/0.108	6.28E-13	1.76E-11	7.85E-11	9.98E-10	2.69E-09	1.92E-08
SS	0.144/0.108	4.51E-15	1.02E-13	1.88E-12	2.09E-11	1.66E-10	1.09E-09
FF	0.144/0.108	8.92E-11	2.28E-09	2.89E-09	3.28E-08	3.58E-08	2.36E-07
SNFP	0.144/0.108	5.07E-14	1.30E-12	1.19E-11	1.44E-10	6.67E-10	4.65E-09
FNSP	0.144/0.108	7.67E-12	2.20E-10	4.97E-10	6.26E-09	1.03E-08	7.19E-08

Table 5-7 Corner window of PMOS Ioff

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IOFF (A)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	9.26E-13	1.01E-12	6.94E-11	7.62E-11	1.67E-09	1.90E-09
SS	9/9	1.21E-13	1.35E-13	1.51E-11	1.65E-11	5.42E-10	6.14E-10
FF	9/9	7.23E-12	7.85E-12	3.17E-10	3.49E-10	5.01E-09	5.72E-09
SNFP	9/9	2.63E-12	2.85E-12	1.50E-10	1.65E-10	2.92E-09	3.33E-09
FNSP	9/9	3.30E-13	3.61E-13	3.21E-11	3.52E-11	9.50E-10	1.08E-09
TT	9/0.108	1.14E-10	3.24E-09	5.55E-09	7.16E-08	9.95E-08	7.14E-07
SS	9/0.108	7.60E-12	1.83E-10	7.22E-10	8.28E-09	2.17E-08	1.43E-07
FF	9/0.108	1.55E-09	4.30E-08	3.89E-08	4.84E-07	4.19E-07	2.94E-06
SNFP	9/0.108	4.37E-10	1.26E-08	1.51E-08	1.96E-07	2.08E-07	1.50E-06
FNSP	9/0.108	2.88E-11	7.71E-10	1.98E-09	2.46E-08	4.64E-08	3.24E-07
TT	0.144/9	4.20E-14	4.74E-14	2.59E-12	2.84E-12	5.37E-11	6.09E-11
SS	0.144/9	2.57E-15	5.05E-15	3.13E-13	3.44E-13	1.14E-11	1.29E-11
FF	0.144/9	7.13E-13	7.71E-13	2.05E-11	2.25E-11	2.36E-10	2.71E-10
SNFP	0.144/9	1.76E-13	1.92E-13	7.43E-12	8.13E-12	1.14E-10	1.30E-10
FNSP	0.144/9	1.01E-14	1.31E-14	8.95E-13	9.80E-13	2.48E-11	2.81E-11
TT	0.144/0.108	2.31E-13	3.39E-12	2.44E-11	2.00E-10	7.89E-10	4.28E-09
SS	0.144/0.108	2.80E-15	3.33E-14	8.49E-13	5.67E-12	6.37E-11	2.92E-10
FF	0.144/0.108	1.80E-11	2.79E-10	6.26E-10	5.27E-09	8.42E-09	4.82E-08
SNFP	0.144/0.108	2.10E-12	3.32E-11	1.27E-10	1.10E-09	2.65E-09	1.53E-08
FNSP	0.144/0.108	2.48E-14	3.32E-13	4.55E-12	3.45E-11	2.26E-10	1.16E-09

6 Shallow Trench Isolation (STI) or LOD Stress Effect

6.1 Simulation results for NMOS at SA=0.38μm (SAREF=2.2μm)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008μm). The dimension is after 90% shrink and 8nm sizing channel length back.

Table 6-1 Simulation results for NMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value

Long channel device (W/L= 10 μ m / 10 μ m)

V_{TLIN}	V_{DS} = 0.05 V	V	0.217
I_{ON}	V_{DS} = 1.2 V	μA/ μm	15.91

Wide and short channel device (W/L= 10 μ m / 0.116 μ m)

V_{TLIN}	V_{DS} = 0.05 V	V	0.370
V_{TSAT}	V_{DS} = 1.2 V	V	0.263
Body Effect V_T shift	V_{DS} = 1.2 V V_{BS}= 0 ~ -1.2 V	V	0.029
DIBL	V_{DS}= 0.05~1.2 V	V	0.107
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	651.87
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	2.4E-09

Narrow and long channel device (W/L= 0.14 μ m / 10 μ m)

V_{TLIN}	V_{DS} = 0.05 V	V	0.175
V_{TSAT}	V_{DS}= 1.2 V	V	0.159
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	18.08

Narrow and short channel device (W/L= 0.14 μm / 0.116 μm)

V_{TLIN}	V_{DS} = 0.05 V	V	0.313
V_{TSAT}	V_{DS}= 1.2 V	V	0.225
I_{ON}	V_{DS}=V_{GS}= 1.2 V	μA/μm	923.14
I_{OFF}	V_{DS}= 1.2 V, V_{GS}=0 V	A/μm	4.4E-09

6.1.1 Model verification plots for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

- ***V_T vs. SA***

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008µm). The dimension is after 90% shrink and 8nm sizing channel length back.

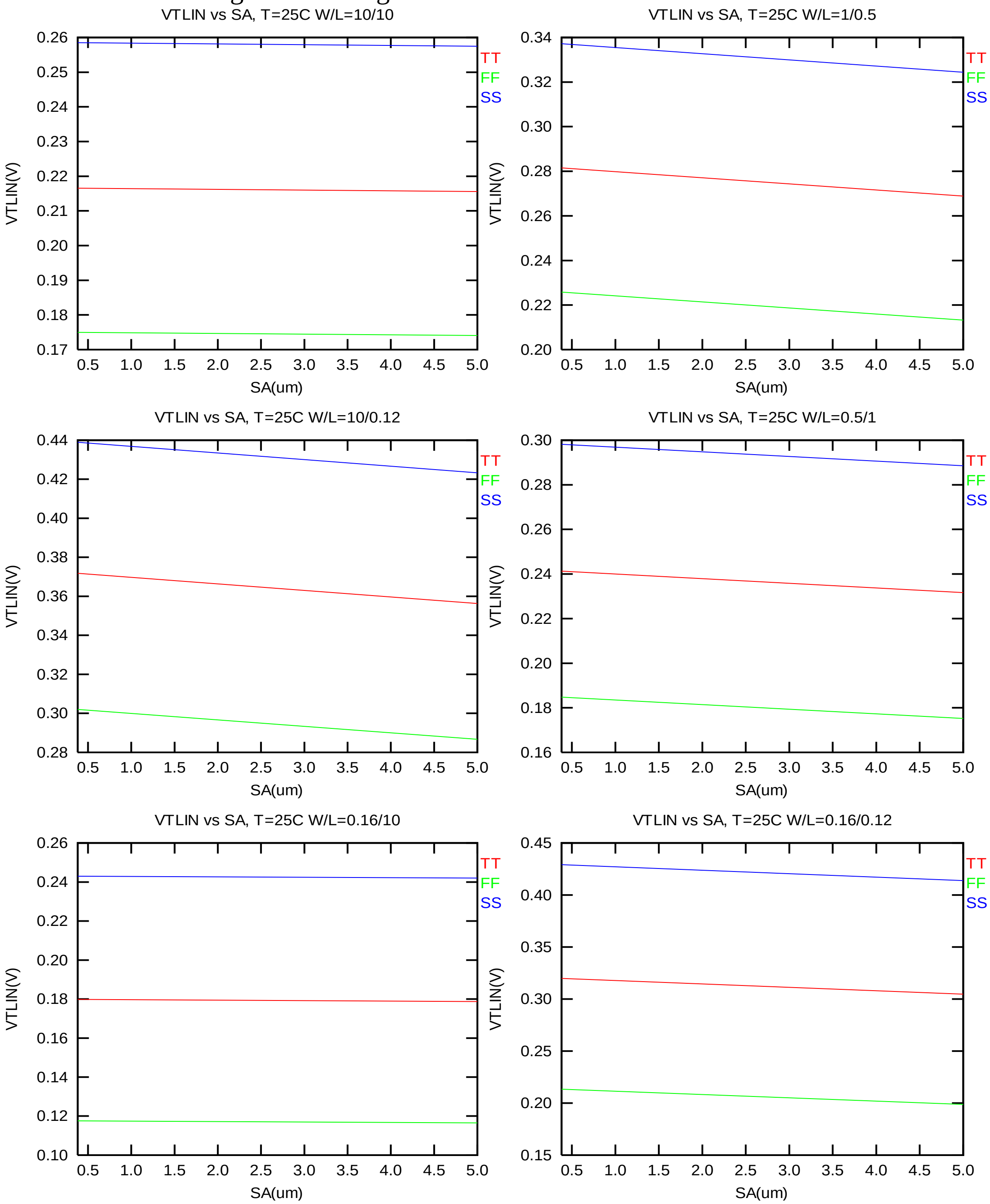


Figure 6-1 V_T vs. SA for low drain bias (V_{DS} = 0.05 V) for NMOS

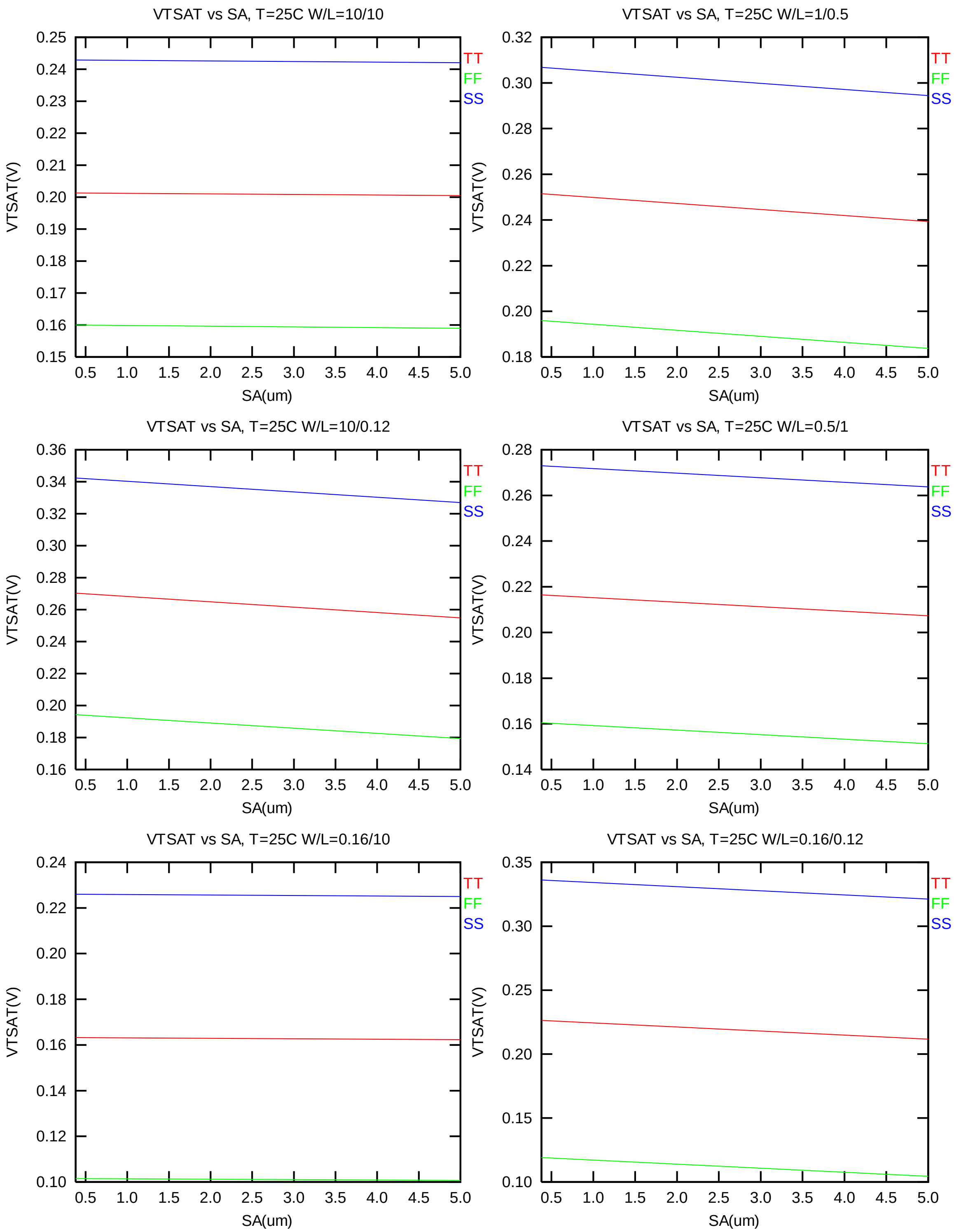


Figure 6-2 V_T vs. SA for high drain bias (V_{DS} = 1.2 V) for NMOS

➢ *I_{ON} vs. SA*

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}=1.2 V for V_{BS}=0 at 25℃.

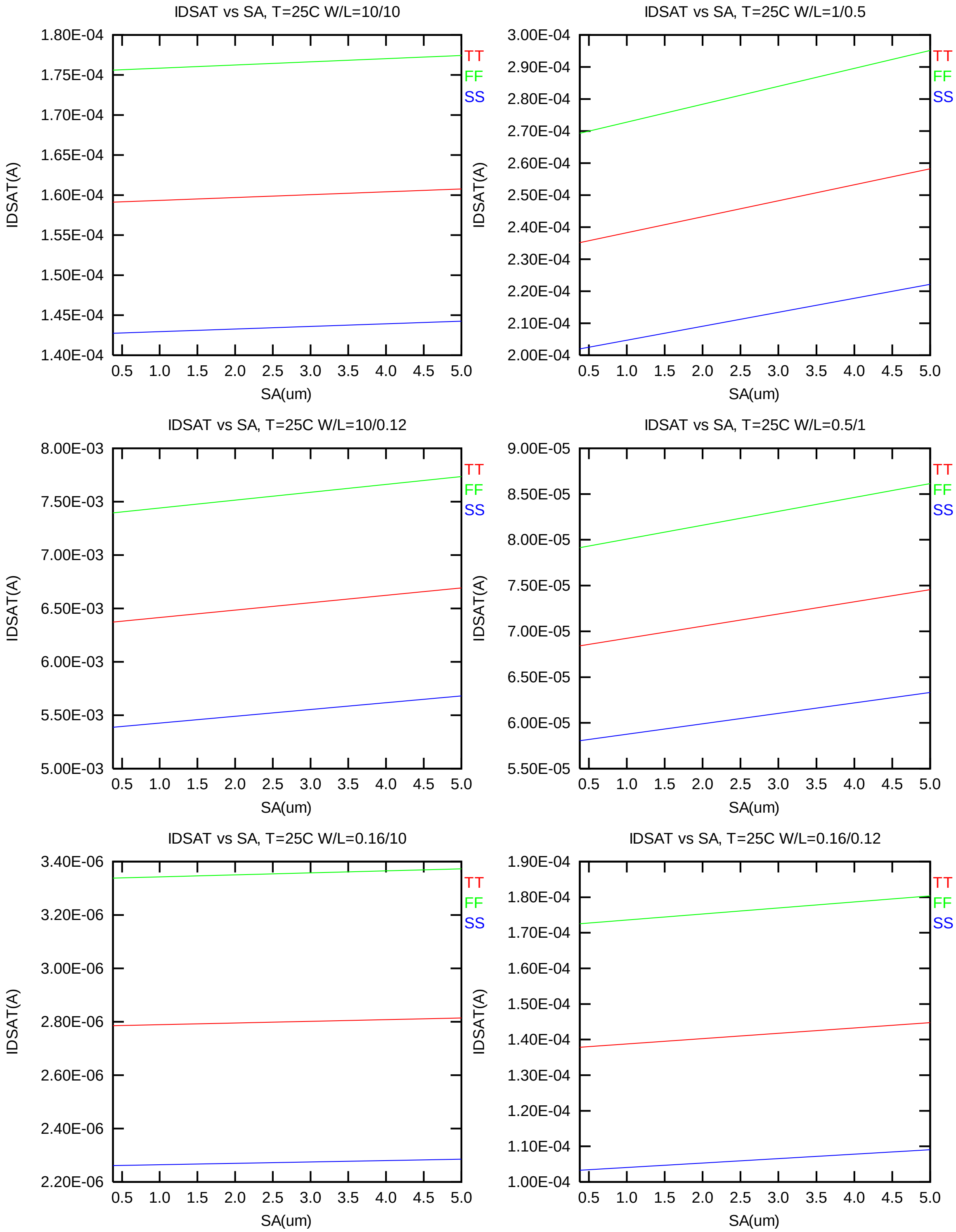


Figure 6-3 Saturation current I_{ON} vs. SA for NMOS

6.2 Simulation results for PMOS at SA=0.38μm (SAREF=2.2μm)

The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at ID =-70nA* (W_{DES} *0.9)/(L_{DES} *0.9+0.008μm). The dimension is after 90% shrink and 8nm sizing channel length back.

Table 6-2 Simulation results for PMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W/L= 10 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = -0.05 V	V	-0.241
I_{ON}	V_{DS} = -1.2 V	μA/ μm	-3.08
Wide and short channel device (W/L= 10 μ m / 0.116 μ m)			
V_{TLIN}	V_{DS} = -0.05 V	V	-0.280
V_{TSAT}	V_{DS} = -1.2 V	V	-0.176
Body Effect V_T shift	V_{DS} = -1.2 V V_{BS}= 0 ~ 1.2 V	V	-0.137
DIBL	V_{DS}= -0.05~-1.2 V	V	-0.104
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-311.83
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-8.8E-09
Narrow and long channel device (W/L= 0.14 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = -0.05 V	V	-0.209
V_{TSAT}	V_{DS}= -1.2 V	V	-0.203
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-4.83
Narrow and short channel device (W/L= 0.14 μm / 0.116 μm)			
V_{TLIN}	V_{DS} = -0.05 V	V	-0.294
V_{TSAT}	V_{DS}= -1.2 V	V	-0.218
I_{ON}	V_{DS}=V_{GS}= -1.2 V	μA/μm	-401.37
I_{OFF}	V_{DS}= -1.2 V, V_{GS}=0 V	A/μm	-1.5E-09

6.2.1 Model verification plots for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

- ***V_T vs. SA***

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+0.008µm). The dimension is after 90% shrink and 8nm sizing channel length back.

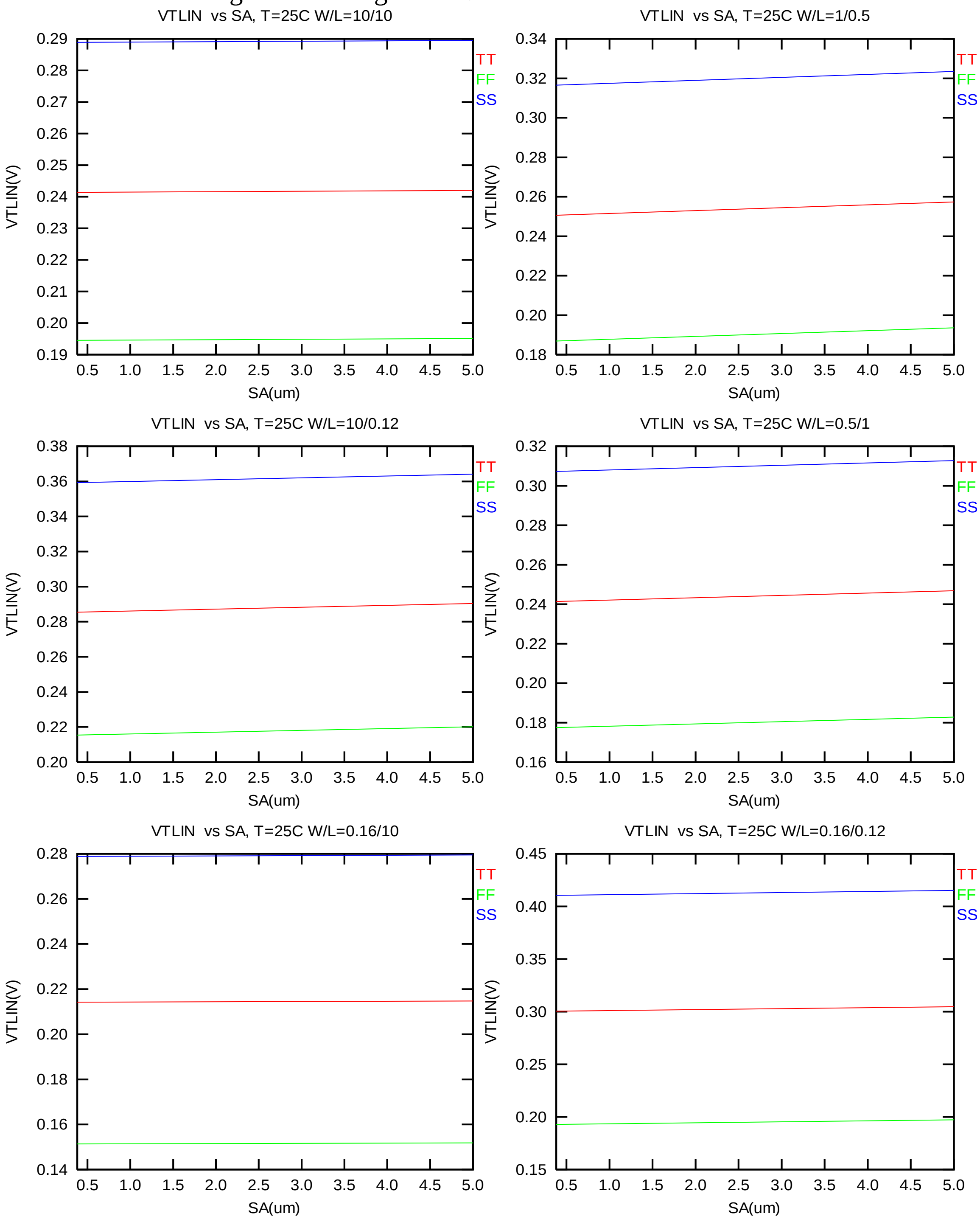


Figure 6-4 V_T vs. SA for low drain bias (V_{DS}= -0.05 V) for PMOS

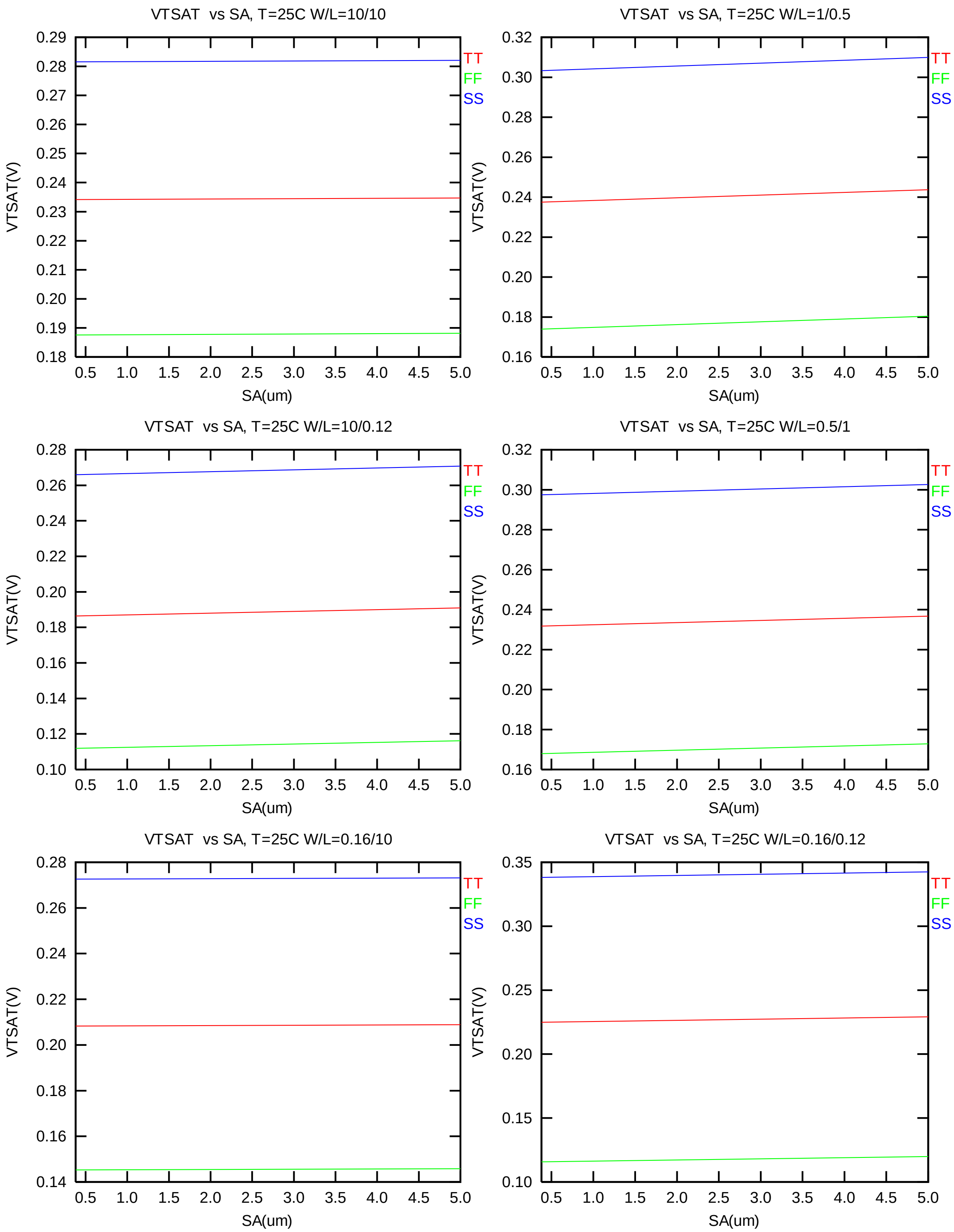


Figure 6-5 V_T vs. SA for high drain bias (V_{DS}= -1.2 V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}= -1.2$ V for $V_{BS}=0$ at 25℃.

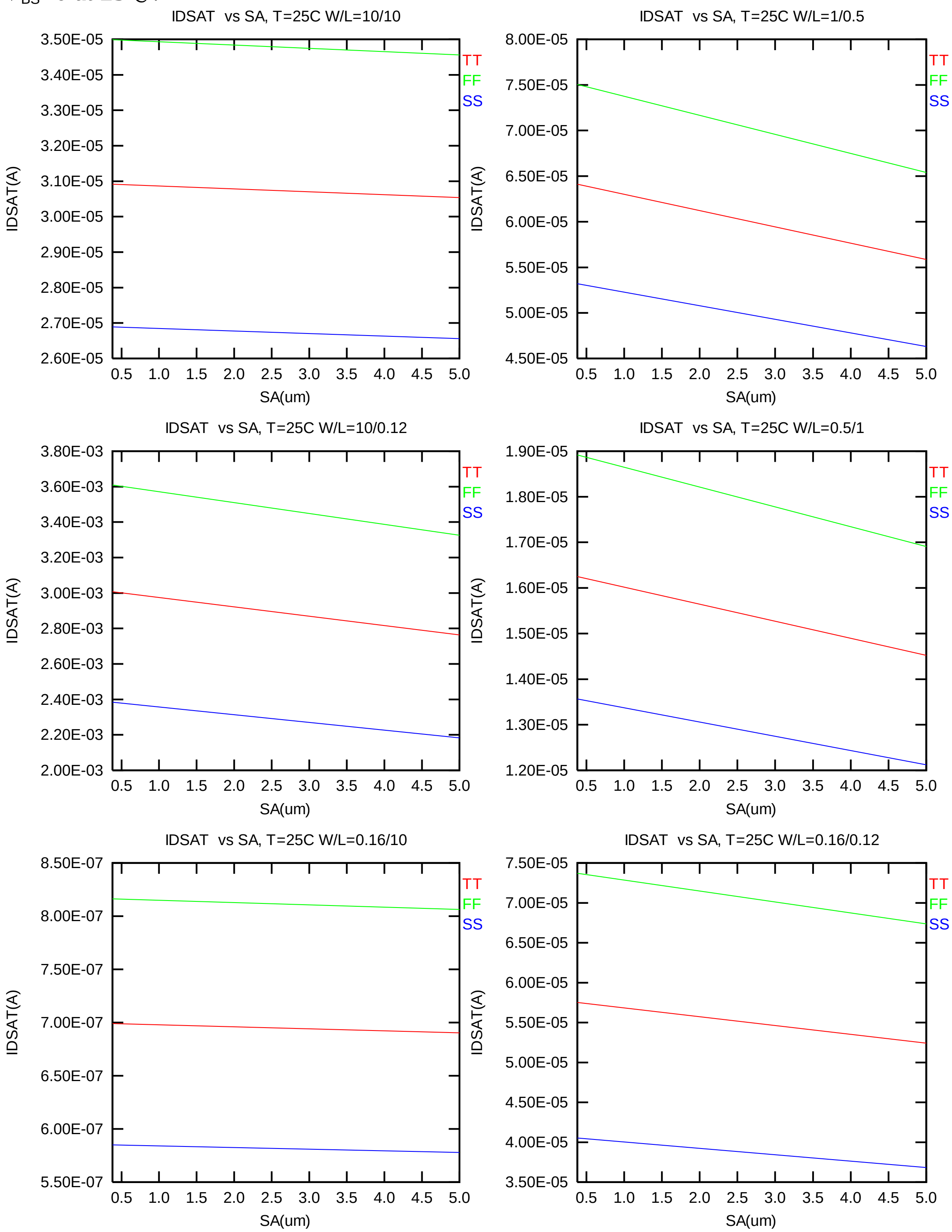


Figure 6-6 Saturation current I_{ON} vs. SA for PMOS

6.3 V_T and I_{ON} variation v. SA for both N/PMOS

Delta V_T is defined as V_T (SA)-V_T (2.2µm) for NMOS and -[V_T (SA)-V_T (2.2µm)] for PMOS. Delta I_{ON} is defined as [I_{ON} (SA)-I_{ON} (2.2µm)]/I_{ON} (2.2µm)*100%.

- ***Delta V_T and I_{ON} vs. SA***

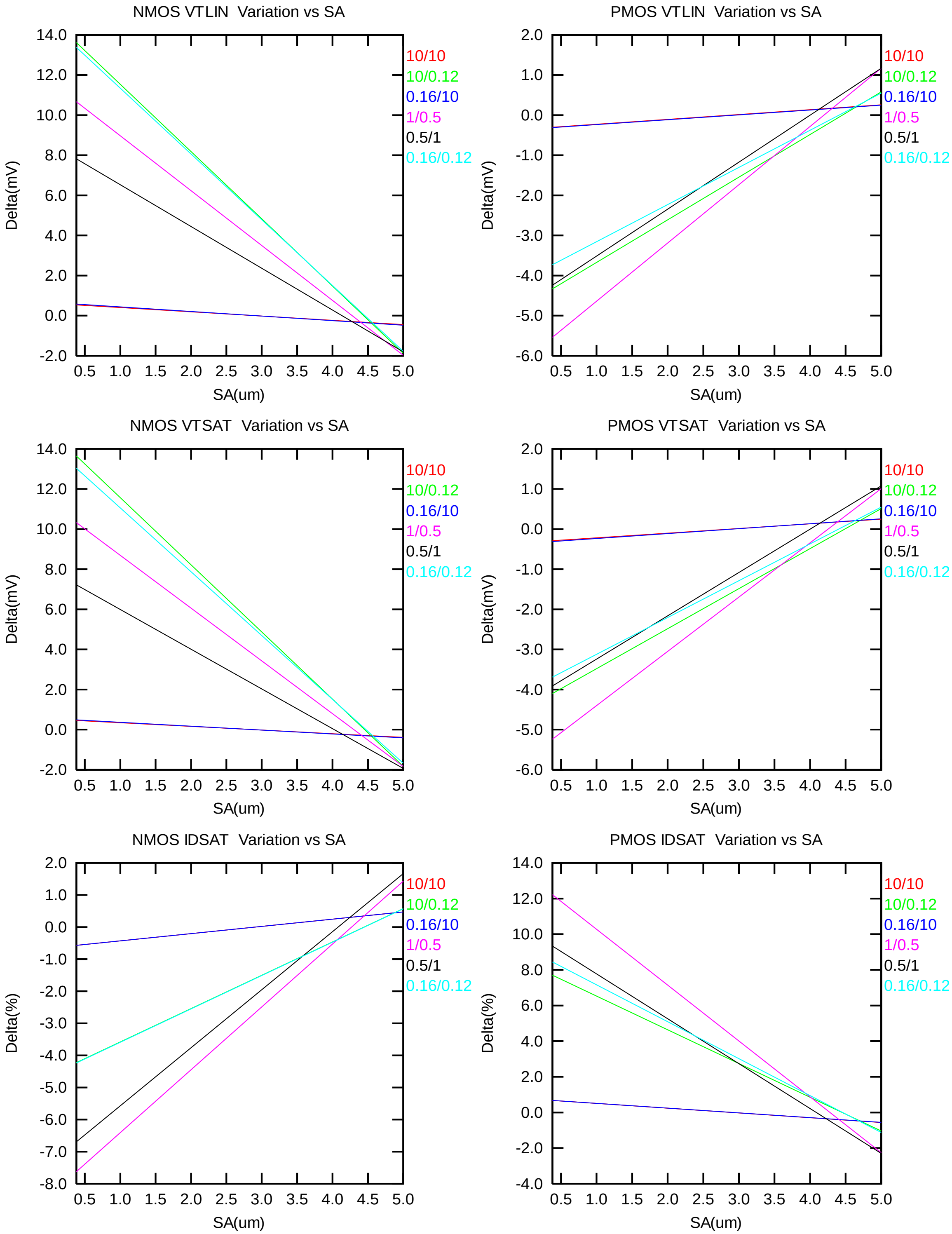
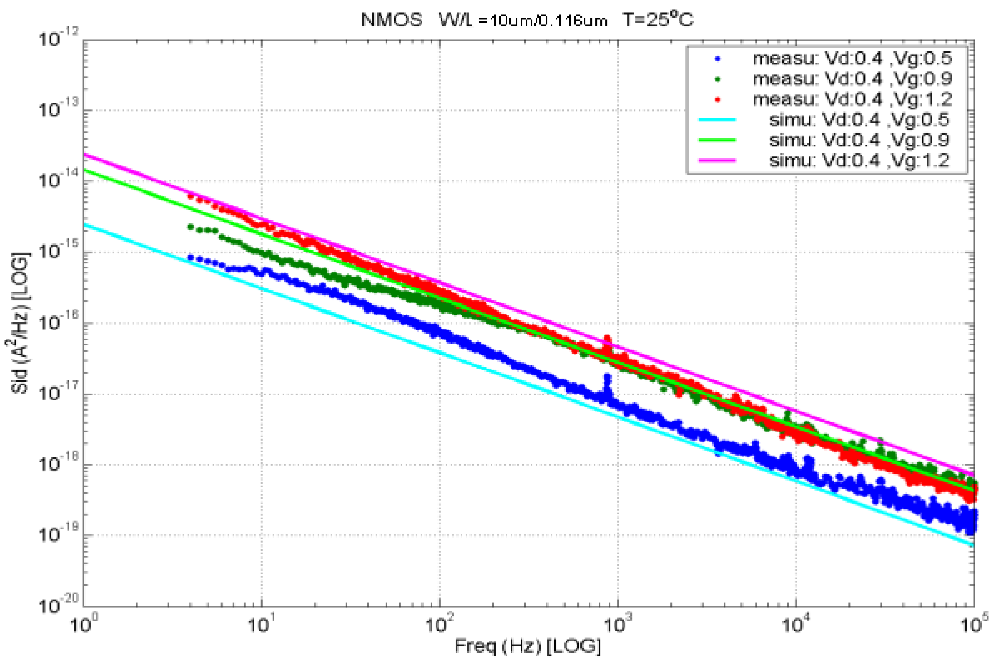


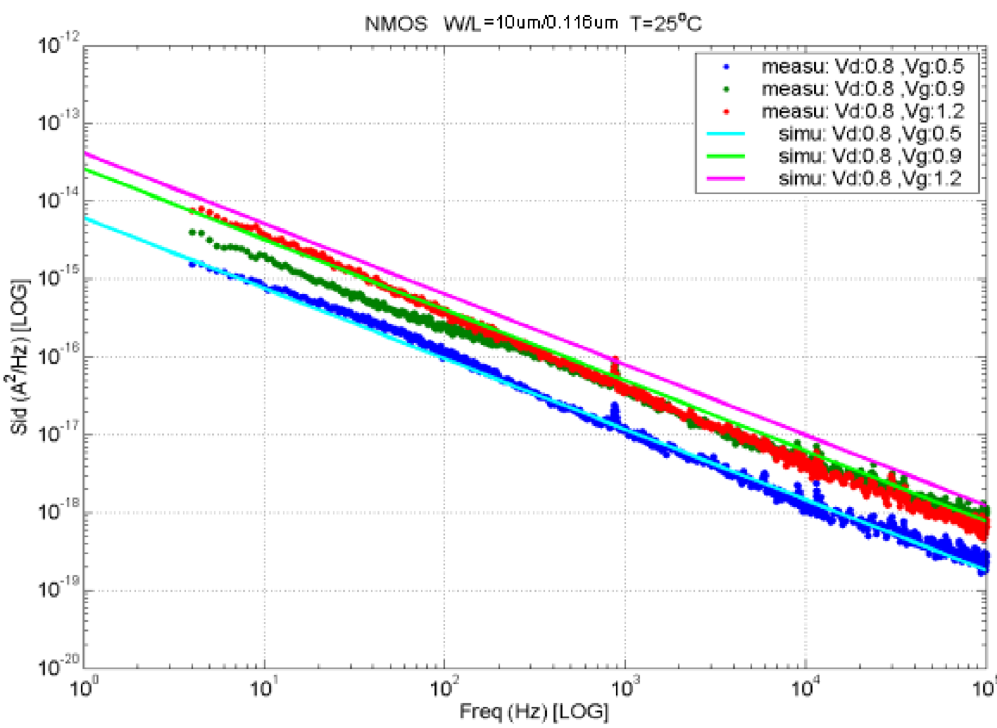
Figure 6-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

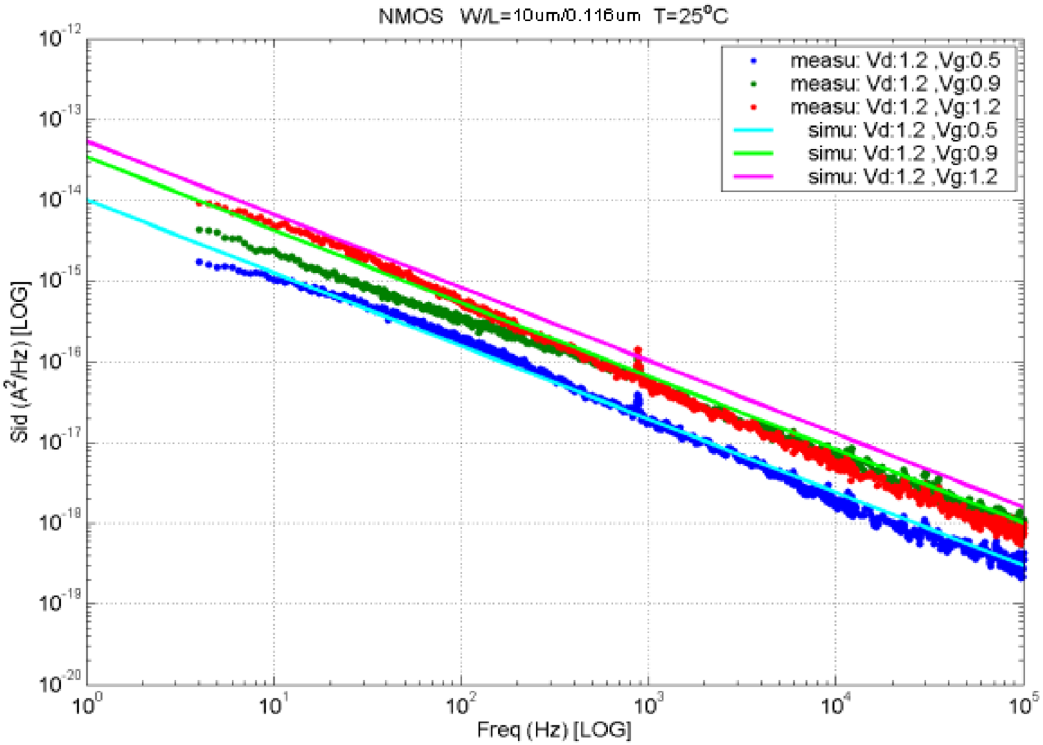
7 Noise Characteristic

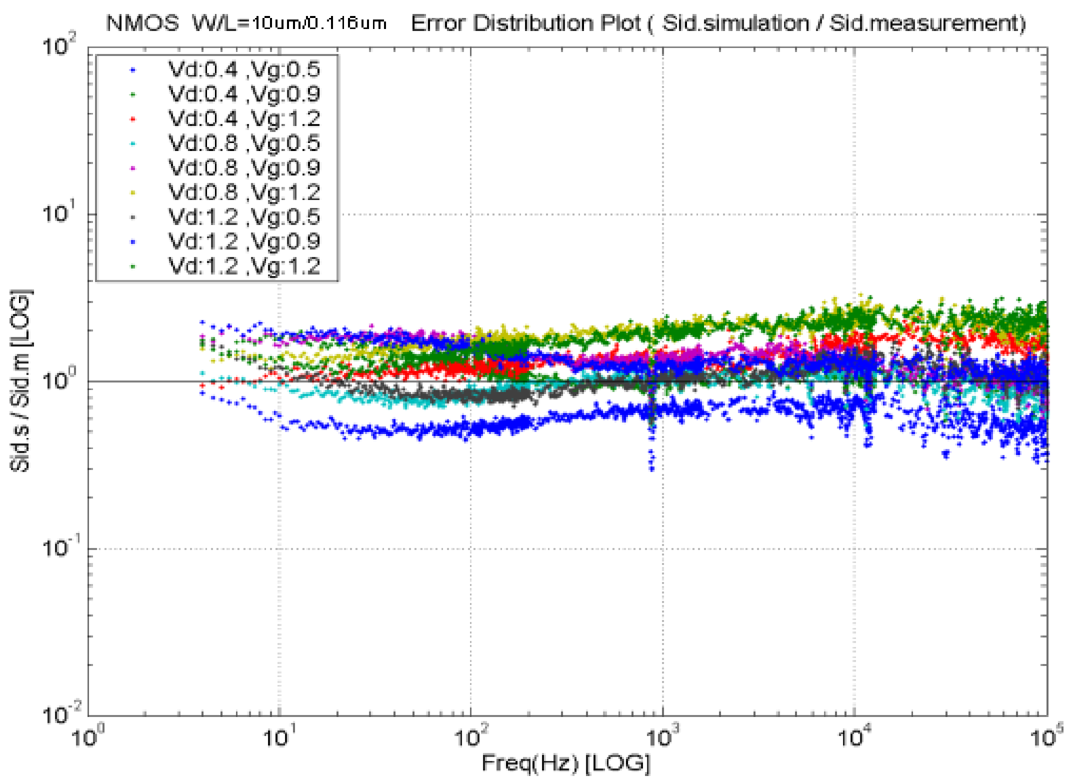
7.1 Verification plots

- ***NMOS, W=10μm L=0.116μm***

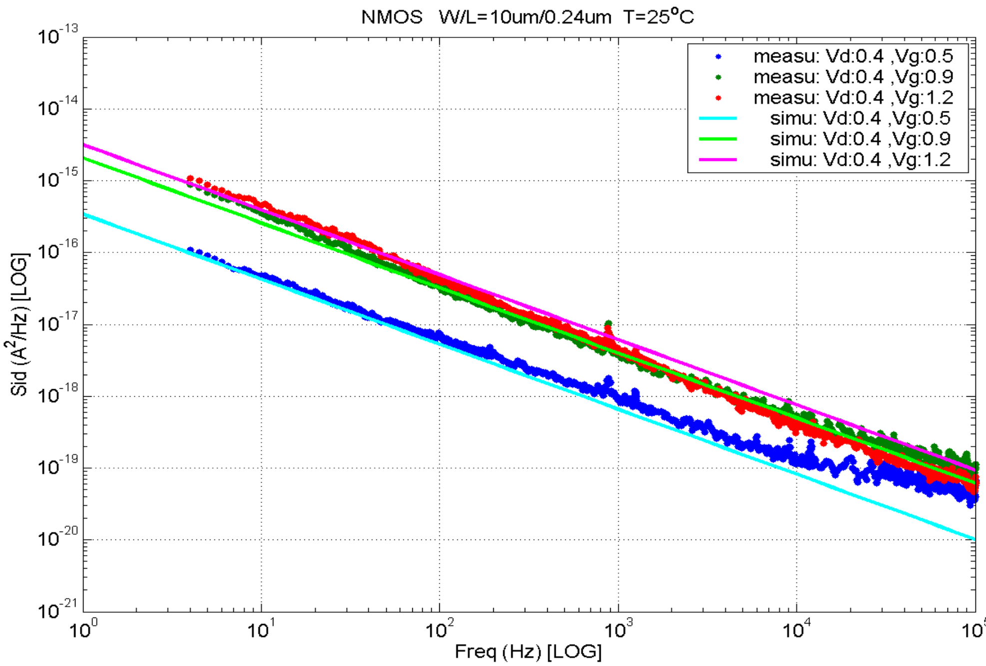


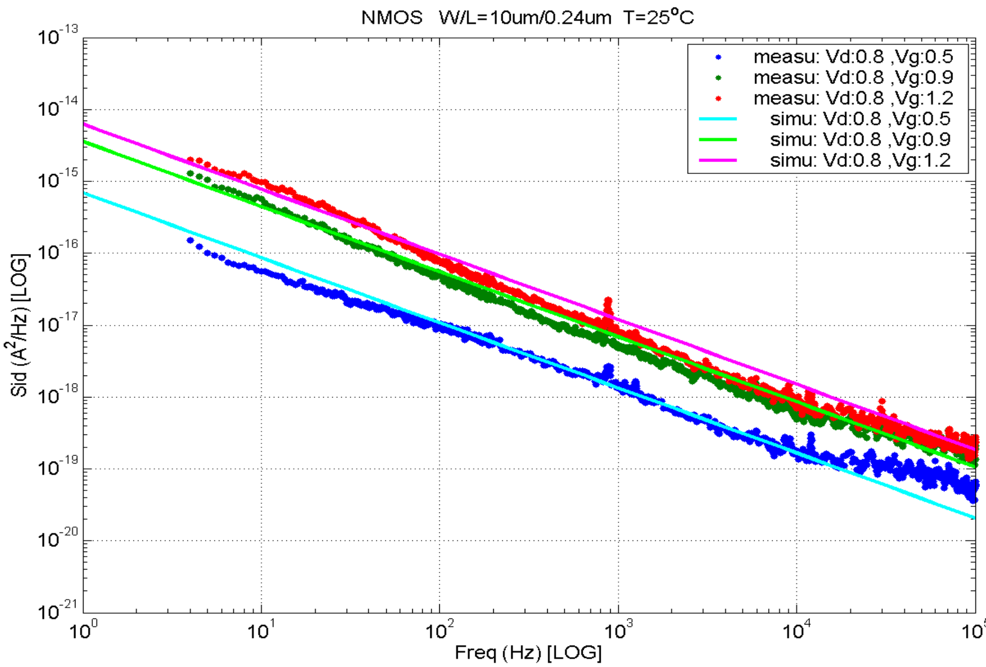


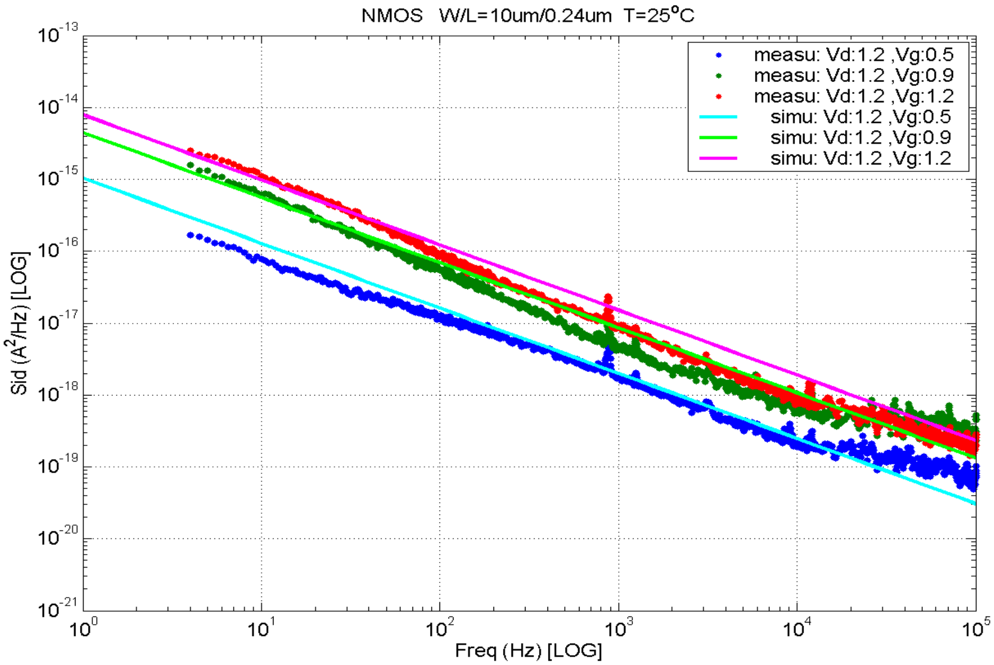

NMOS W/L=10um/0.116um T=25°C
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]
measu: Vd:1.2 ,Vg:0.5
measu: Vd:1.2 ,Vg:0.9
measu: Vd:1.2 ,Vg:1.2
simu: Vd:1.2 ,Vg:0.5
simu: Vd:1.2 ,Vg:0.9
simu: Vd:1.2 ,Vg:1.2

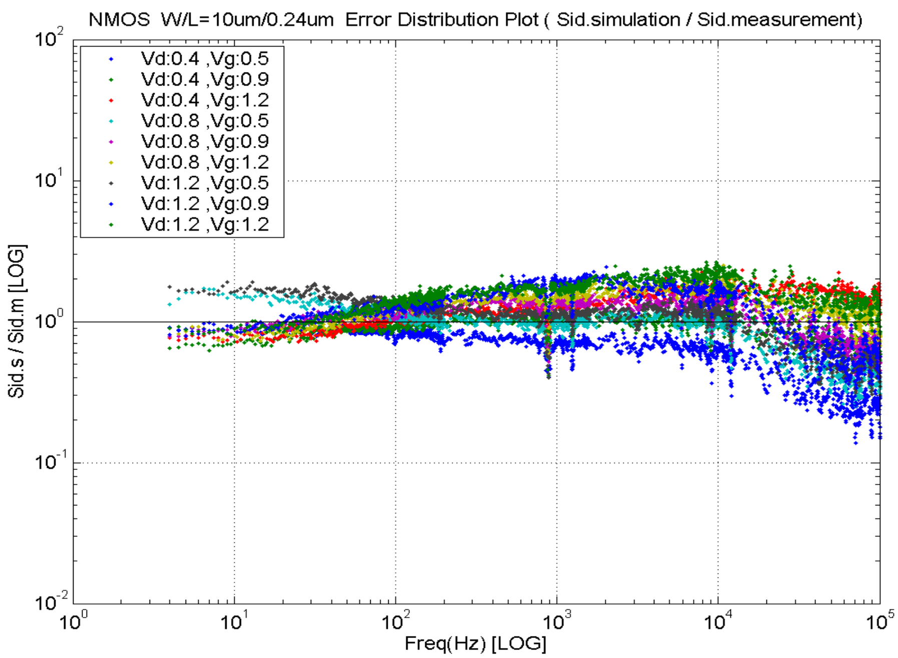

NMOS W/L=10um/0.116um Error Distribution Plot (Sid.simulation / Sid.measurement)
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]
Vd:0.4 ,Vg:0.5
Vd:0.4 ,Vg:0.9
Vd:0.4 ,Vg:1.2
Vd:0.8 ,Vg:0.5
Vd:0.8 ,Vg:0.9
Vd:0.8 ,Vg:1.2
Vd:1.2 ,Vg:0.5
Vd:1.2 ,Vg:0.9
Vd:1.2 ,Vg:1.2

➢ *NMOS, W=10µm L=0.24µm*

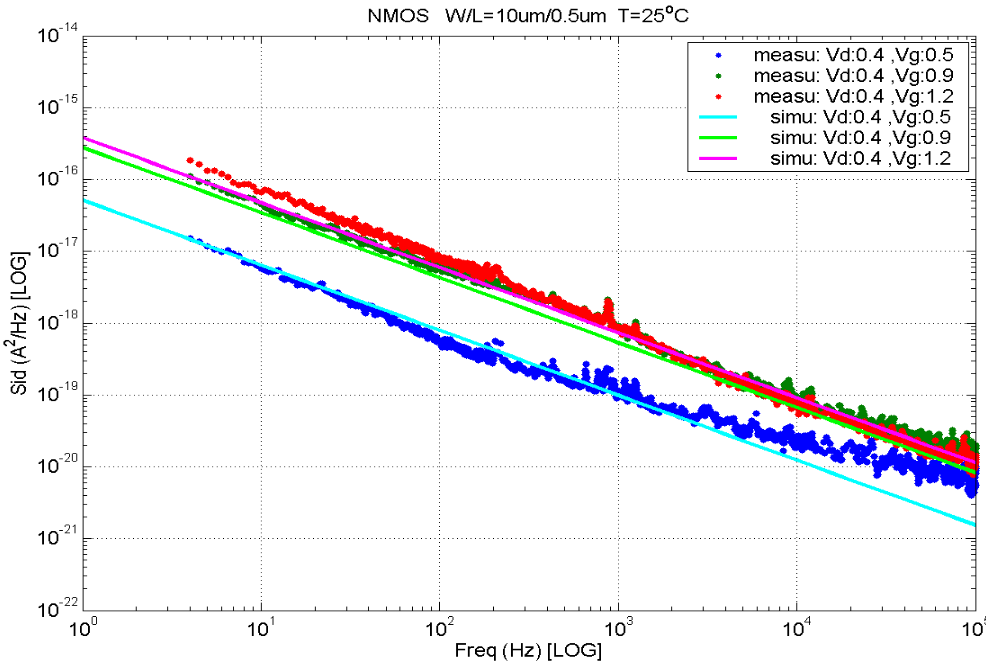


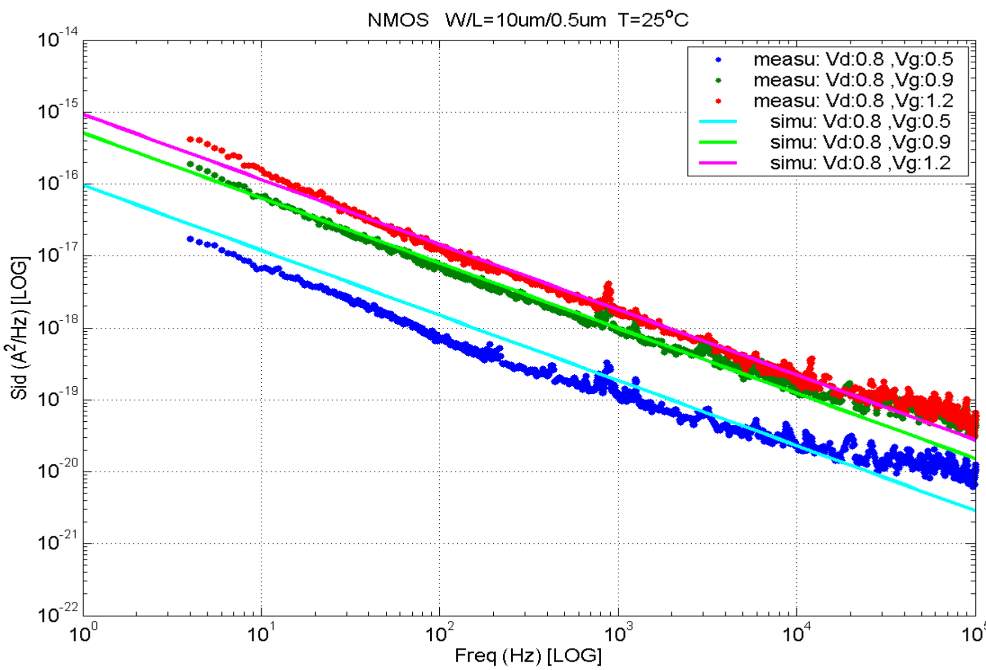


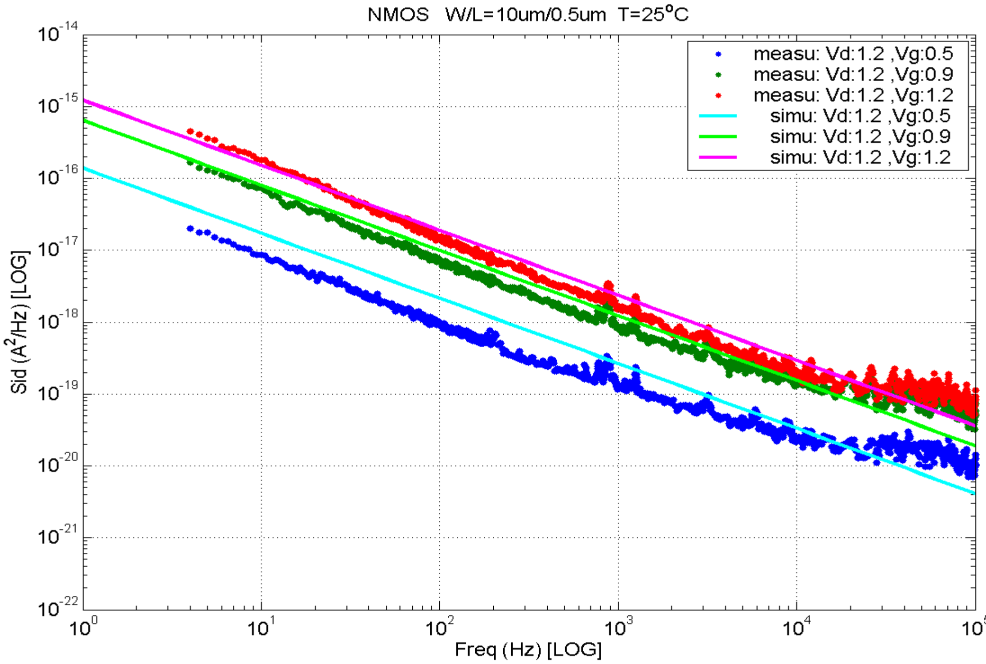

NMOS W/L=10um/0.24um T=25°C
measu: Vd:1.2 ,Vg:0.5
measu: Vd:1.2 ,Vg:0.9
measu: Vd:1.2 ,Vg:1.2
simu: Vd:1.2 ,Vg:0.5
simu: Vd:1.2 ,Vg:0.9
simu: Vd:1.2 ,Vg:1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

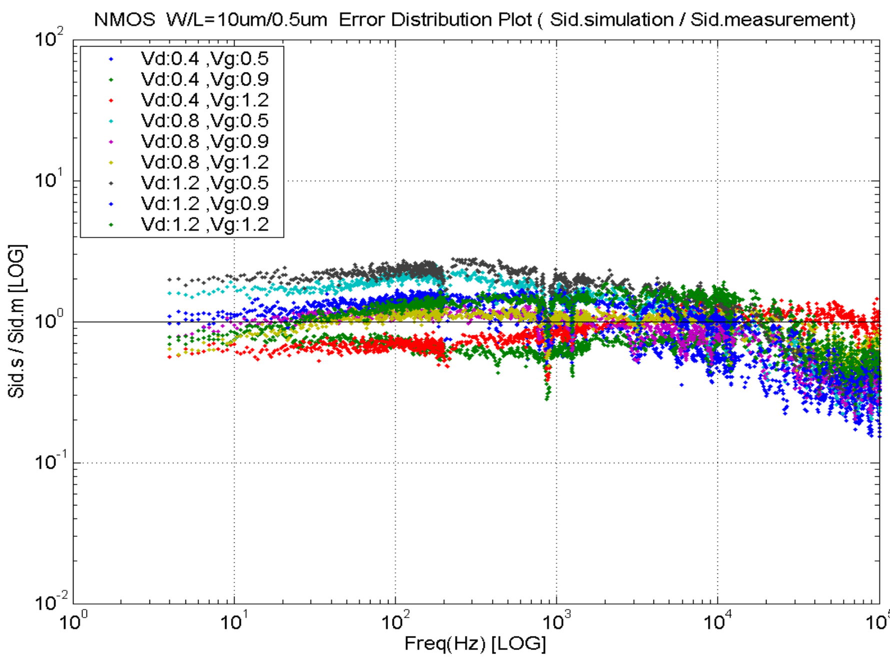

NMOS W/L=10um/0.24um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:0.4 ,Vg:0.5
Vd:0.4 ,Vg:0.9
Vd:0.4 ,Vg:1.2
Vd:0.8 ,Vg:0.5
Vd:0.8 ,Vg:0.9
Vd:0.8 ,Vg:1.2
Vd:1.2 ,Vg:0.5
Vd:1.2 ,Vg:0.9
Vd:1.2 ,Vg:1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ ***NMOS, W=10μm L=0.5μm***

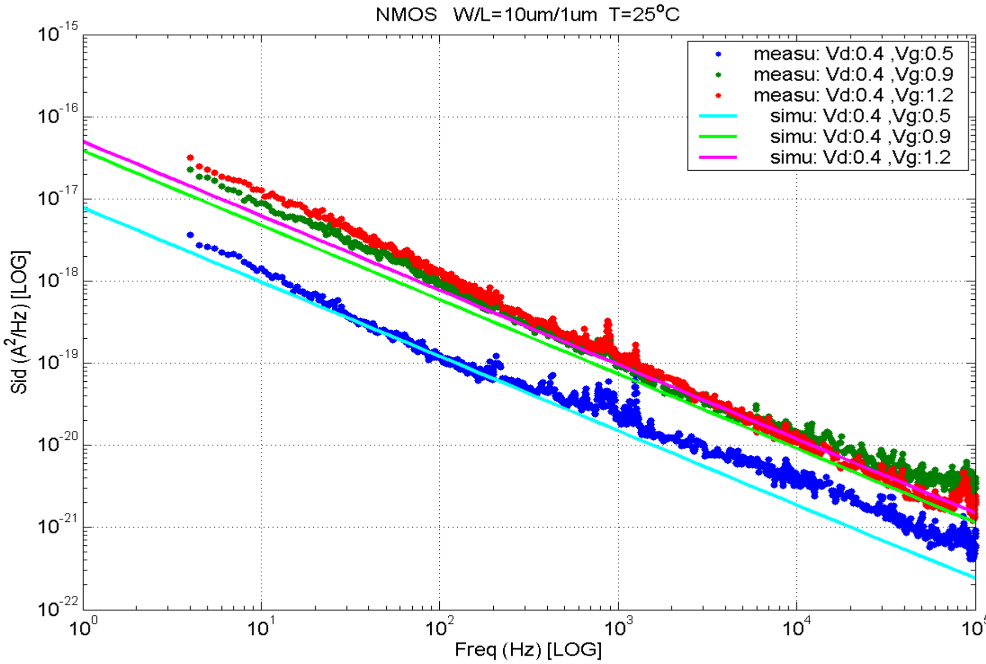


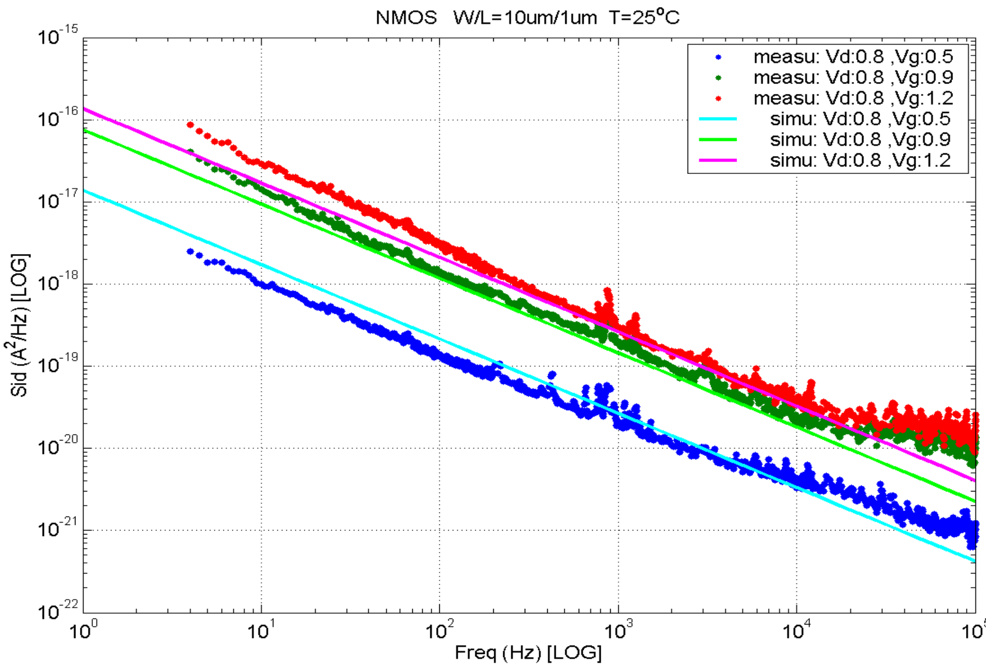


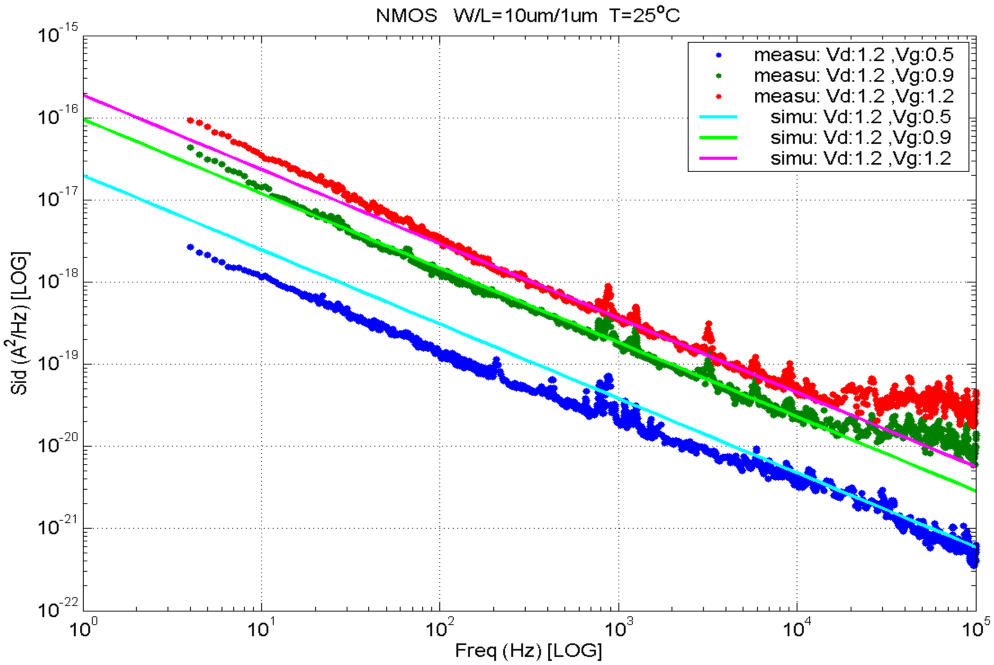

NMOS W/L=10um/0.5um T=25°C
measu: Vd:1.2 ,Vg:0.5
measu: Vd:1.2 ,Vg:0.9
measu: Vd:1.2 ,Vg:1.2
simu: Vd:1.2 ,Vg:0.5
simu: Vd:1.2 ,Vg:0.9
simu: Vd:1.2 ,Vg:1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

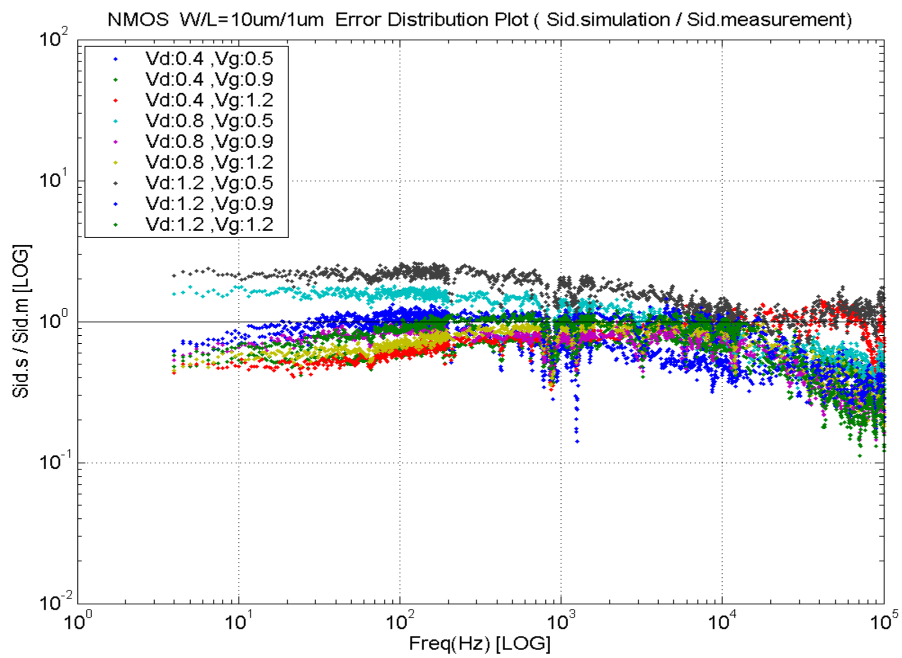

NMOS W/L=10um/0.5um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:0.4 ,Vg:0.5
Vd:0.4 ,Vg:0.9
Vd:0.4 ,Vg:1.2
Vd:0.8 ,Vg:0.5
Vd:0.8 ,Vg:0.9
Vd:0.8 ,Vg:1.2
Vd:1.2 ,Vg:0.5
Vd:1.2 ,Vg:0.9
Vd:1.2 ,Vg:1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ *NMOS, W=10µm L=1µm*

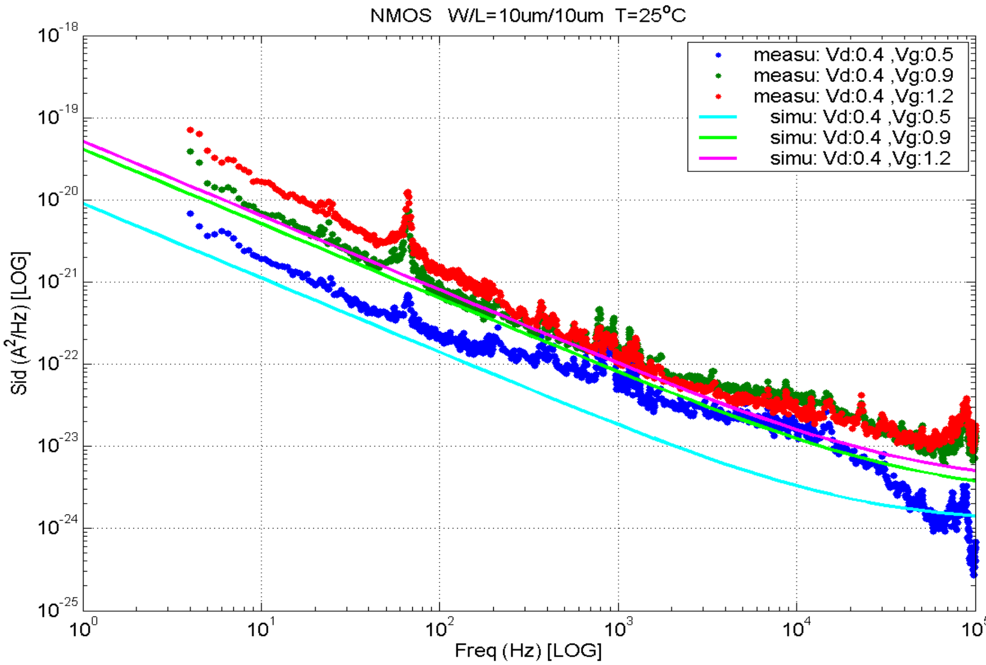


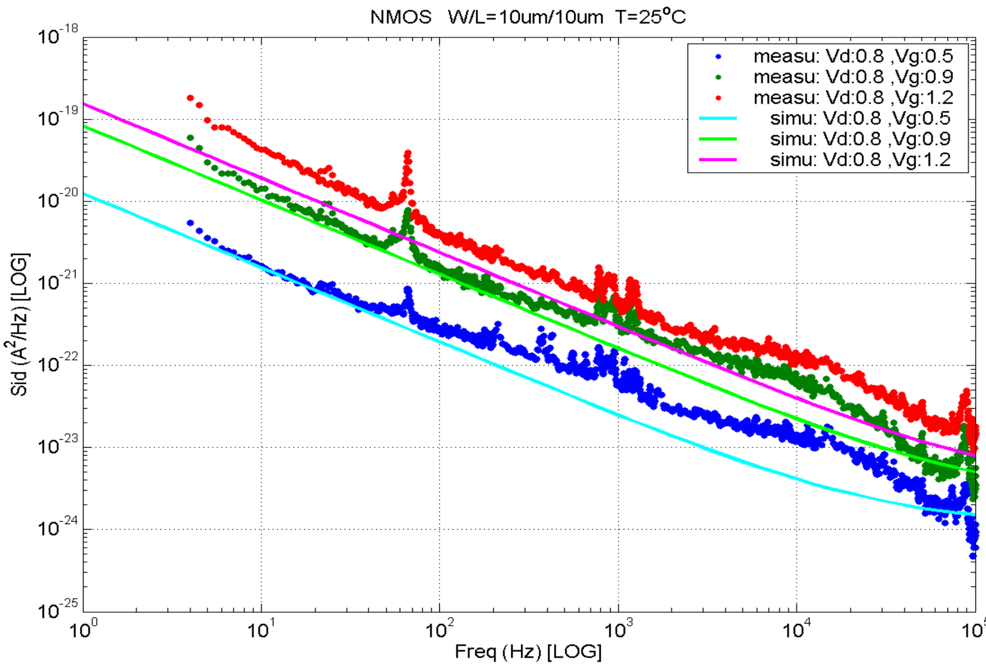


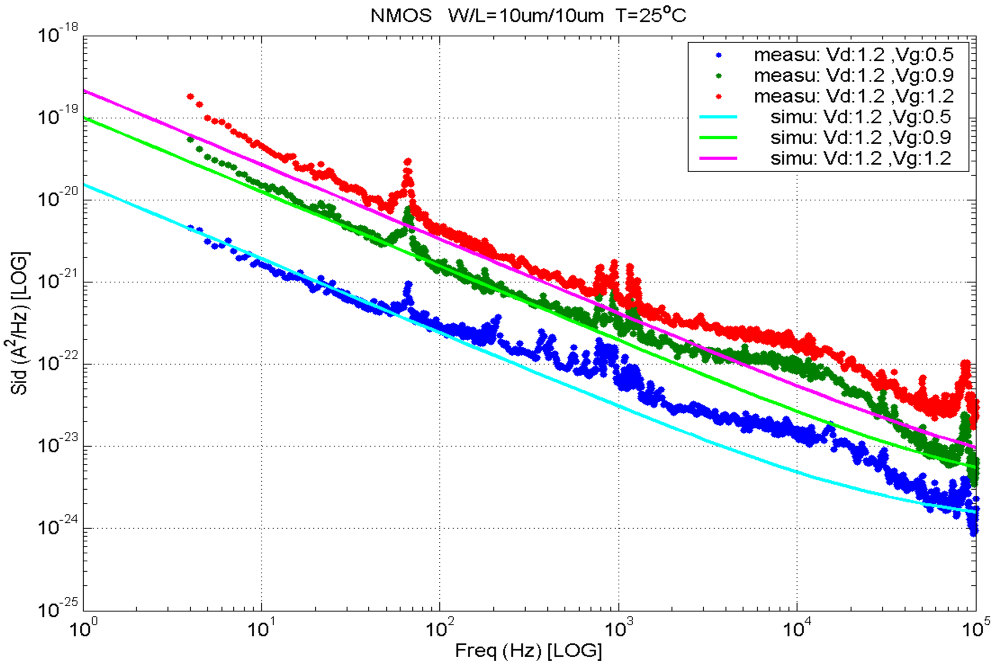

NMOS W/L=10um/1um T=25°C
measu: Vd:1.2 ,Vg:0.5
measu: Vd:1.2 ,Vg:0.9
measu: Vd:1.2 ,Vg:1.2
simu: Vd:1.2 ,Vg:0.5
simu: Vd:1.2 ,Vg:0.9
simu: Vd:1.2 ,Vg:1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

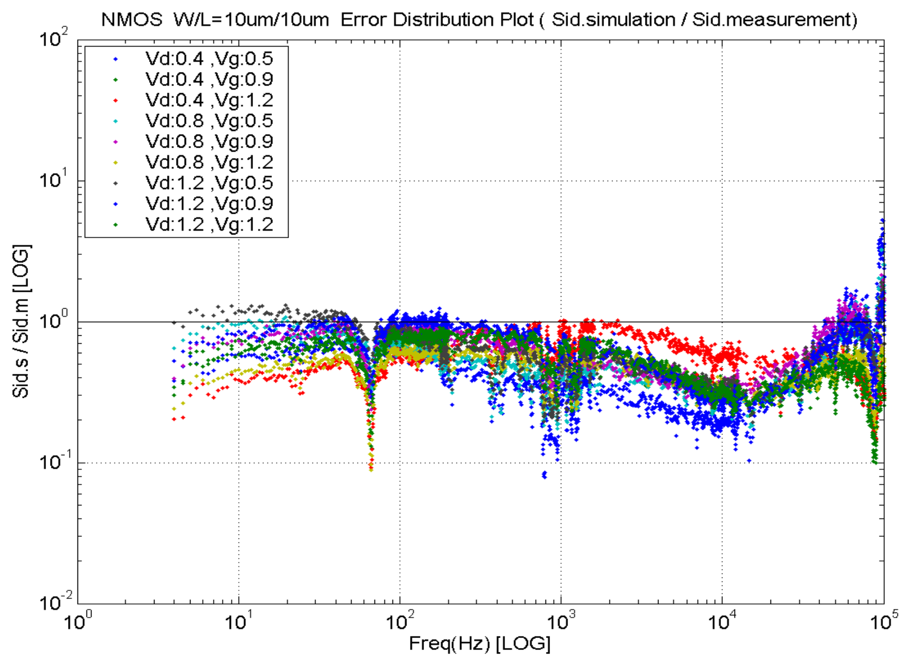

NMOS W/L=10um/1um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:0.4 ,Vg:0.5
Vd:0.4 ,Vg:0.9
Vd:0.4 ,Vg:1.2
Vd:0.8 ,Vg:0.5
Vd:0.8 ,Vg:0.9
Vd:0.8 ,Vg:1.2
Vd:1.2 ,Vg:0.5
Vd:1.2 ,Vg:0.9
Vd:1.2 ,Vg:1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ ***NMOS, W=10μm L=10μm***

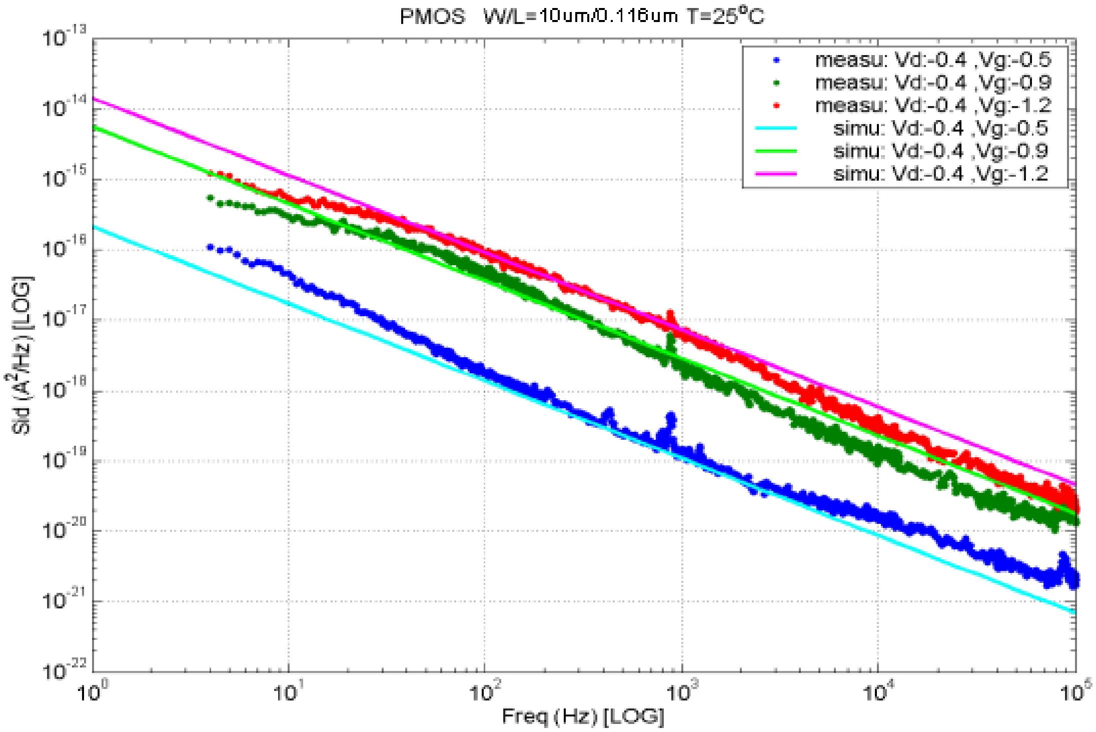


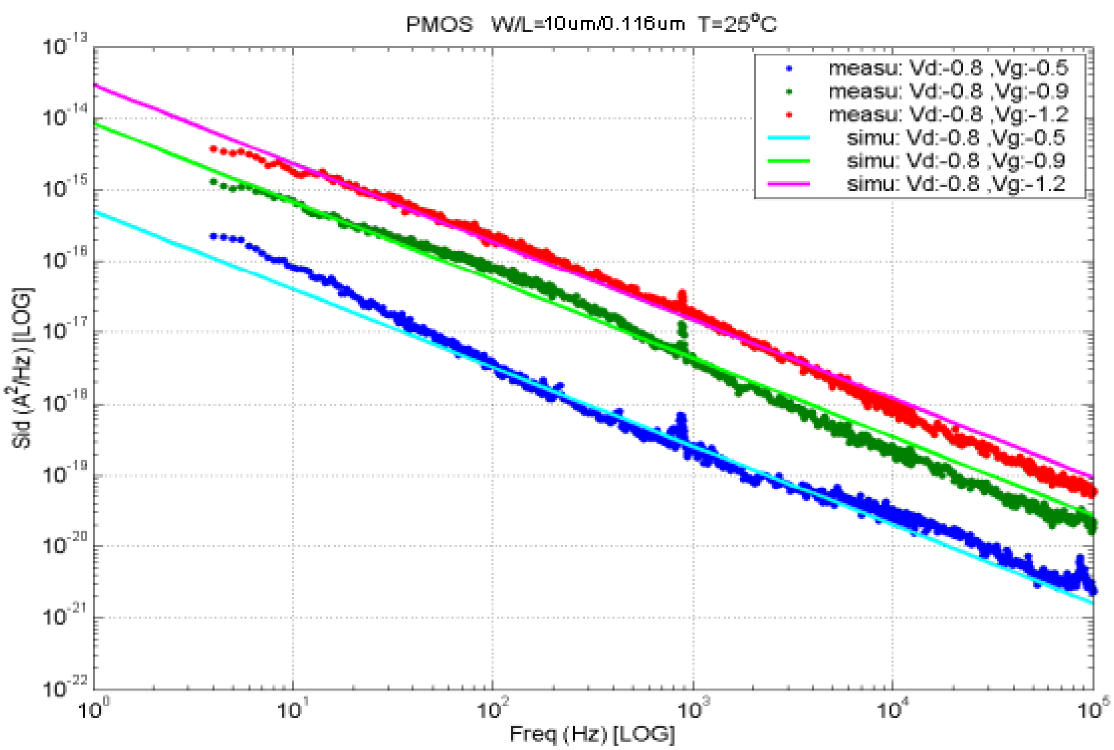


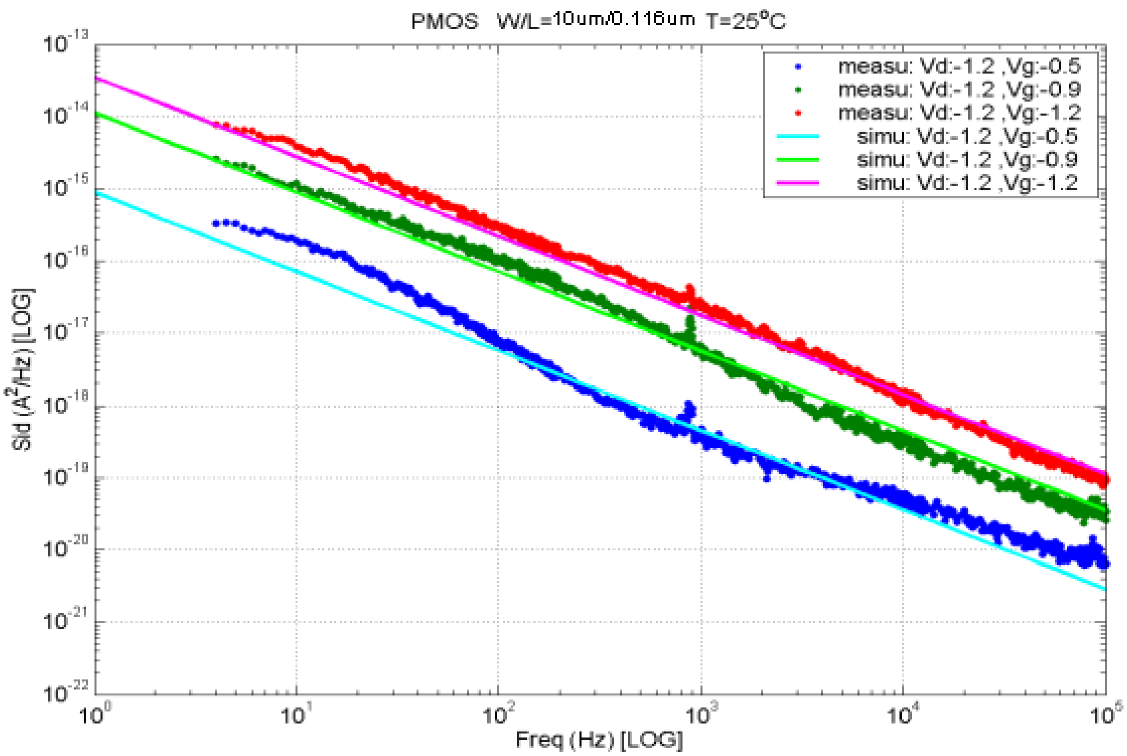

NMOS W/L=10um/10um T=25°C
measu: Vd:1.2 ,Vg:0.5
measu: Vd:1.2 ,Vg:0.9
measu: Vd:1.2 ,Vg:1.2
simu: Vd:1.2 ,Vg:0.5
simu: Vd:1.2 ,Vg:0.9
simu: Vd:1.2 ,Vg:1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

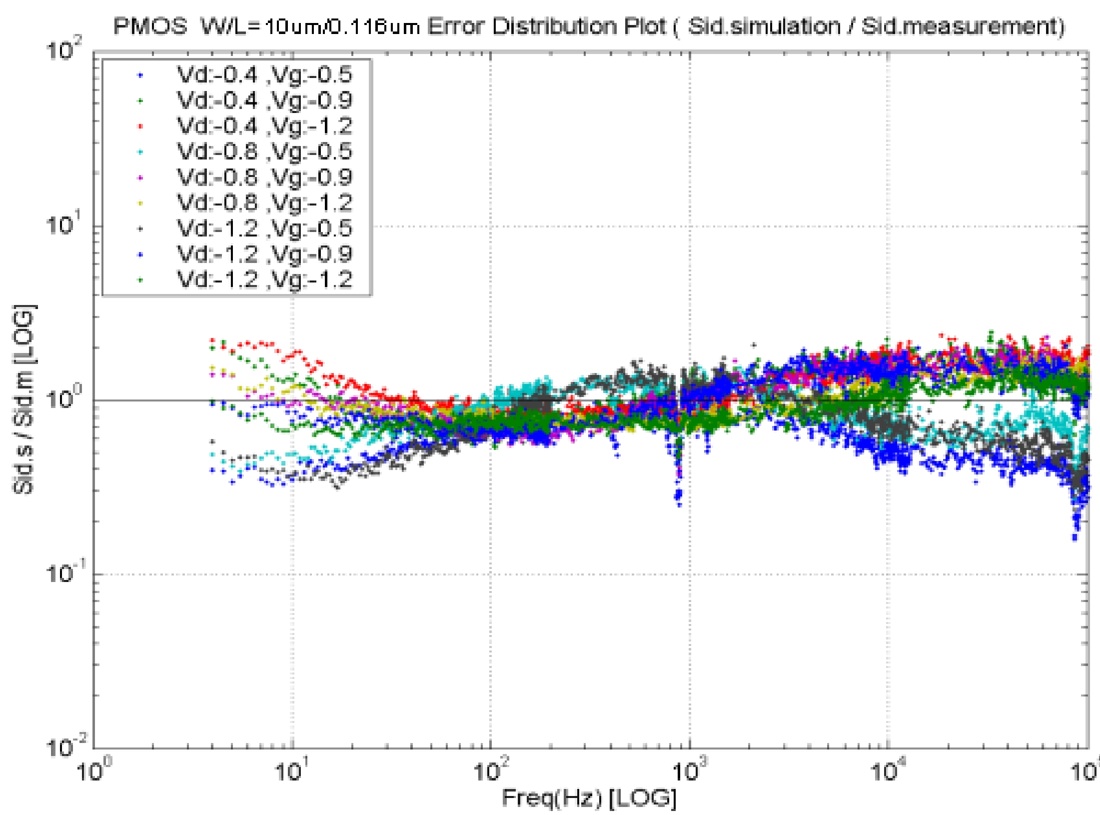

NMOS W/L=10um/10um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:0.4 ,Vg:0.5
Vd:0.4 ,Vg:0.9
Vd:0.4 ,Vg:1.2
Vd:0.8 ,Vg:0.5
Vd:0.8 ,Vg:0.9
Vd:0.8 ,Vg:1.2
Vd:1.2 ,Vg:0.5
Vd:1.2 ,Vg:0.9
Vd:1.2 ,Vg:1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ ***PMOS, W=10μm L=0.116μm***

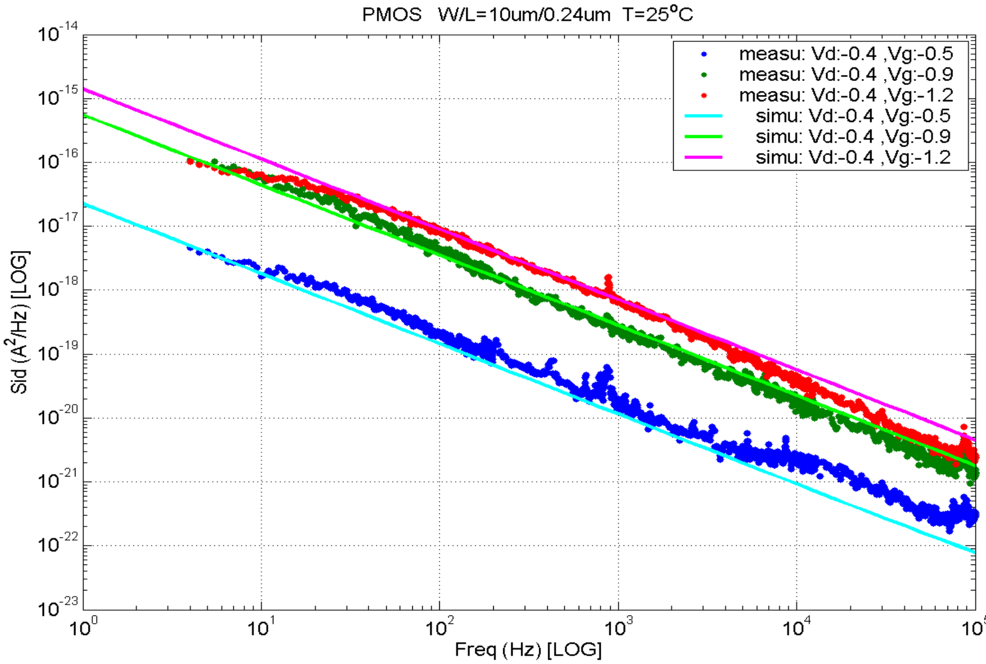


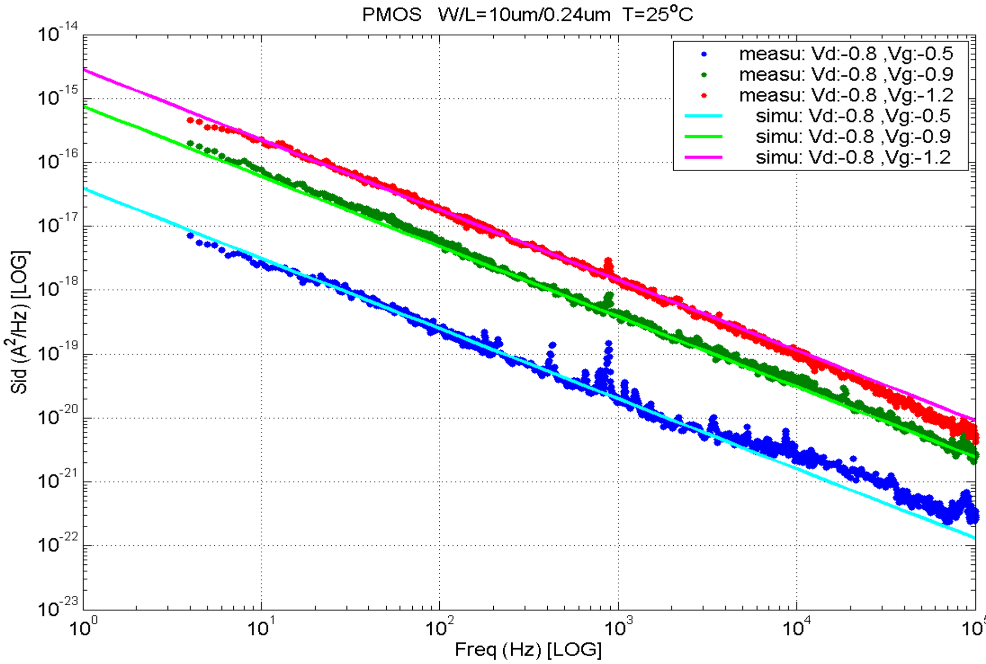


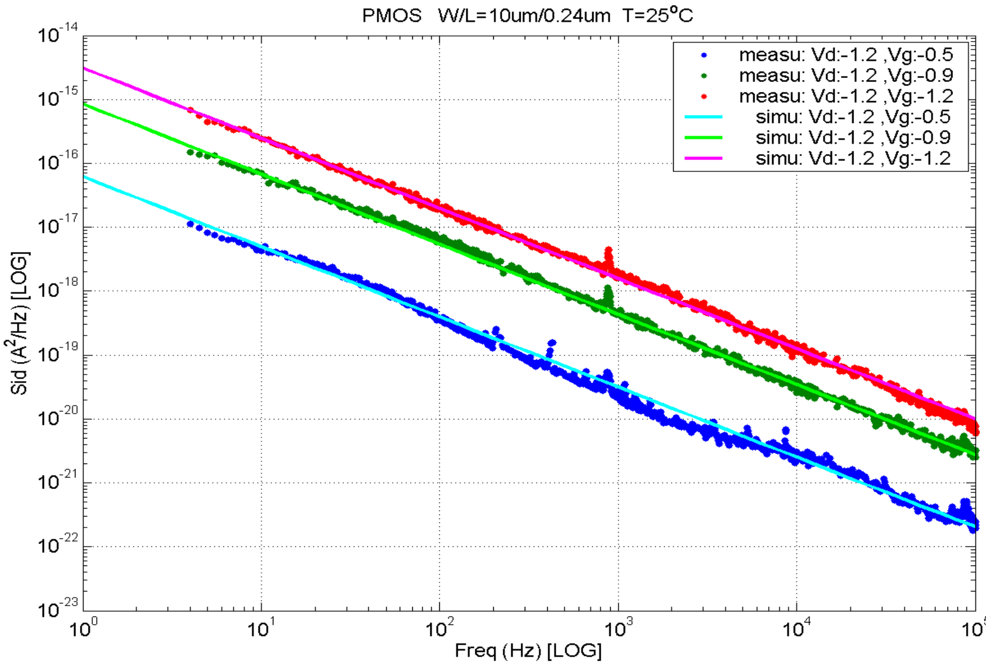

PMOS W/L=10um/0.116um T=25°C
measu: Vd:-1.2 ,Vg:-0.5
measu: Vd:-1.2 ,Vg:-0.9
measu: Vd:-1.2 ,Vg:-1.2
simu: Vd:-1.2 ,Vg:-0.5
simu: Vd:-1.2 ,Vg:-0.9
simu: Vd:-1.2 ,Vg:-1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

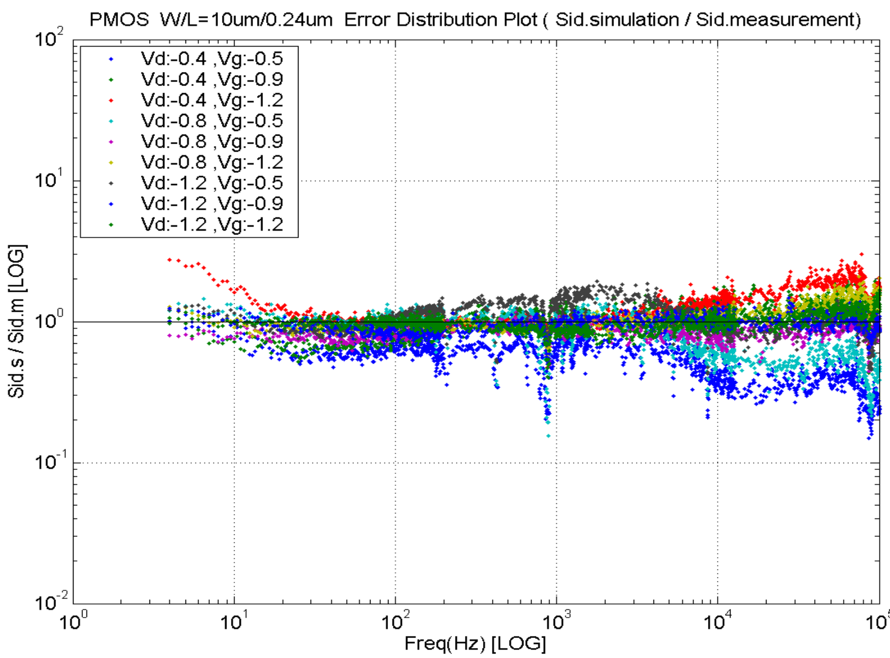

PMOS W/L=10um/0.116um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:-0.4 ,Vg:-0.5
Vd:-0.4 ,Vg:-0.9
Vd:-0.4 ,Vg:-1.2
Vd:-0.8 ,Vg:-0.5
Vd:-0.8 ,Vg:-0.9
Vd:-0.8 ,Vg:-1.2
Vd:-1.2 ,Vg:-0.5
Vd:-1.2 ,Vg:-0.9
Vd:-1.2 ,Vg:-1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ *PMOS, W=10µm L=0.24µm*

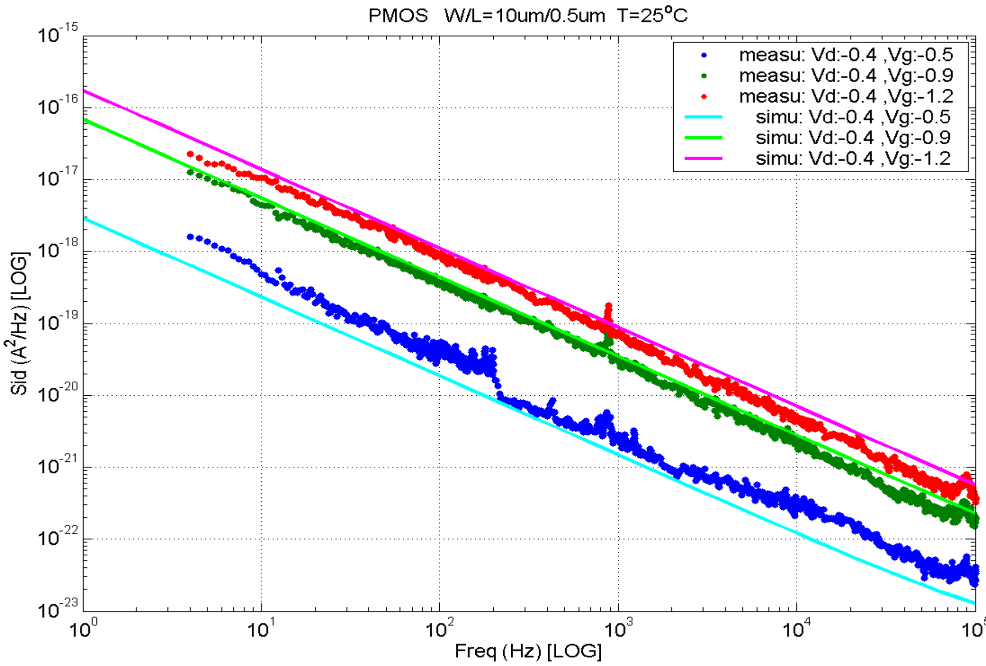


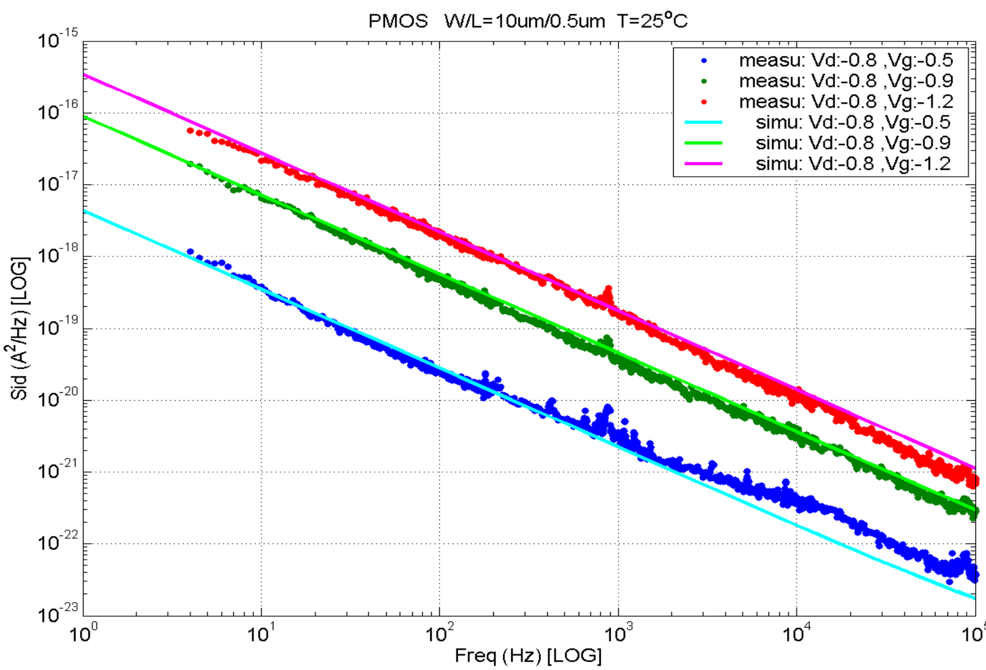


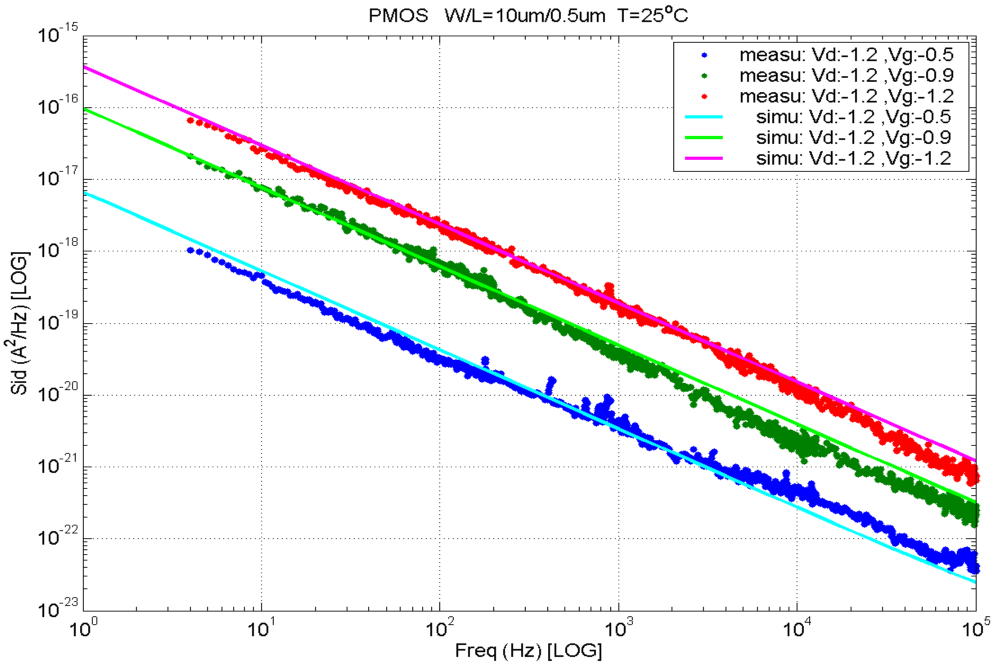

PMOS W/L=10um/0.24um T=25°C
measu: Vd:-1.2 ,Vg:-0.5
measu: Vd:-1.2 ,Vg:-0.9
measu: Vd:-1.2 ,Vg:-1.2
simu: Vd:-1.2 ,Vg:-0.5
simu: Vd:-1.2 ,Vg:-0.9
simu: Vd:-1.2 ,Vg:-1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

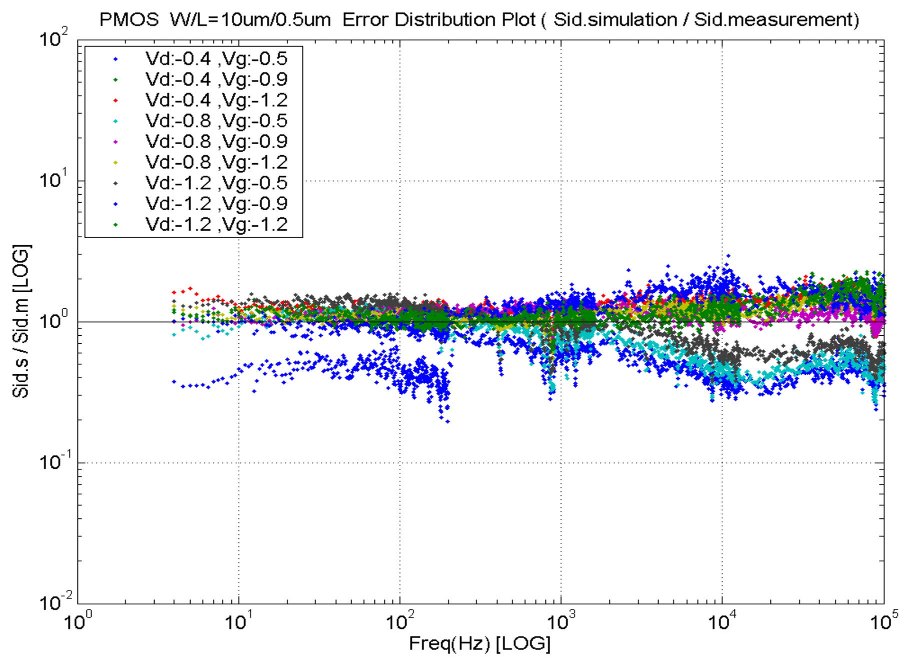

PMOS W/L=10um/0.24um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:-0.4 ,Vg:-0.5
Vd:-0.4 ,Vg:-0.9
Vd:-0.4 ,Vg:-1.2
Vd:-0.8 ,Vg:-0.5
Vd:-0.8 ,Vg:-0.9
Vd:-0.8 ,Vg:-1.2
Vd:-1.2 ,Vg:-0.5
Vd:-1.2 ,Vg:-0.9
Vd:-1.2 ,Vg:-1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ *PMOS, W=10μm L=0.5μm*

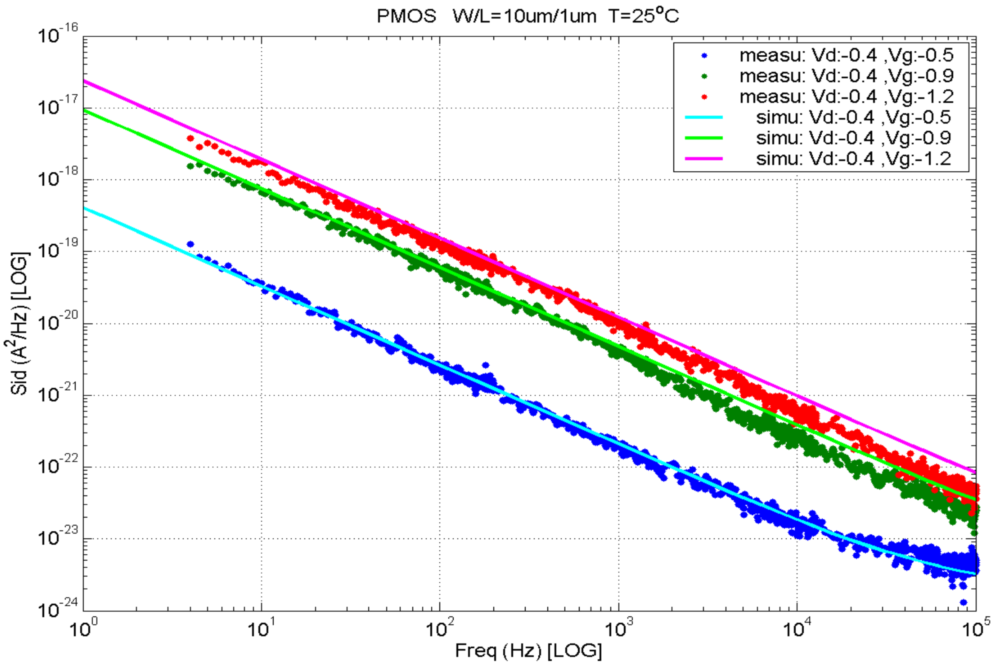


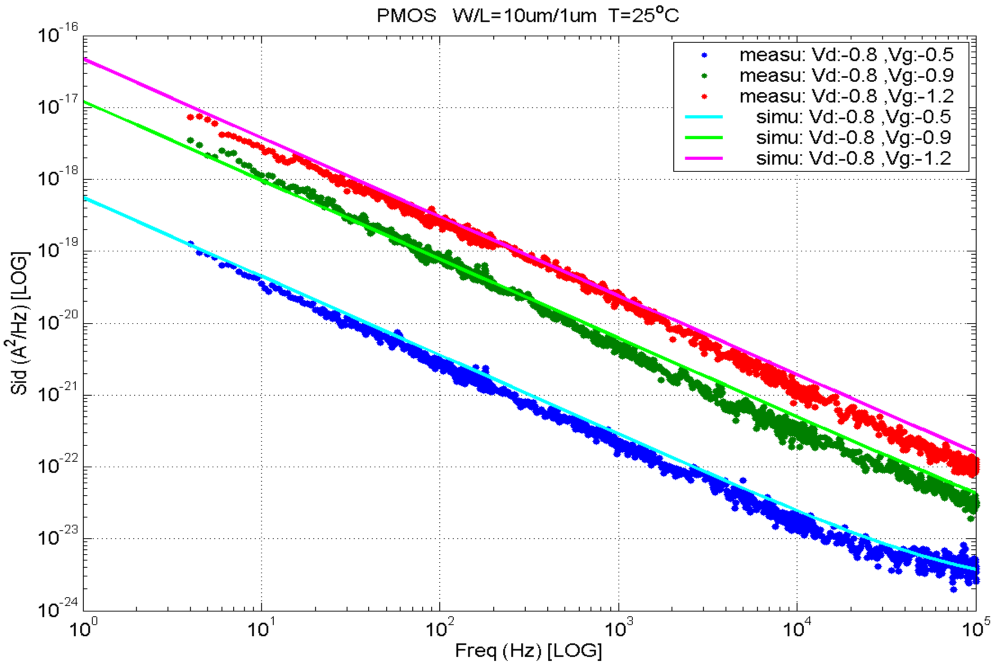


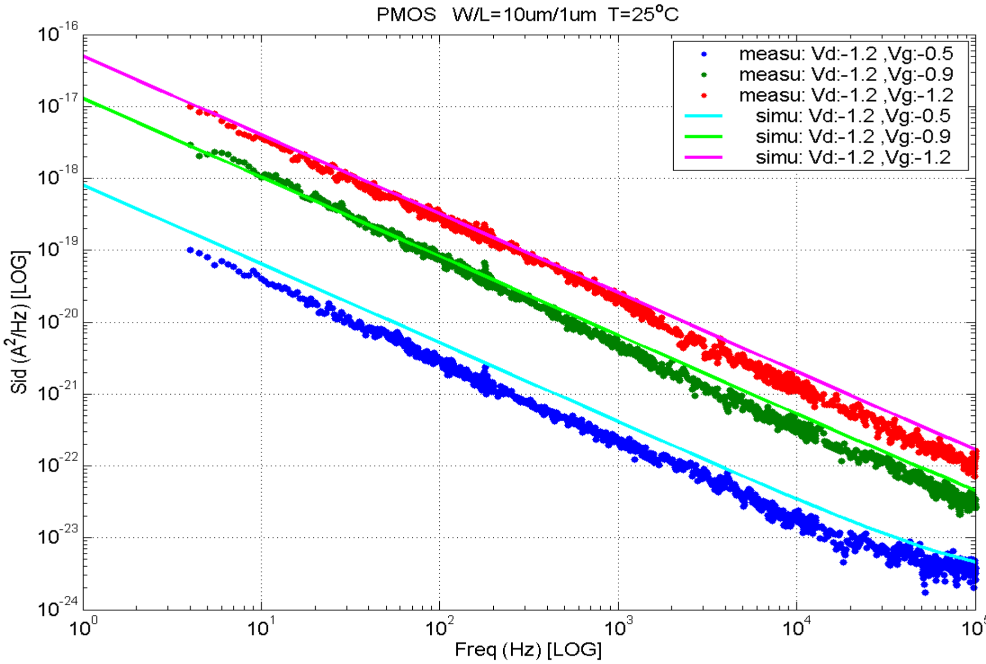

PMOS W/L=10um/0.5um T=25°C
measu: Vd:-1.2 ,Vg:-0.5
measu: Vd:-1.2 ,Vg:-0.9
measu: Vd:-1.2 ,Vg:-1.2
simu: Vd:-1.2 ,Vg:-0.5
simu: Vd:-1.2 ,Vg:-0.9
simu: Vd:-1.2 ,Vg:-1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

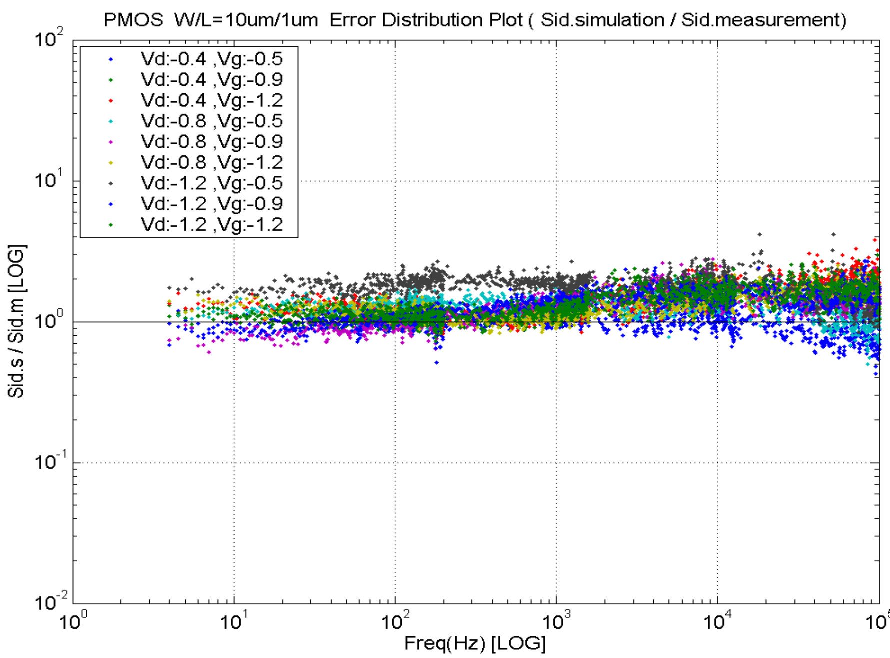

PMOS W/L=10um/0.5um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:-0.4 ,Vg:-0.5
Vd:-0.4 ,Vg:-0.9
Vd:-0.4 ,Vg:-1.2
Vd:-0.8 ,Vg:-0.5
Vd:-0.8 ,Vg:-0.9
Vd:-0.8 ,Vg:-1.2
Vd:-1.2 ,Vg:-0.5
Vd:-1.2 ,Vg:-0.9
Vd:-1.2 ,Vg:-1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ *PMOS, W=10μm L=1μm*

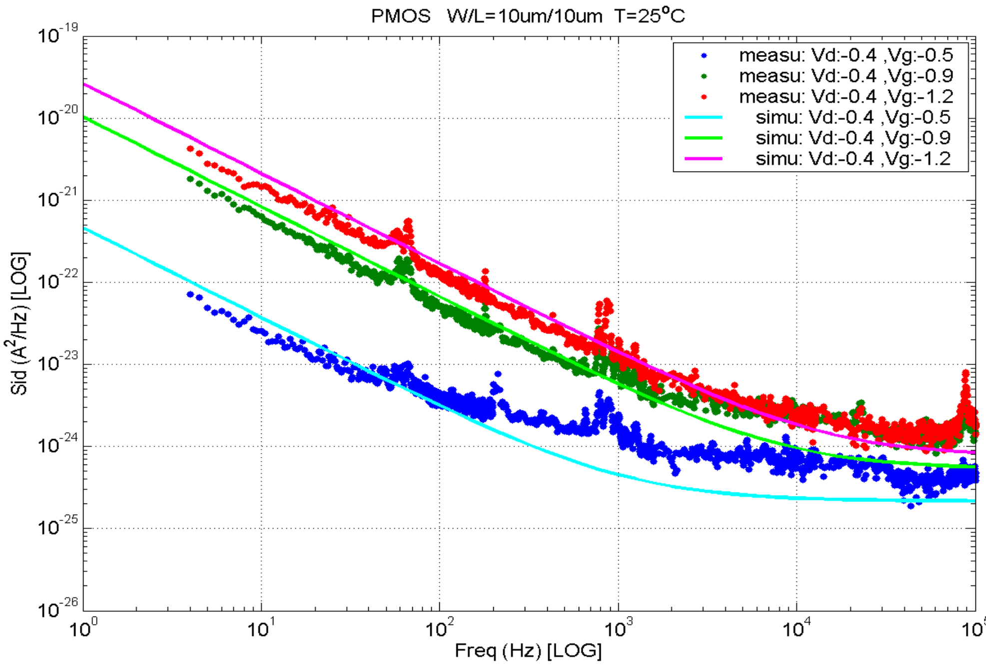


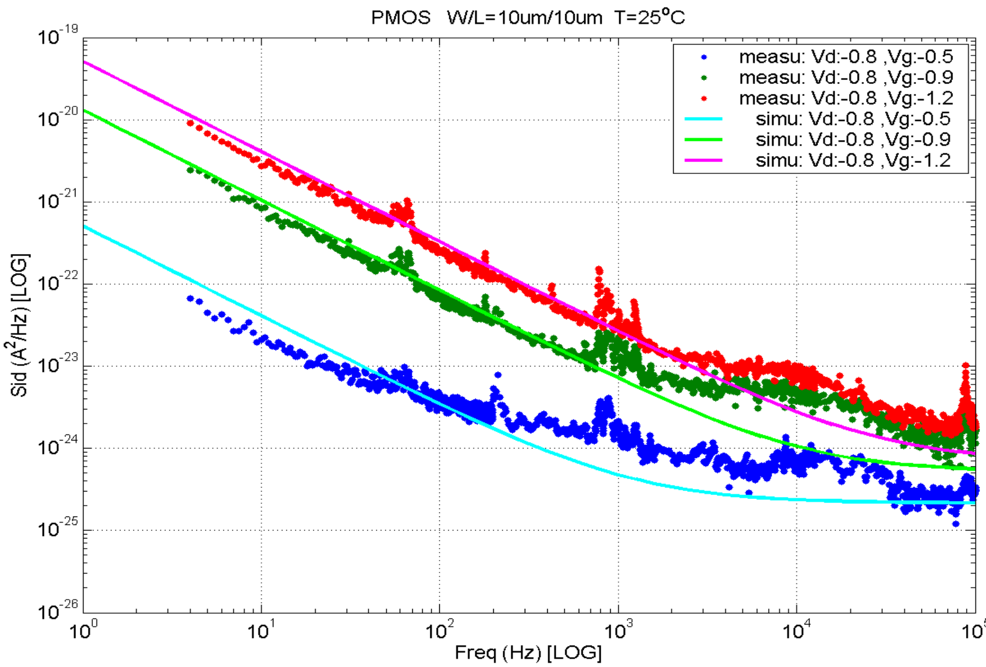


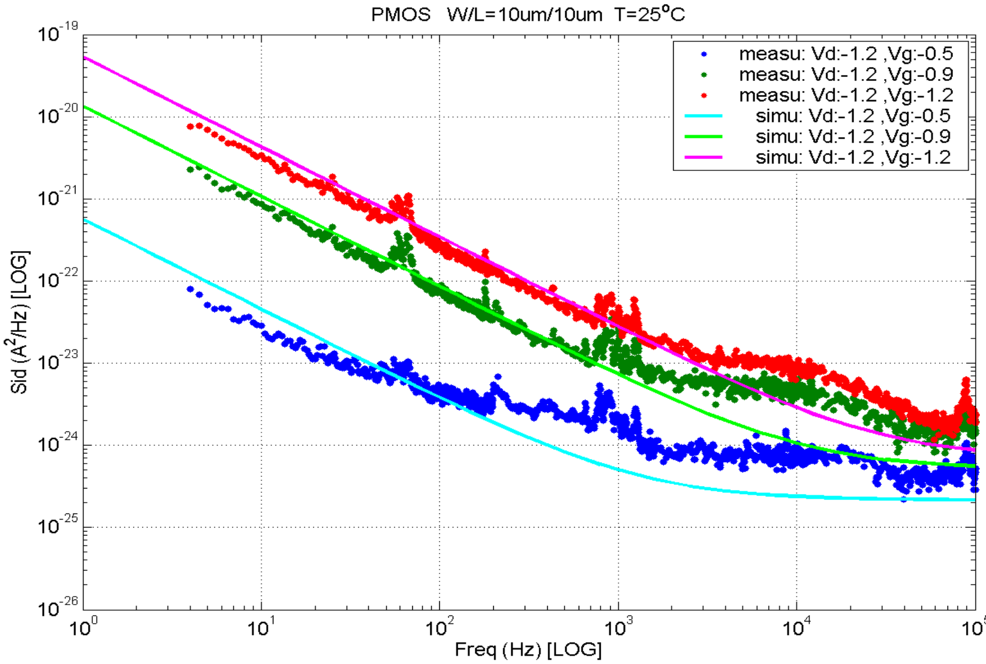

PMOS W/L=10um/1um T=25°C
measu: Vd:-1.2 ,Vg:-0.5
measu: Vd:-1.2 ,Vg:-0.9
measu: Vd:-1.2 ,Vg:-1.2
simu: Vd:-1.2 ,Vg:-0.5
simu: Vd:-1.2 ,Vg:-0.9
simu: Vd:-1.2 ,Vg:-1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

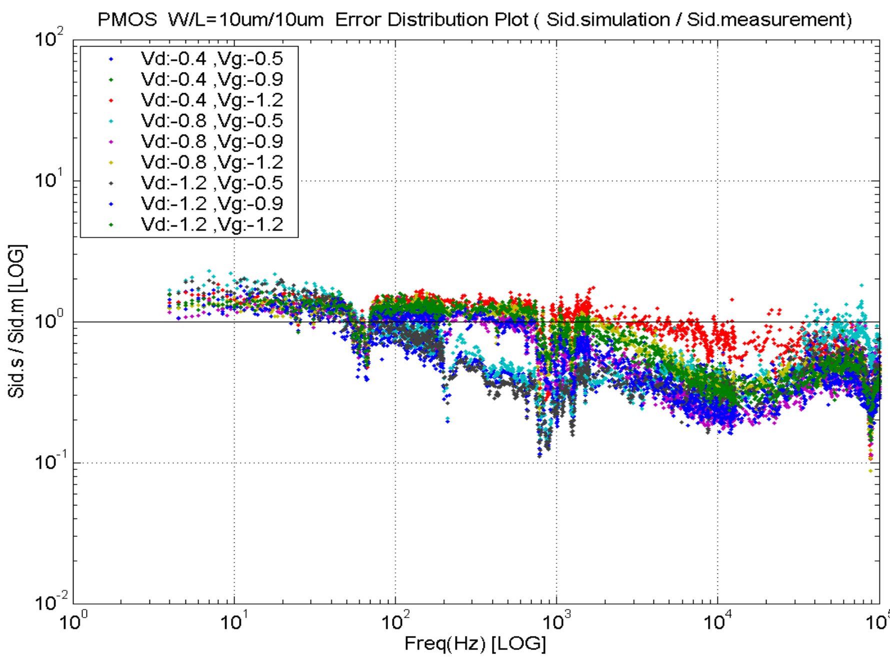

PMOS W/L=10um/1um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:-0.4 ,Vg:-0.5
Vd:-0.4 ,Vg:-0.9
Vd:-0.4 ,Vg:-1.2
Vd:-0.8 ,Vg:-0.5
Vd:-0.8 ,Vg:-0.9
Vd:-0.8 ,Vg:-1.2
Vd:-1.2 ,Vg:-0.5
Vd:-1.2 ,Vg:-0.9
Vd:-1.2 ,Vg:-1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

➢ *PMOS, W=10μm L=10μm*



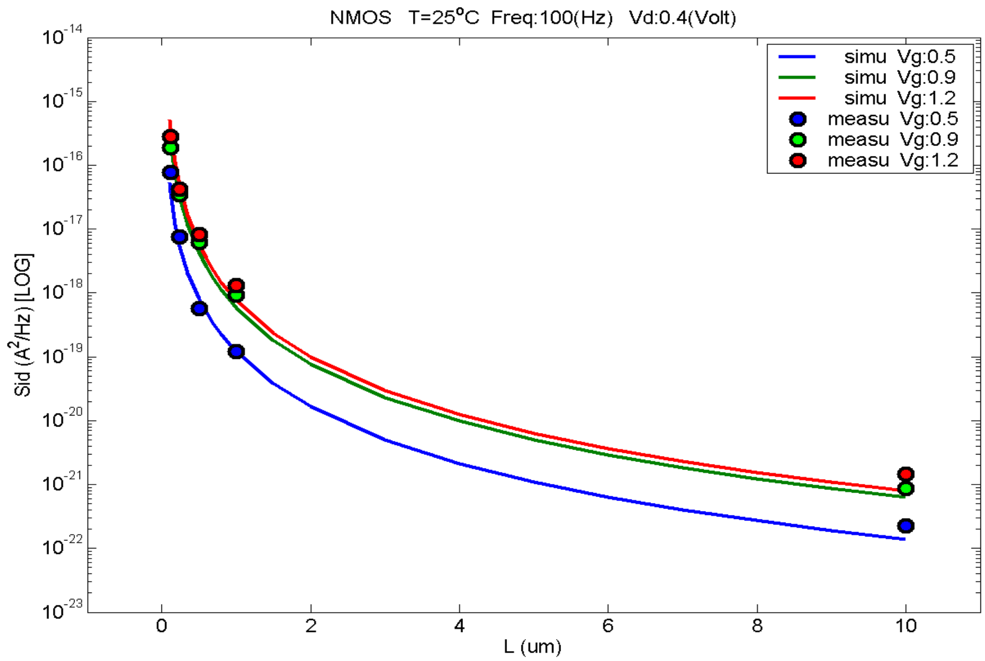


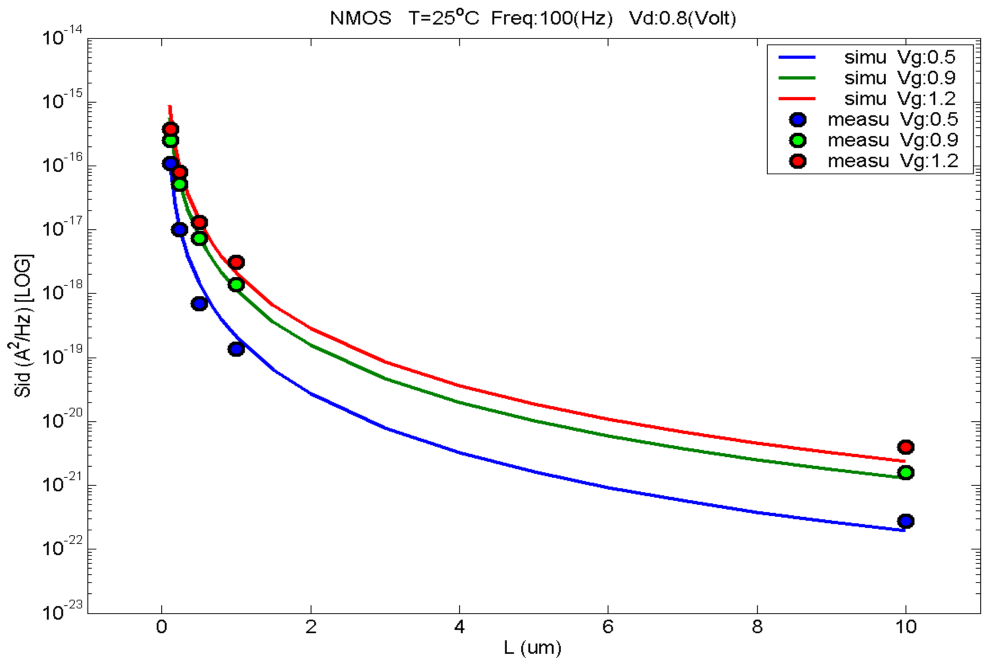

PMOS W/L=10um/10um T=25°C
measu: Vd:-1.2 ,Vg:-0.5
measu: Vd:-1.2 ,Vg:-0.9
measu: Vd:-1.2 ,Vg:-1.2
simu: Vd:-1.2 ,Vg:-0.5
simu: Vd:-1.2 ,Vg:-0.9
simu: Vd:-1.2 ,Vg:-1.2
Sid (A²/Hz) [LOG]
Freq (Hz) [LOG]

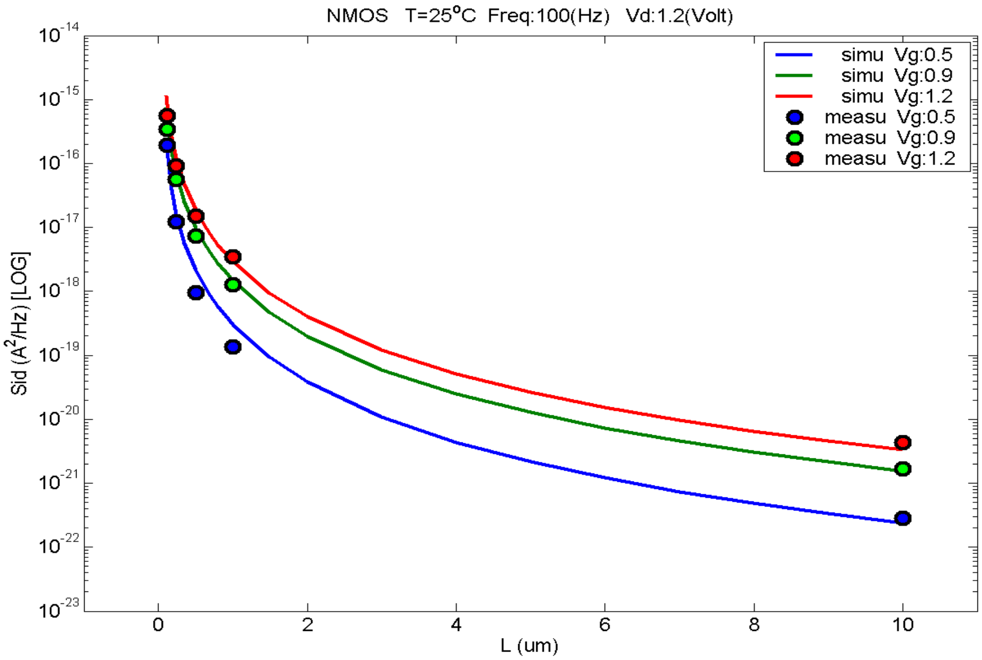

PMOS W/L=10um/10um Error Distribution Plot (Sid.simulation / Sid.measurement)
Vd:-0.4 ,Vg:-0.5
Vd:-0.4 ,Vg:-0.9
Vd:-0.4 ,Vg:-1.2
Vd:-0.8 ,Vg:-0.5
Vd:-0.8 ,Vg:-0.9
Vd:-0.8 ,Vg:-1.2
Vd:-1.2 ,Vg:-0.5
Vd:-1.2 ,Vg:-0.9
Vd:-1.2 ,Vg:-1.2
Sid.s / Sid.m [LOG]
Freq(Hz) [LOG]

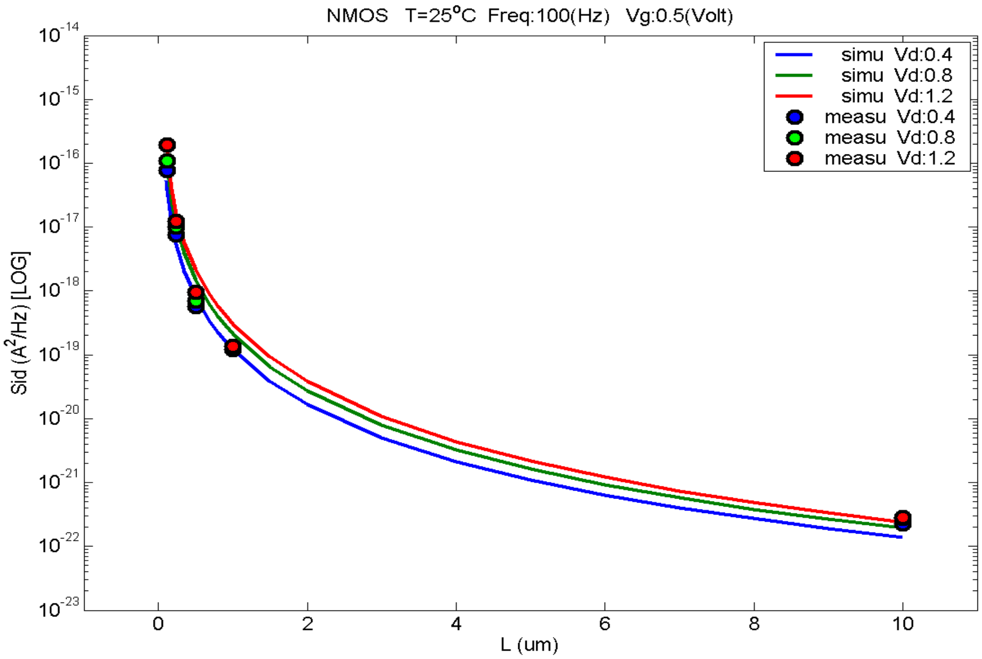
7.2 Trend plots at frequency=100Hz

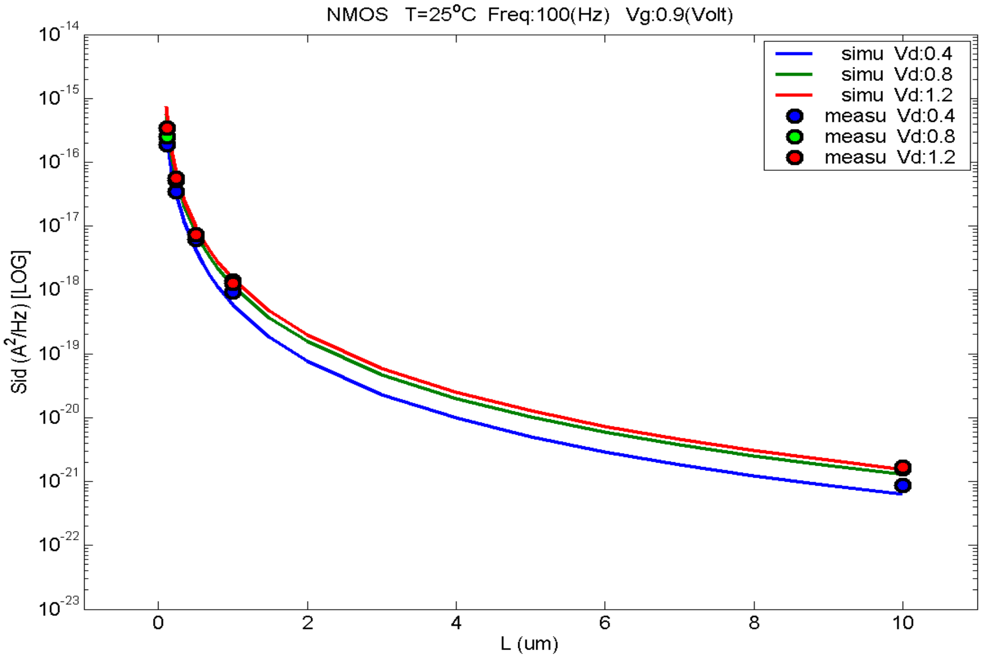
A. NMOS

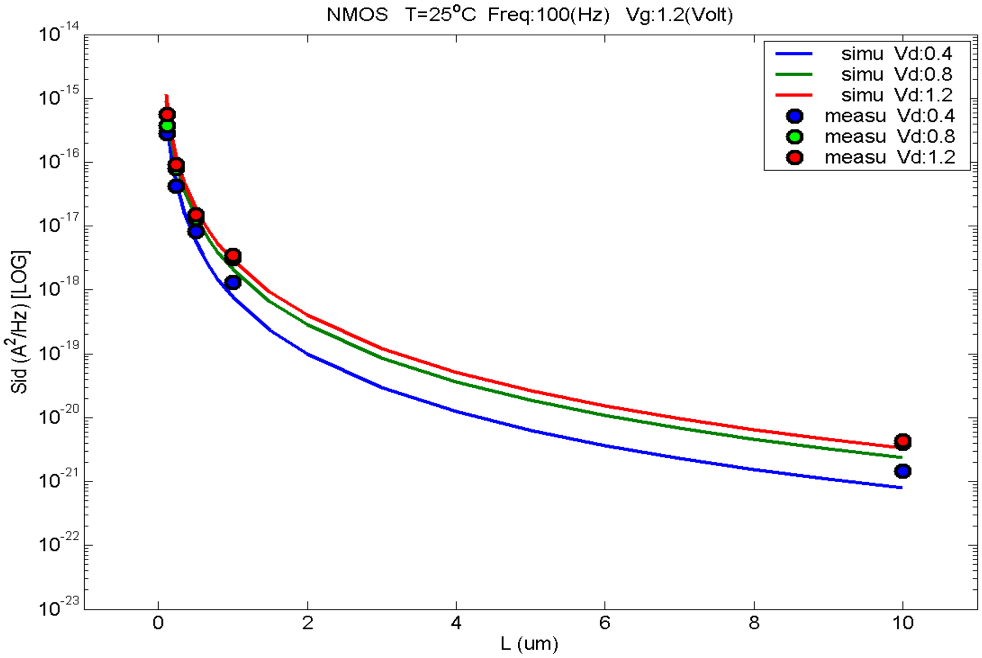


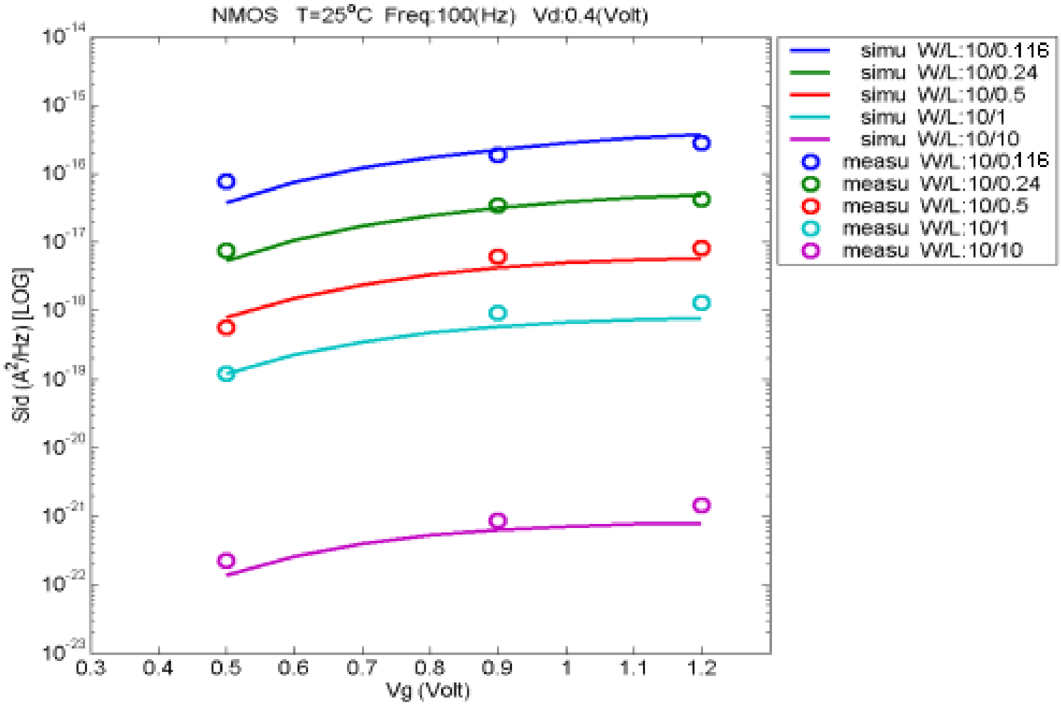


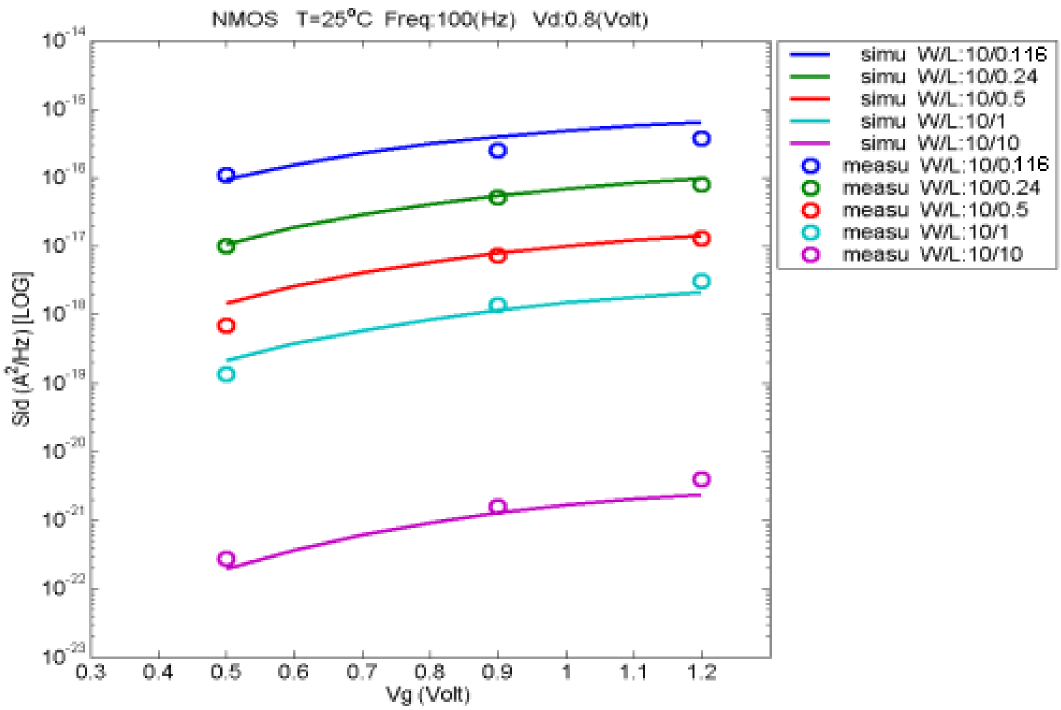

NMOS T=25°C Freq:100(Hz) Vd:1.2(Volt)
simu Vg:0.5
simu Vg:0.9
simu Vg:1.2
measu Vg:0.5
measu Vg:0.9
measu Vg:1.2
Sid (A²/Hz) [LOG]
L (um)

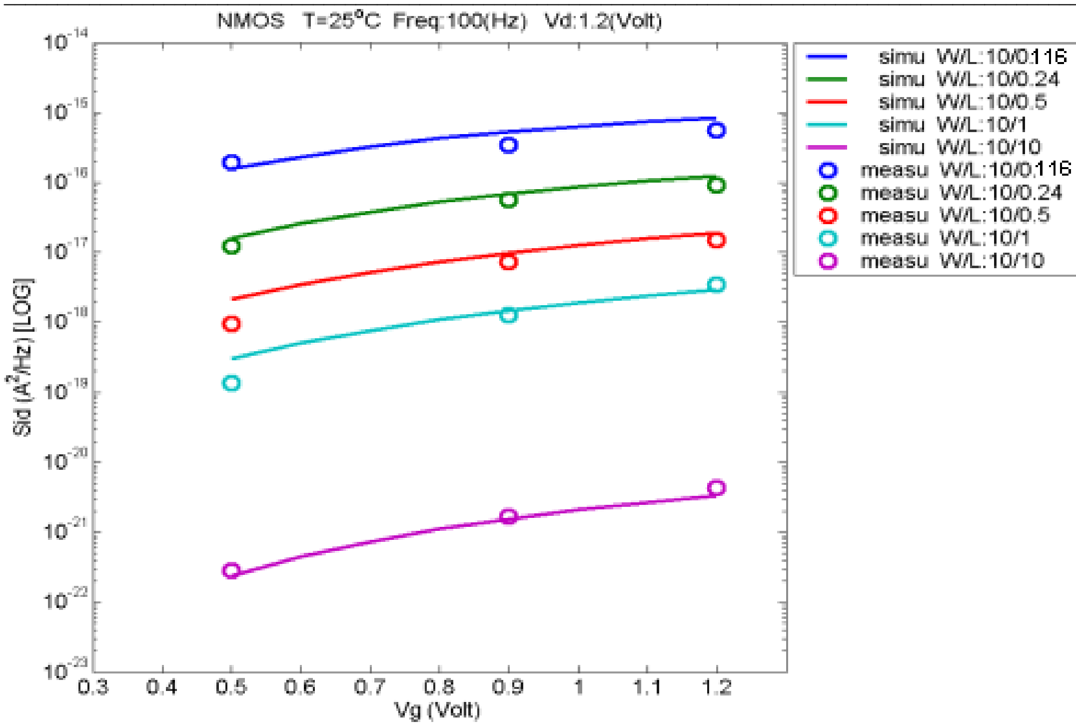

NMOS T=25°C Freq:100(Hz) Vg:0.5(Volt)
simu Vd:0.4
simu Vd:0.8
simu Vd:1.2
measu Vd:0.4
measu Vd:0.8
measu Vd:1.2
Sid (A²/Hz) [LOG]
L (um)

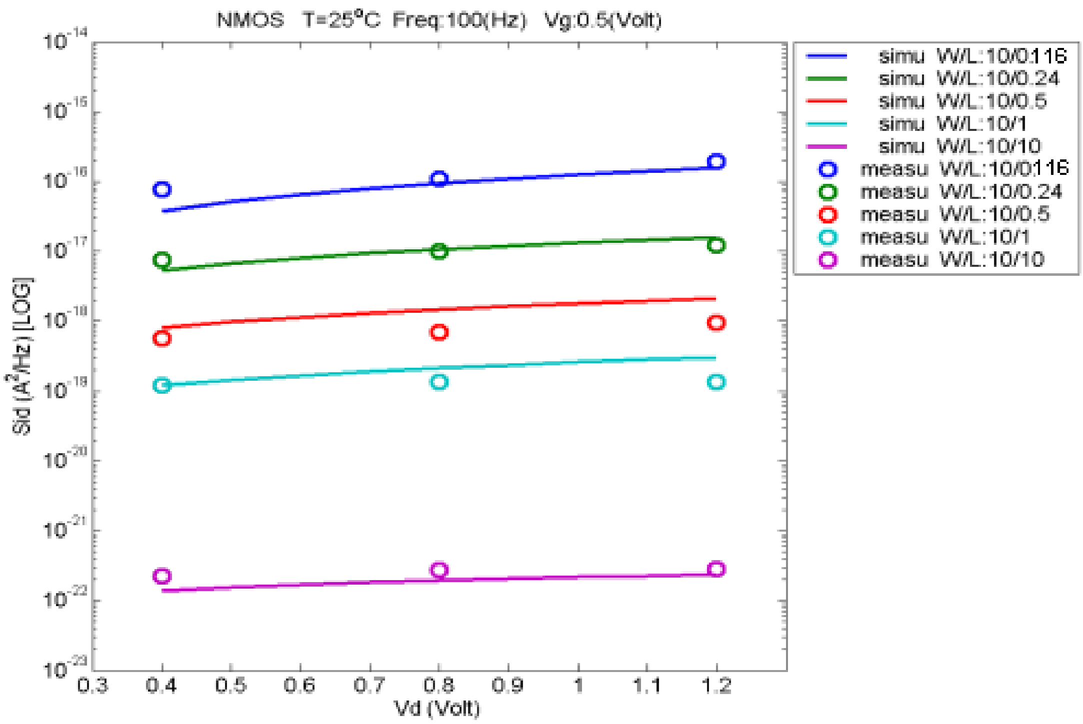

NMOS T=25°C Freq:100(Hz) Vg:0.9(Volt)
simu Vd:0.4
simu Vd:0.8
simu Vd:1.2
measu Vd:0.4
measu Vd:0.8
measu Vd:1.2
Sid (A²/Hz) [LOG]
L (um)

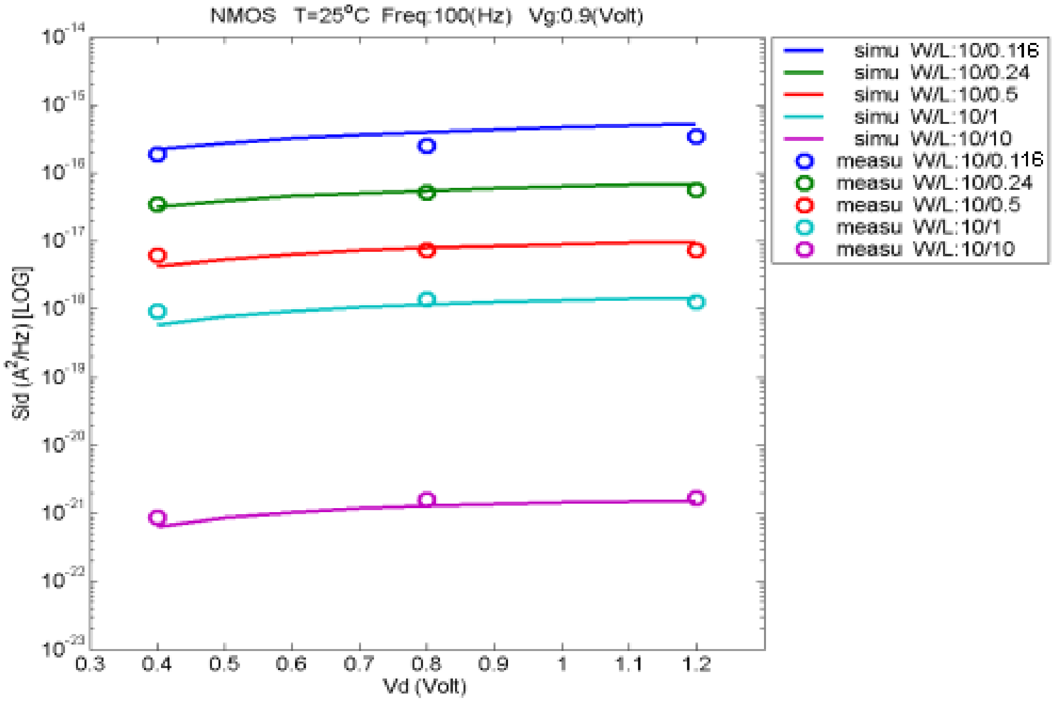

NMOS T=25°C Freq:100(Hz) Vg:1.2(Volt)
simu Vd:0.4
simu Vd:0.8
simu Vd:1.2
measu Vd:0.4
measu Vd:0.8
measu Vd:1.2
Sid (A²/Hz) [LOG]
L (um)

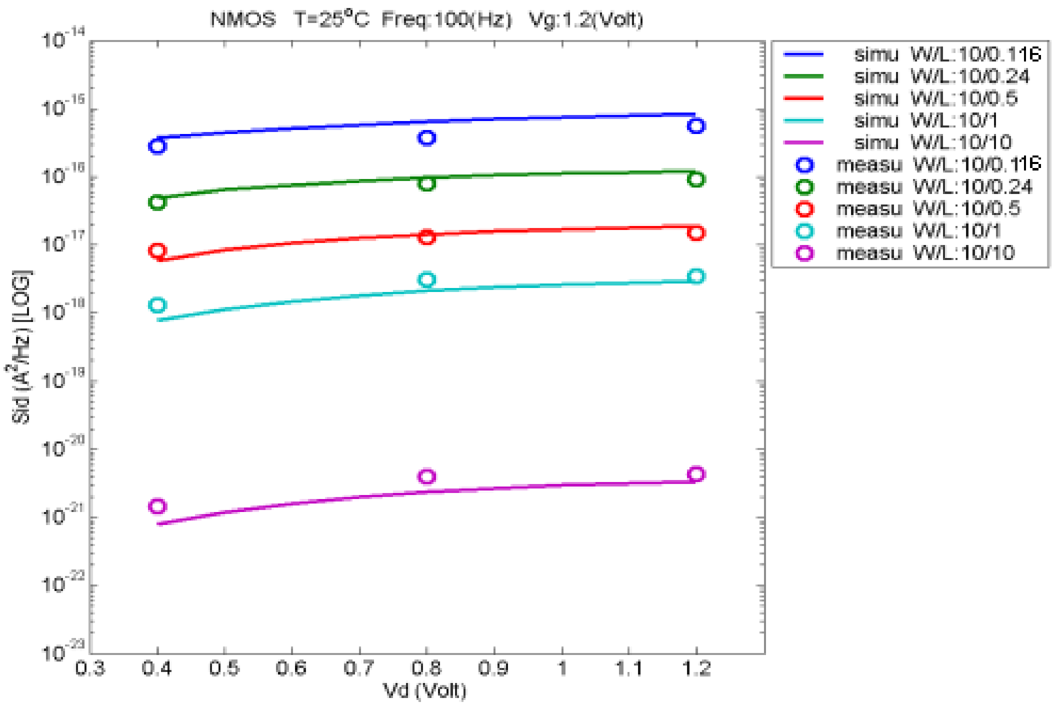

NMOS T=25°C Freq:100(Hz) Vd:0.4(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


NMOS T=25°C Freq:100(Hz) Vd:0.8(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

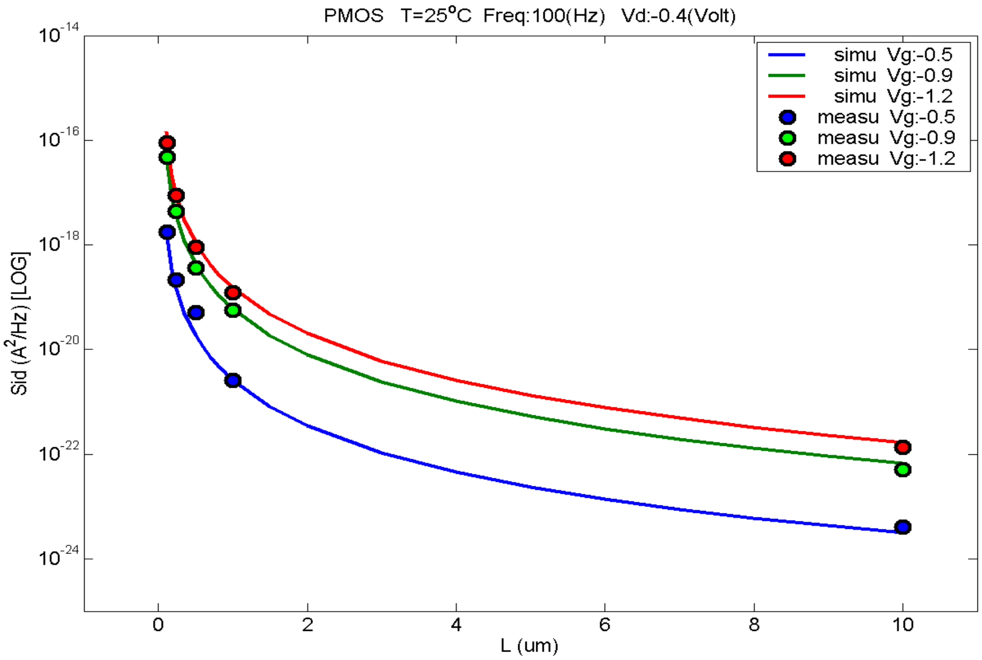

NMOS T=25°C Freq:100(Hz) Vd:1.2(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

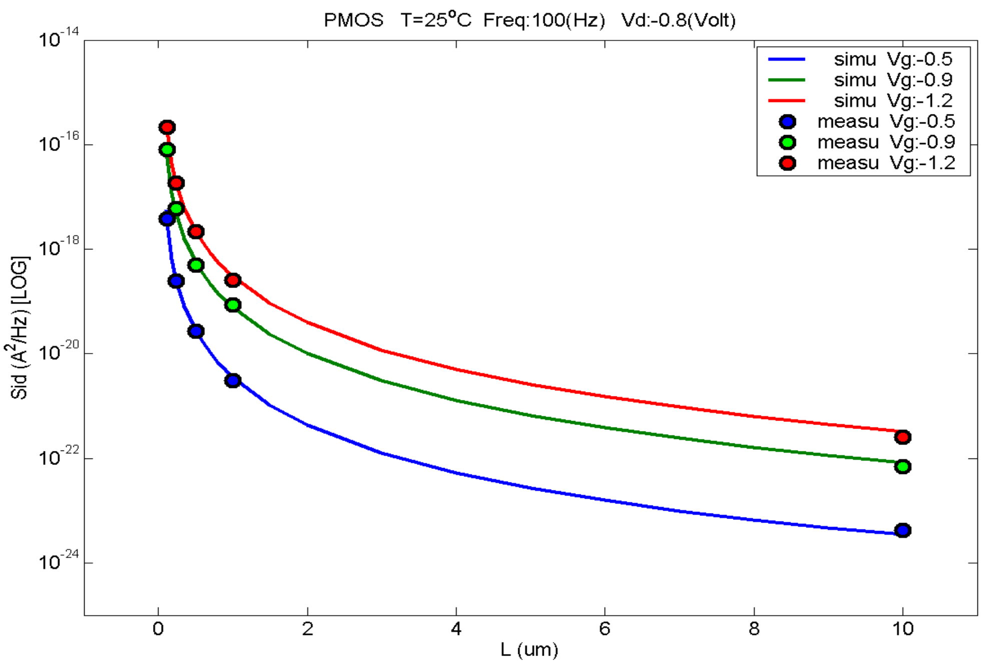

NMOS T=25°C Freq:100(Hz) Vg:0.5(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

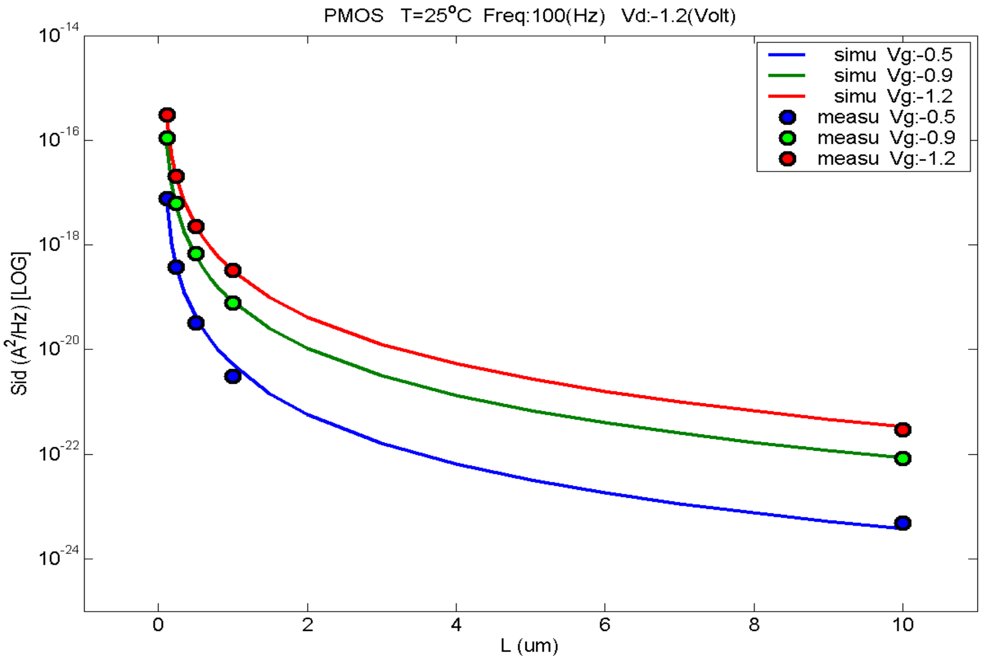

NMOS T=25°C Freq:100(Hz) Vg:0.9(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

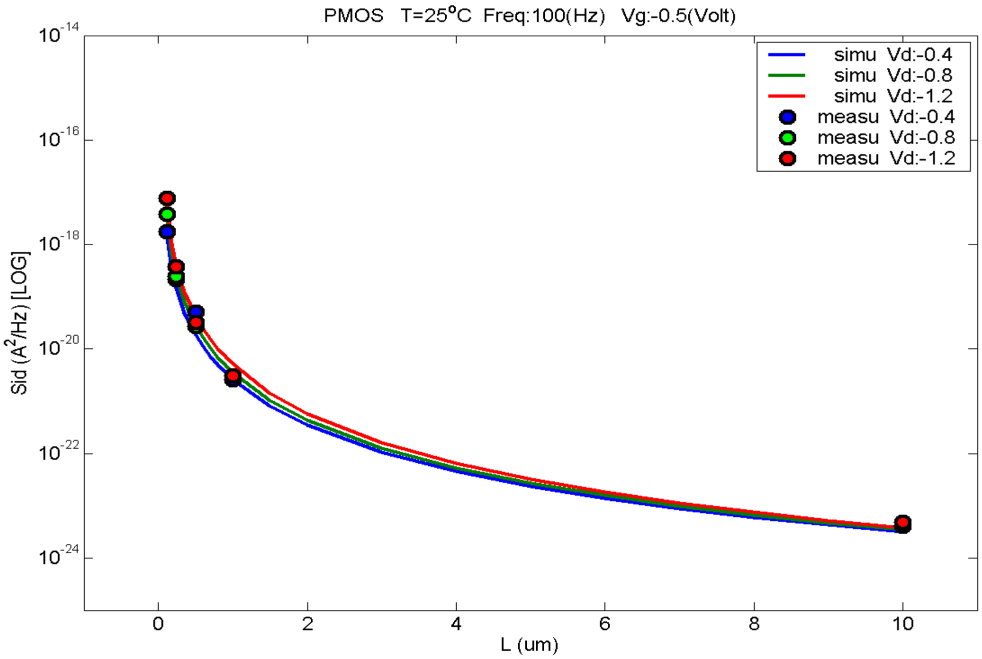

NMOS T=25°C Freq:100(Hz) Vg:1.2(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

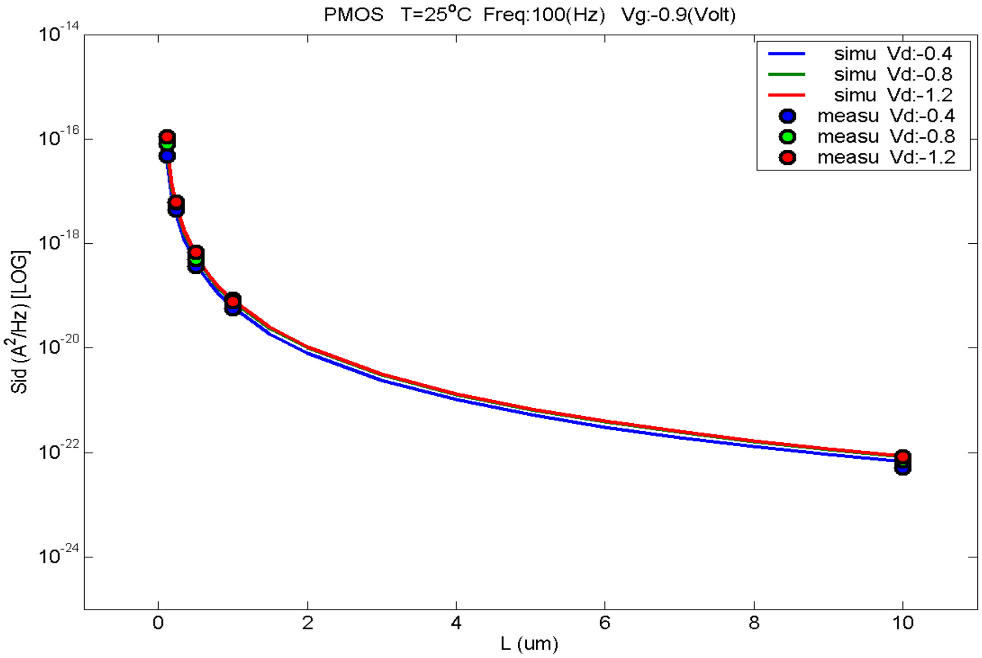
B. PMOS

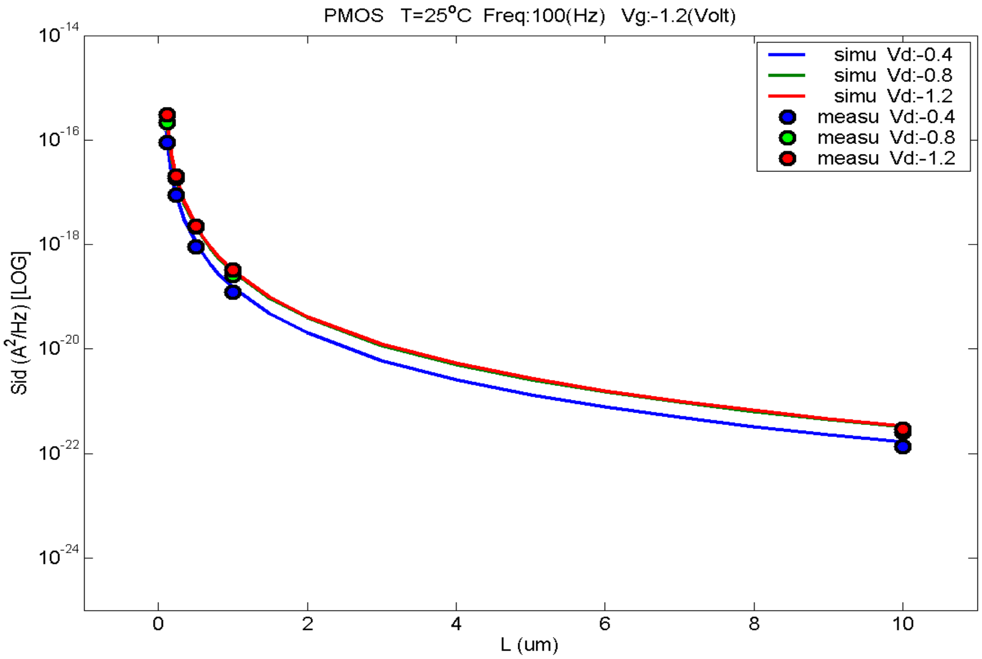

PMOS T=25°C Freq:100(Hz) Vd:-0.4(Volt)
simu Vg:-0.5
simu Vg:-0.9
simu Vg:-1.2
measu Vg:-0.5
measu Vg:-0.9
measu Vg:-1.2
Sid (A²/Hz) [LOG]
10^-14
10^-16
10^-18
10^-20
10^-22
10^-24
0
2
4
6
8
10
L (um)

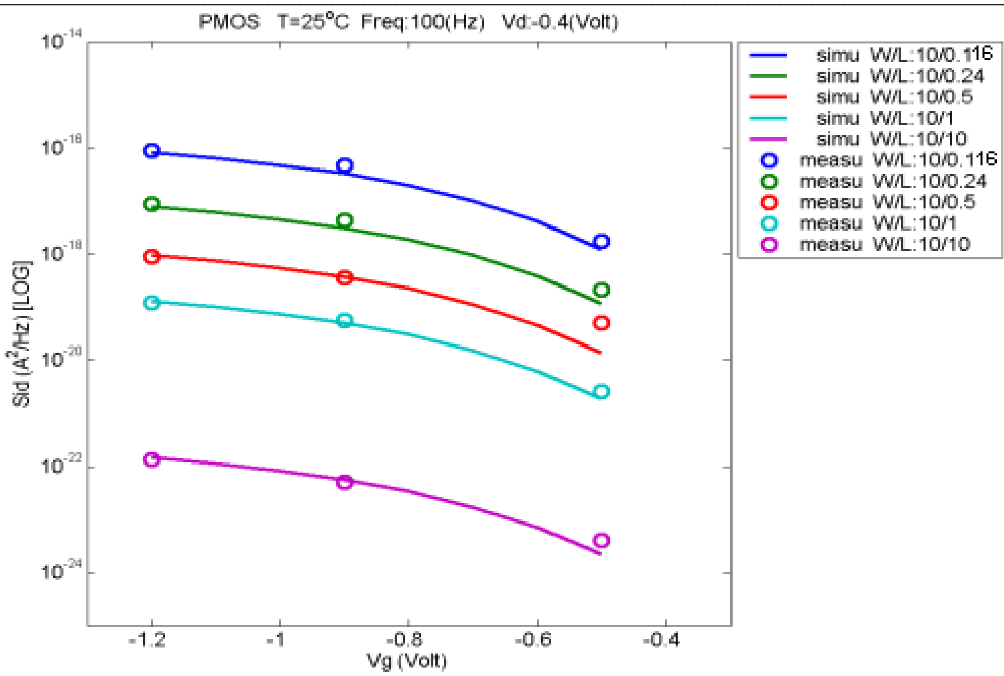

PMOS T=25°C Freq:100(Hz) Vd:-0.8(Volt)
simu Vg:-0.5
simu Vg:-0.9
simu Vg:-1.2
measu Vg:-0.5
measu Vg:-0.9
measu Vg:-1.2
Sid (A²/Hz) [LOG]
10^-14
10^-16
10^-18
10^-20
10^-22
10^-24
0
2
4
6
8
10
L (um)

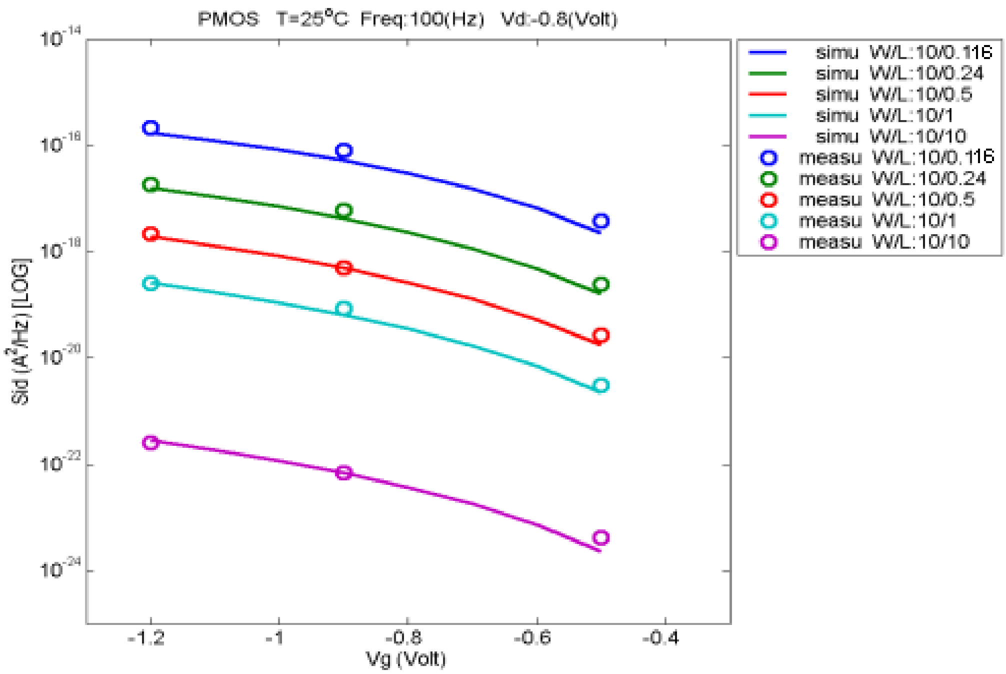

PMOS T=25°C Freq:100(Hz) Vd:-1.2(Volt)
simu Vg:-0.5
simu Vg:-0.9
simu Vg:-1.2
measu Vg:-0.5
measu Vg:-0.9
measu Vg:-1.2
Sid (A²/Hz) [LOG]
L (um)

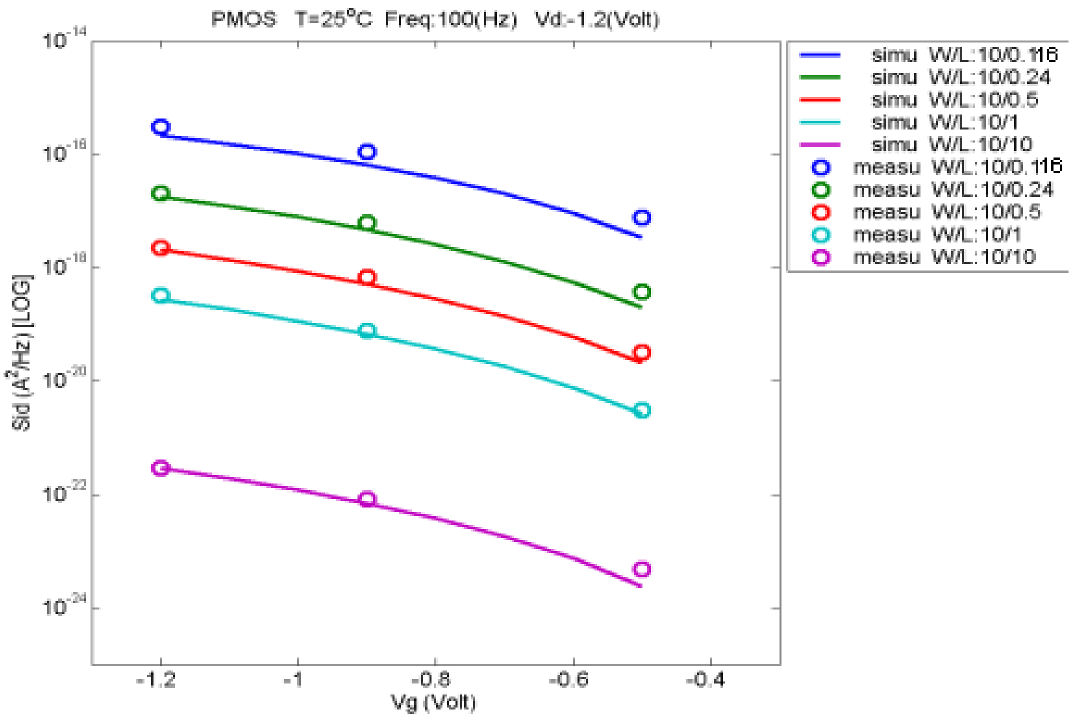

PMOS T=25°C Freq:100(Hz) Vg:-0.5(Volt)
simu Vd:-0.4
simu Vd:-0.8
simu Vd:-1.2
measu Vd:-0.4
measu Vd:-0.8
measu Vd:-1.2
Sid (A²/Hz) [LOG]
L (um)

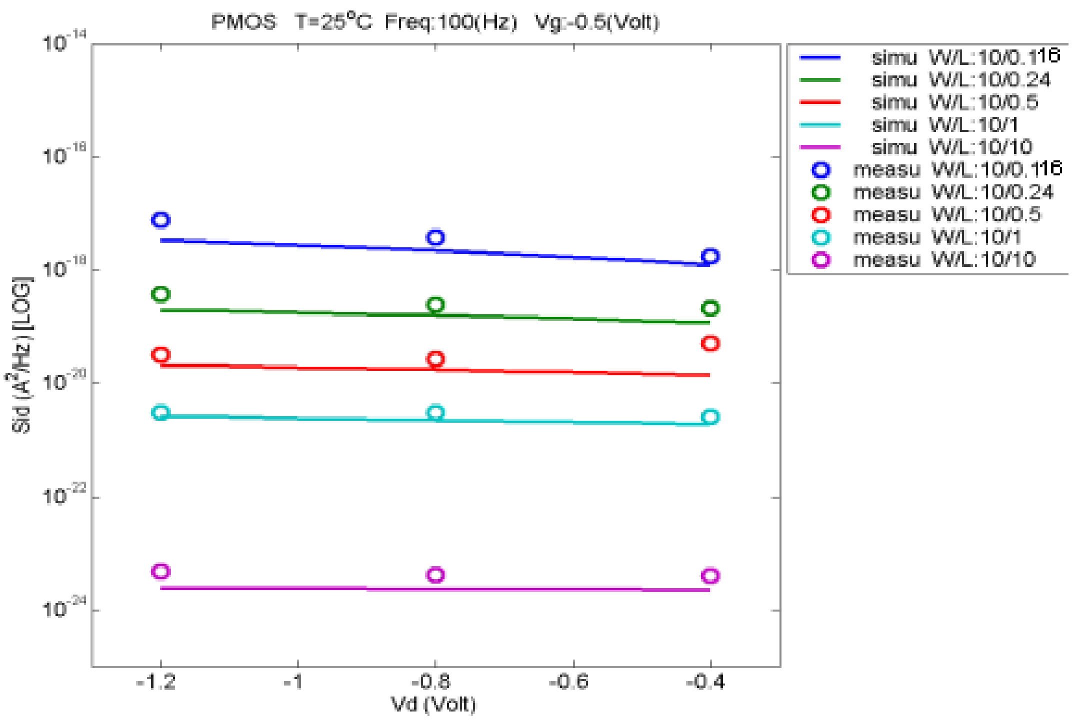

PMOS T=25°C Freq:100(Hz) Vg:-0.9(Volt)
simu Vd:-0.4
simu Vd:-0.8
simu Vd:-1.2
measu Vd:-0.4
measu Vd:-0.8
measu Vd:-1.2
Sid (A²/Hz) [LOG]
L (um)
10^-14
10^-16
10^-18
10^-20
10^-22
10^-24
0
2
4
6
8
10

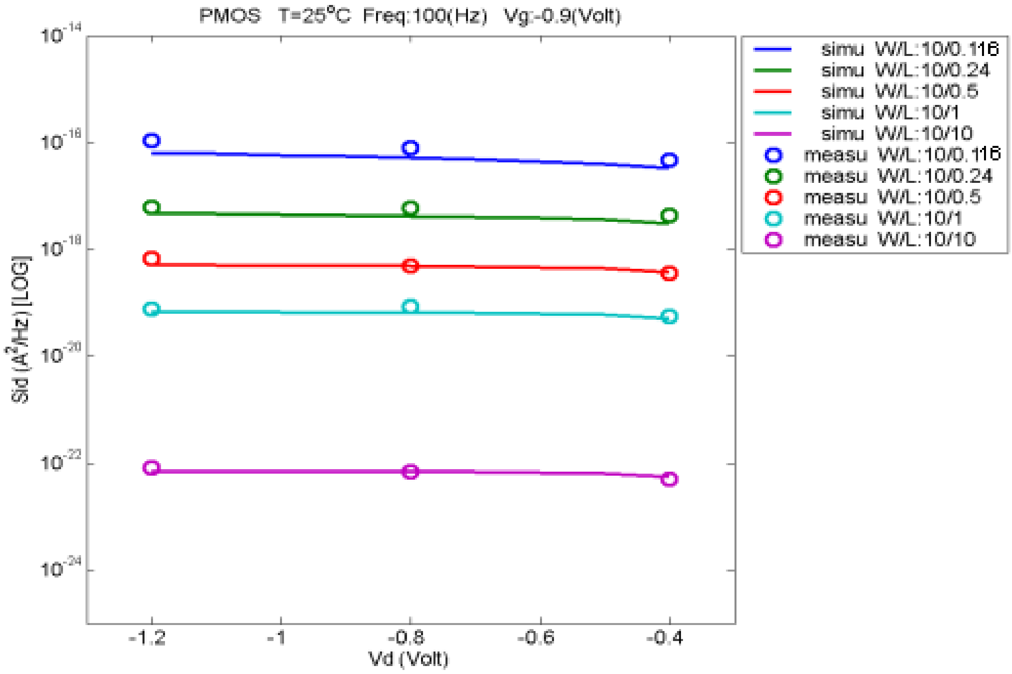

PMOS T=25°C Freq:100(Hz) Vg:-1.2(Volt)
simu Vd:-0.4
simu Vd:-0.8
simu Vd:-1.2
measu Vd:-0.4
measu Vd:-0.8
measu Vd:-1.2
Sid (A²/Hz) [LOG]
L (um)
10^-14
10^-16
10^-18
10^-20
10^-22
10^-24
0
2
4
6
8
10

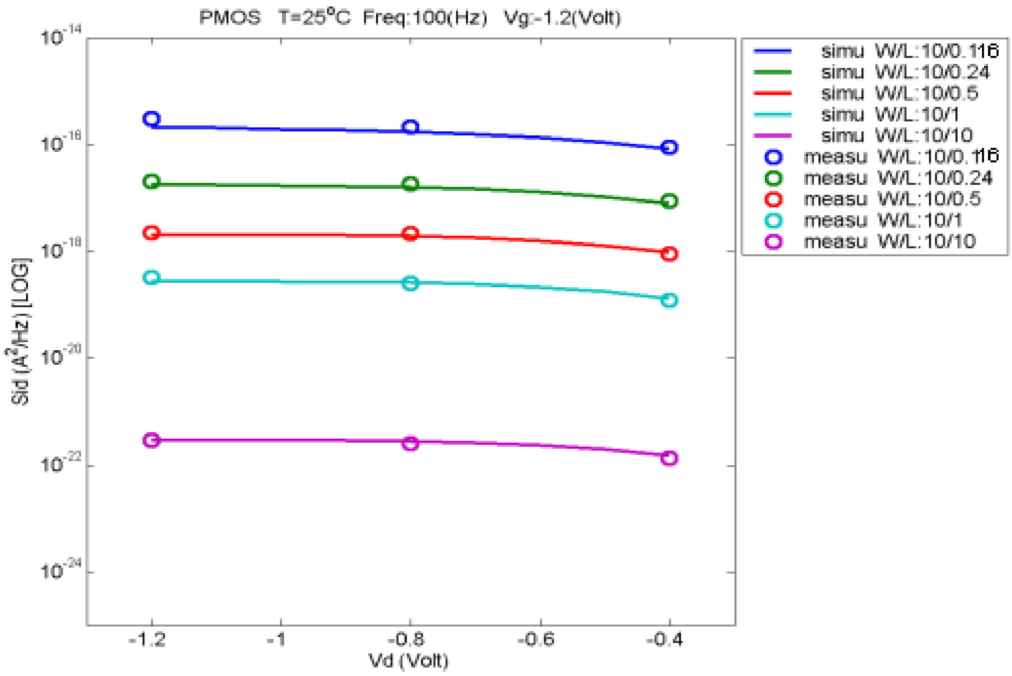

PMOS T=25°C Freq:100(Hz) Vd:-0.4(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


PMOS T=25°C Freq:100(Hz) Vd:-0.8(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


PMOS T=25°C Freq:100(Hz) Vd:-1.2(Volt)
Sid (A²/Hz) [LOG]
Vg (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


PMOS T=25°C Freq:100(Hz) Vg:-0.5(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


PMOS T=25°C Freq:100(Hz) Vg:-0.9(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10


PMOS T=25°C Freq:100(Hz) Vg:-1.2(Volt)
Sid (A²/Hz) [LOG]
Vd (Volt)
simu W/L:10/0.116
simu W/L:10/0.24
simu W/L:10/0.5
simu W/L:10/1
simu W/L:10/10
measu W/L:10/0.116
measu W/L:10/0.24
measu W/L:10/0.5
measu W/L:10/1
measu W/L:10/10

8 Appendix

8.1 Verify corner plot.

Table 8-1 Verify wafer information

Mask ID	A11001	A11001	A11001
Lot ID	D60K6	D5N0S	D63LJ
Wafer Number	#1 and #3	#1 and #2	#8 and #9

Threshold voltage in linear region V_{TLN}

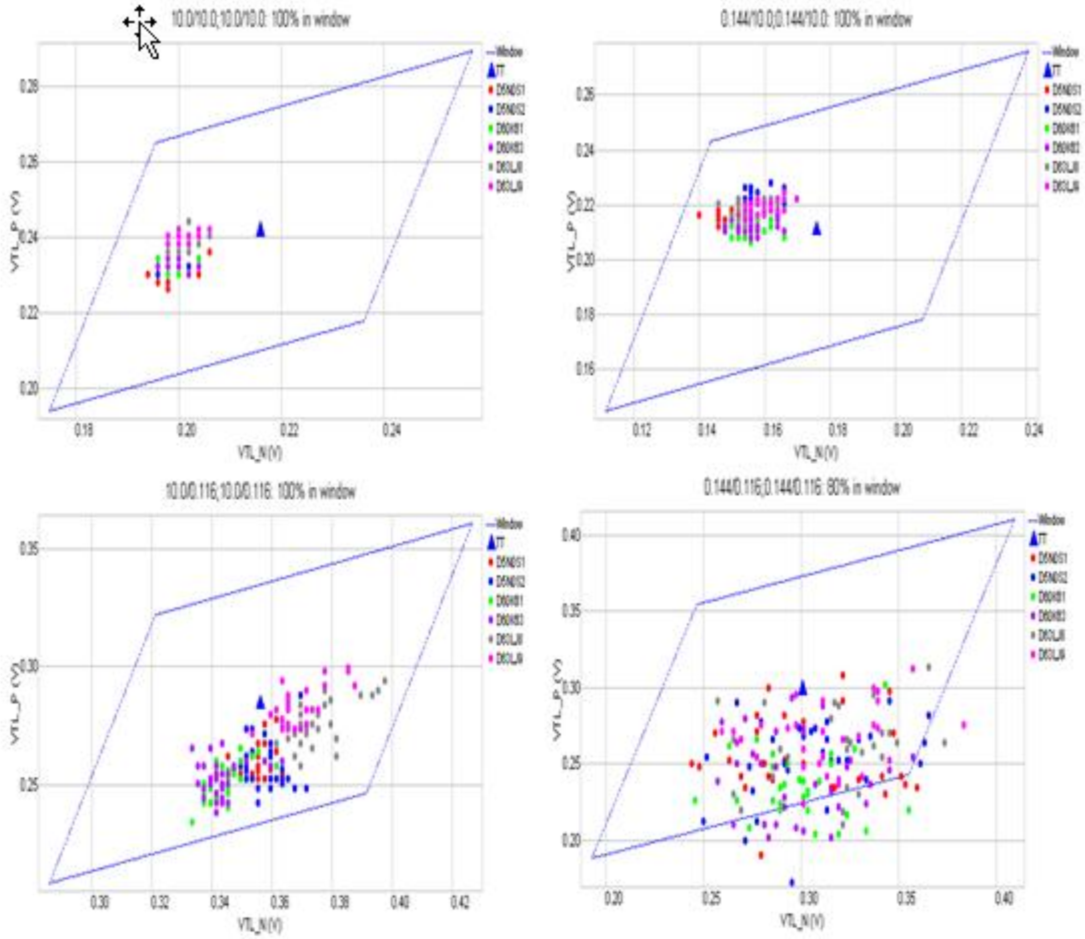


Figure 8-1 Corner graph of threshold voltage at low drain bias

Threshold voltage in Saturation region V_{Tsat}

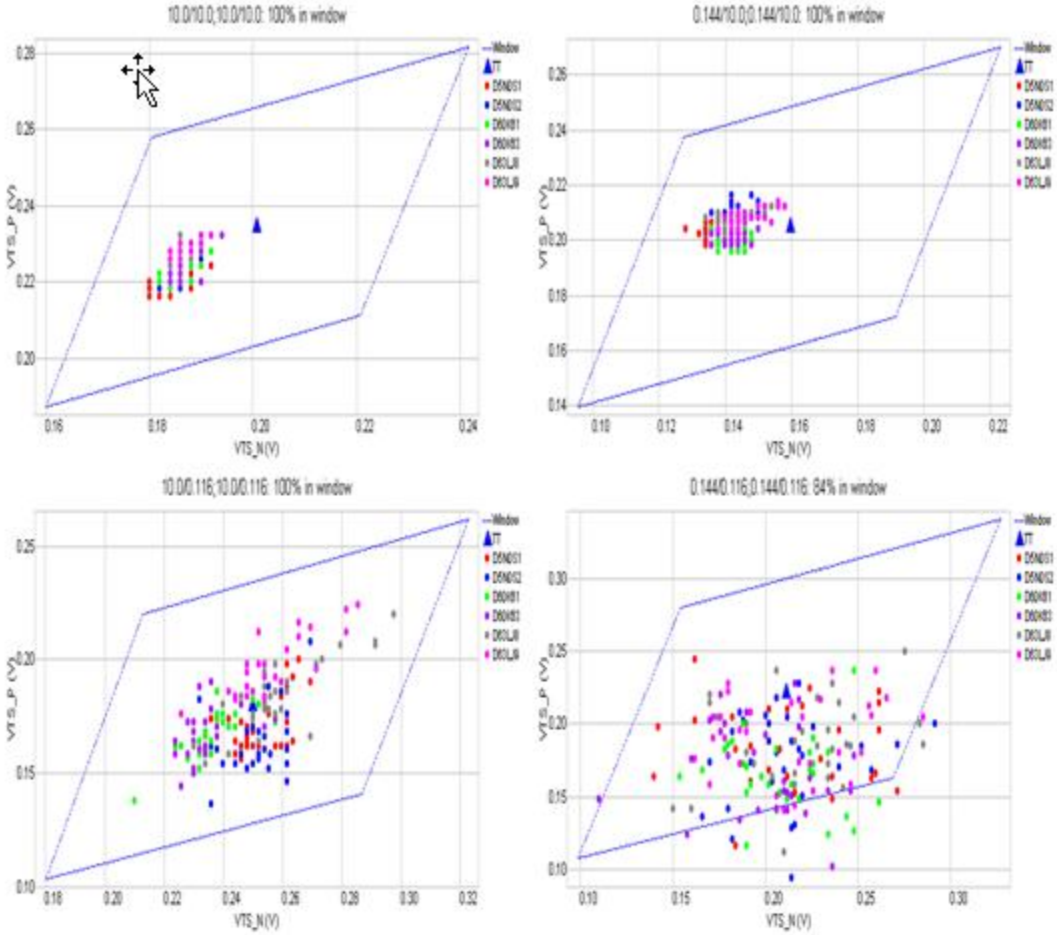


Figure 8-2 Corner graph of threshold voltage at high drain bias

Linear Current I_{LIN}

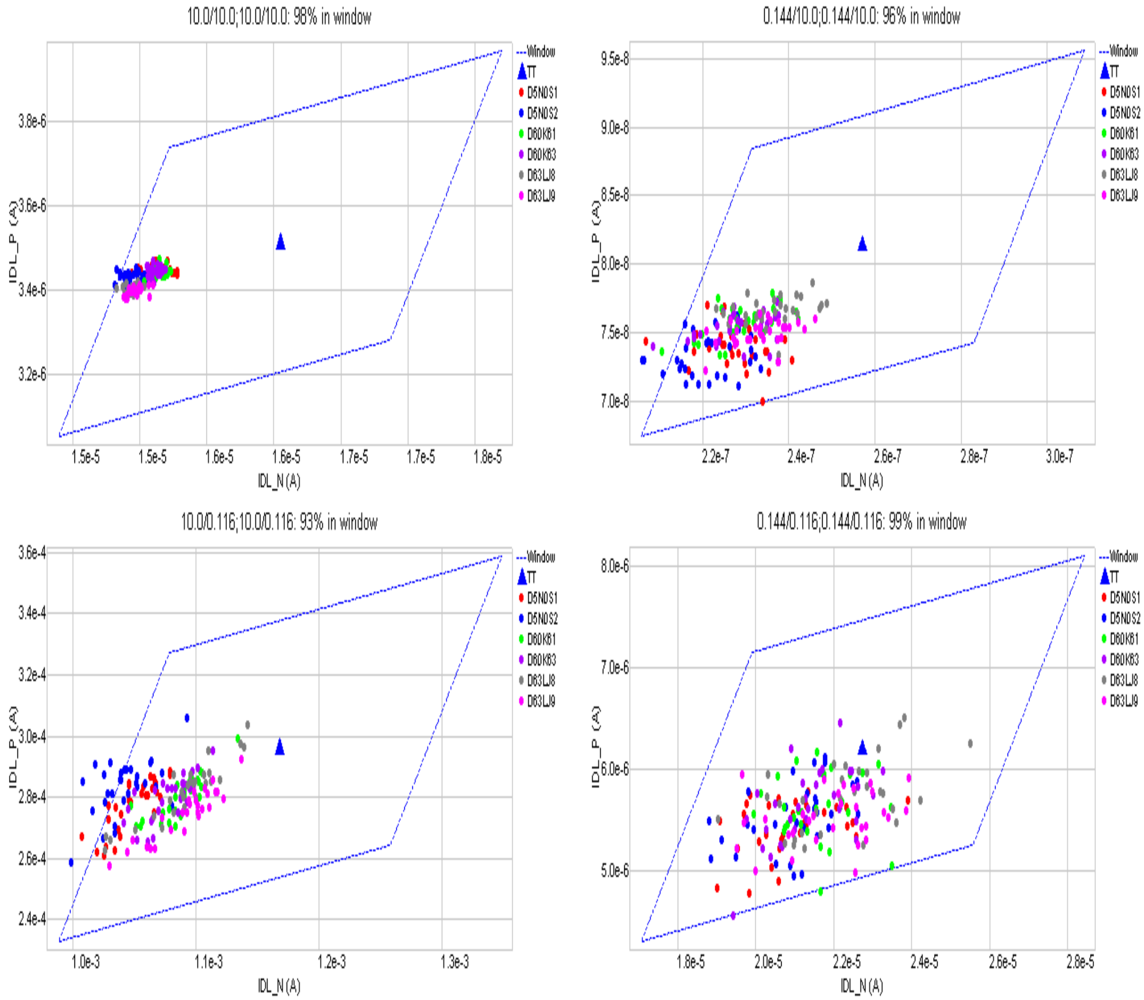


Figure 8-3 Corner graph of linear current

Saturation Current I_{ON}

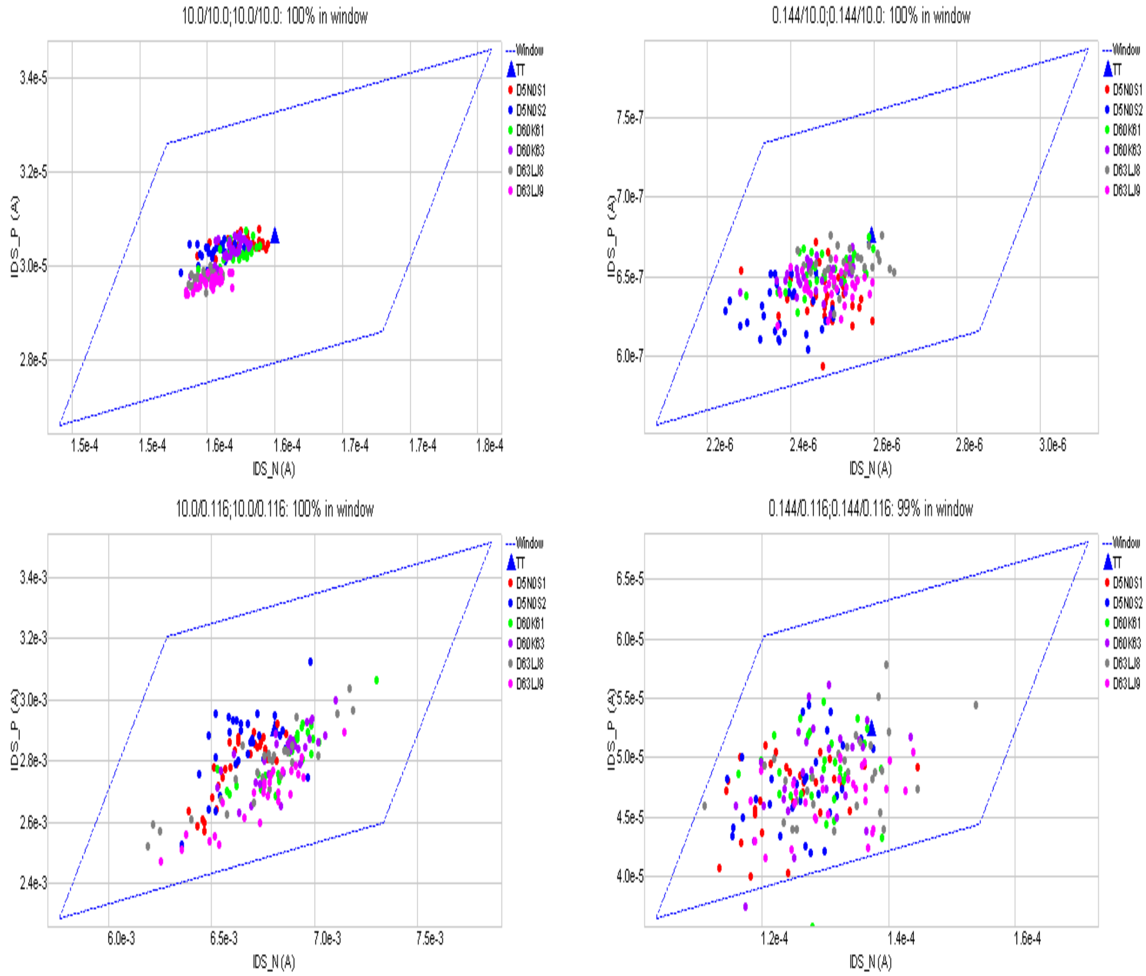


Figure 8-4 Corner graph of on-current

8.2 HSPICE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

8.3 SPECTRE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

8.4 ELDO Warning Message

Item	Warning Message	Explanation
1.	**P_12_HSL110E:** **Rds at current temperature = -5.73415 is negative. Set to zero.**	**Please ignore the warning in theELDO simulation**